



Discrete Mathematics and Its Applications

7th Edition

By Kenneth Rosen

Chapter 10

Graph

- 10.1 Graphs and Graph Models
- 10.2 Graph Terminology and Special
Types of Graphs
- 10.3 Representing Graphs and Graph
- 10.4 Connectivity
- 10.5 Euler and Hamilton Paths
- 10.6 Shortest-Path Problems
- 10.7 Planar Graphs
- 10.8 Graph Coloring

§10.1 Graphs and Graph Models

Def 1. A **graph** $G = (V, E)$ consists of V , a nonempty set of **vertices** (or **nodes**), and E , a set of **edges**. Each edge has either one or two vertices associated with it, called its **endpoints**. An edge is said to **connect** its endpoints.

$G=(V, E)$, where

$V=\{v_1, v_2, \dots, v_7\}$

$E=\{\{v_1, v_2\}, \{v_1, v_3\}, \{v_2, v_3\}$
 $\{v_3, v_4\}, \{v_4, v_5\}, \{v_4, v_6\}$
 $\{v_4, v_7\}, \{v_5, v_6\}, \{v_6, v_7\}\}$

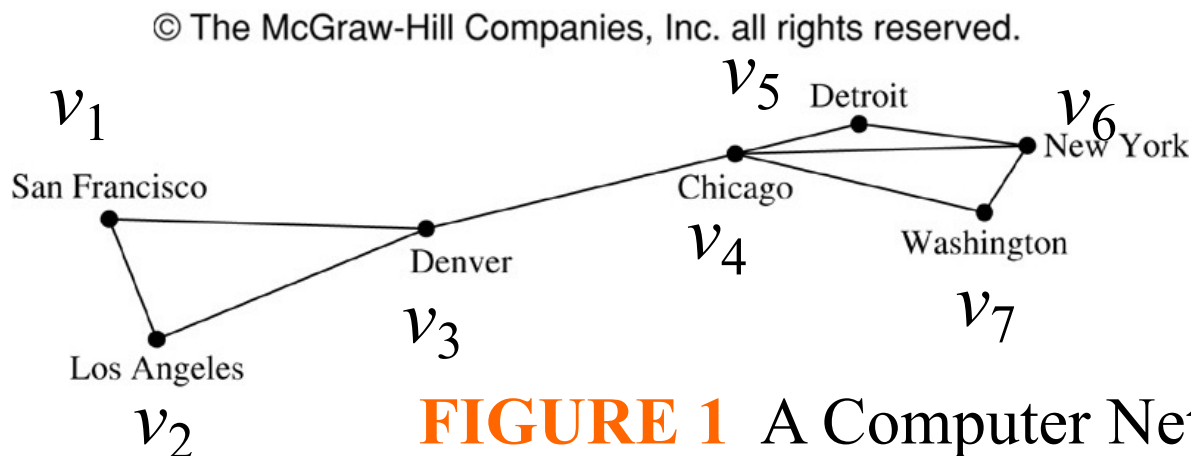


FIGURE 1 A Computer Network.

Def A graph in which each edge connects two different vertices and where no two edges connect the same pair of vertices is called a **simple graph**.

Def **Multigraph**:

simple graph + multiple edges (**multiedges**)
(兩點間允許多條邊)

Multigraphs may have multiple edges connecting the same two vertices. When m different edges connect the vertices u and v , we say that $\{u, v\}$ is an edge of *multiplicity* m .

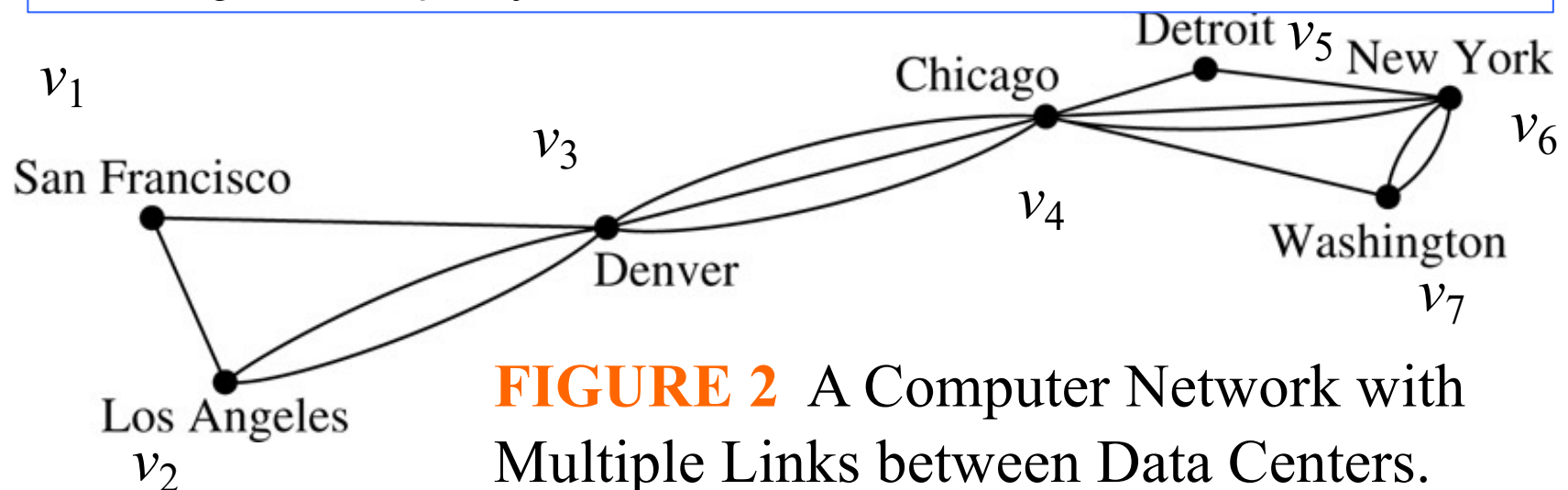
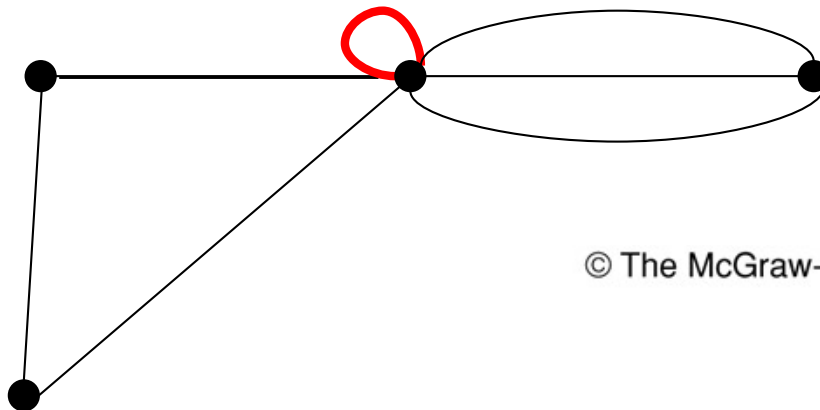


FIGURE 2 A Computer Network with Multiple Links between Data Centers.

Def. Pseudograph:

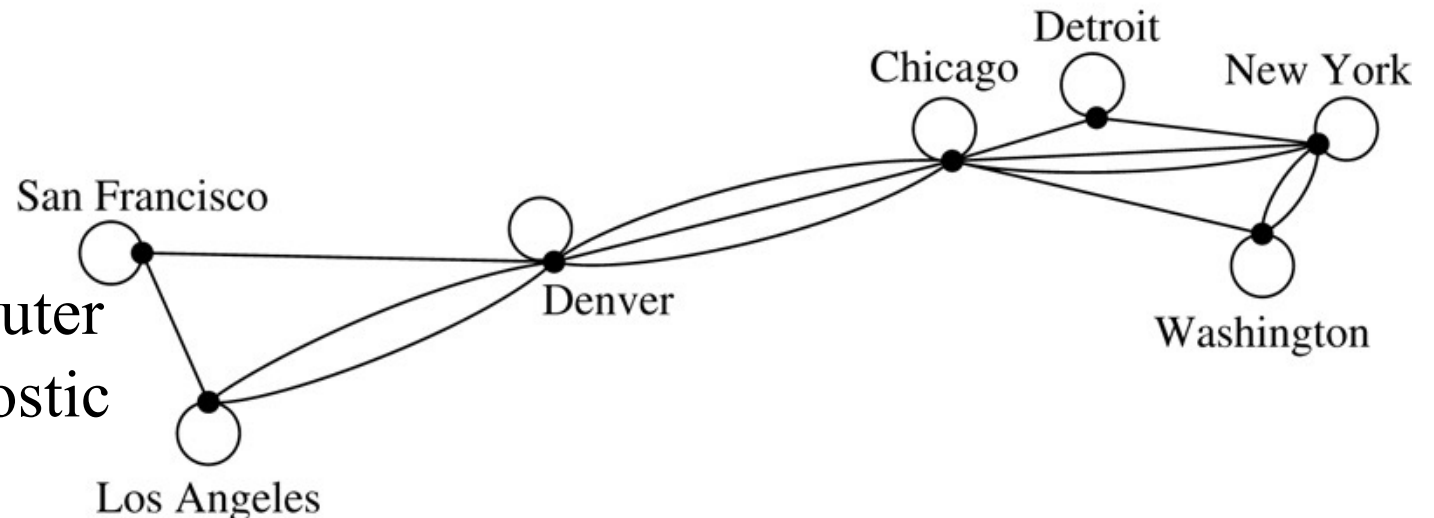
simple graph + multiedge
+ loop
(a loop: )

eg.

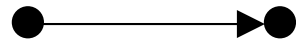


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FIGURE 3 A Computer Network with Diagnostic Links.

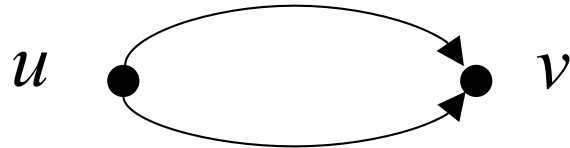


Def 2. Directed graph (digraph):
simple graph with each edge directed

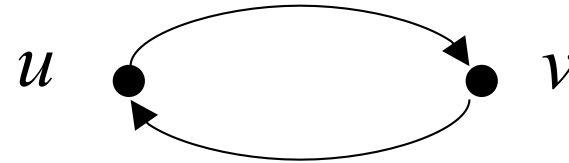


Note:  is allowed in a directed graph

Note:



The two edges $(u,v), (u,v)$
are multiedges.



The two edges $(u,v), (v,u)$
are not multiedges.

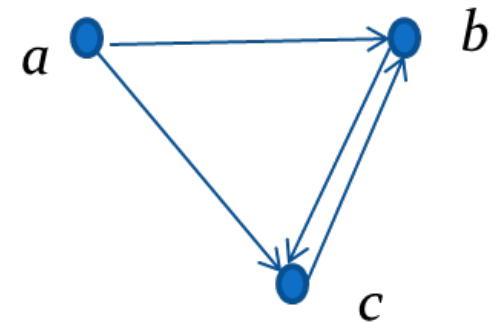
Def. Directed multigraph: digraph+multiedges

Some Terminology

A *simple directed graph* has no loops and no multiple edges.

Example:

This is a directed graph with three vertices and four edges.



A *directed multigraph* may have multiple directed edges. When there are m directed edges from the vertex u to the vertex v , we say that (u,v) is an edge of *multiplicity* m .

Example:

In this directed multigraph the multiplicity of (a,b) is 1 and the multiplicity of (b,c) is 2.

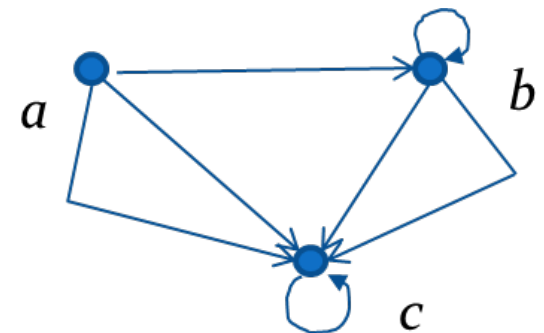


FIGURE 4 (10.1)

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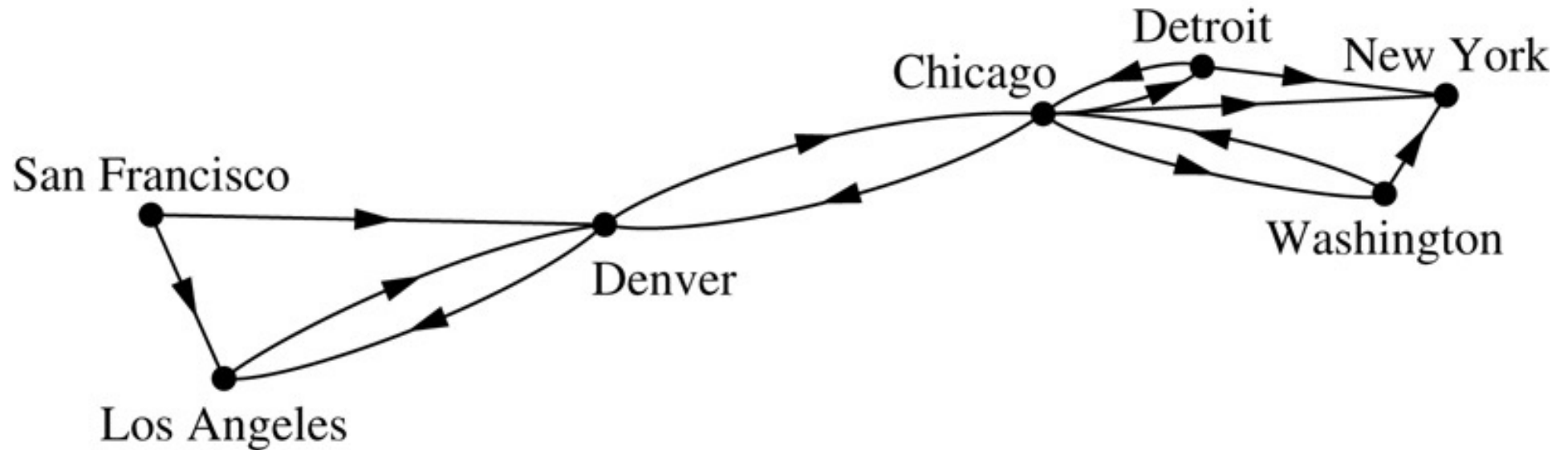


FIGURE 4 A Communications Network with **One-Way** Communications Links.

FIGURE 5 (10.1)

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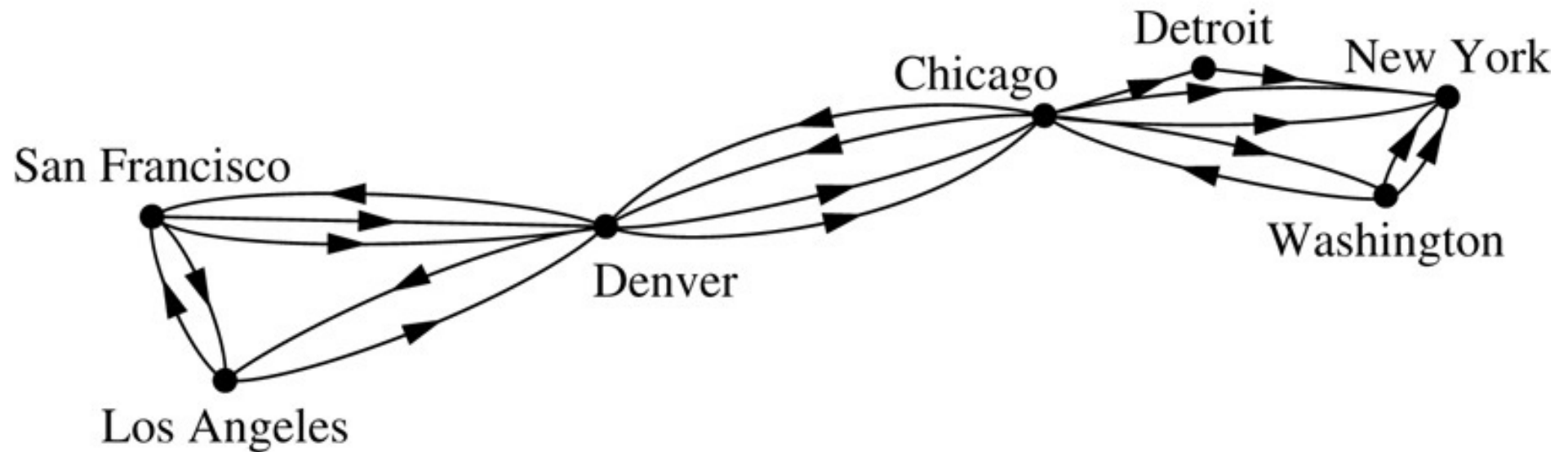


FIGURE 5 A Computer Network with **Multiple One-Way** Links.

Graph Terminology: Summary

To understand the structure of a graph and to build a graph model, we ask these questions:

- Are the edges of the graph undirected or directed (or both)?
- If the edges are undirected, are multiple edges present that connect the same pair of vertices? If the edges are directed, are multiple directed edges present?
- Are loops present?

TABLE 1 Graph Terminology.

<i>Type</i>	<i>Edges</i>	<i>Multiple Edges Allowed?</i>	<i>Loops Allowed?</i>
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

Graph Models

Example 1. (Acquaintanceship graphs)

We can use a simple graph to represent whether two people know each other. Each **person** is represented by a **vertex**. An **undirected edge** is used to connect two people when these people **know each other**.

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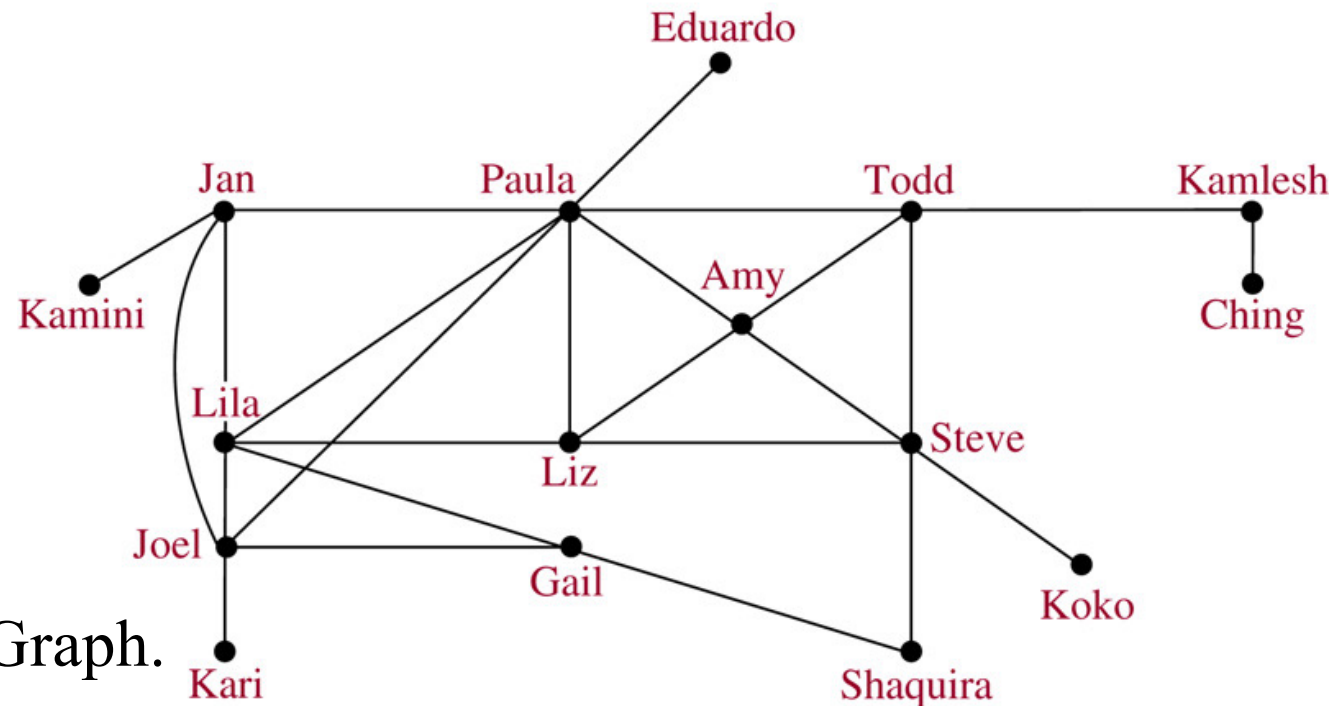


FIGURE 6

Acquaintanceship Graph.

Example 2. (Influence graphs)

In studies of group behavior it is observed that certain people can influence the thinking of others. Simple digraph $\Rightarrow$ Each **person** of the group is represented by a **vertex**. There is a **directed edge** from vertex a to vertex b when the person a **influences** the person b .

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FIGURE 7

Influence Graph.

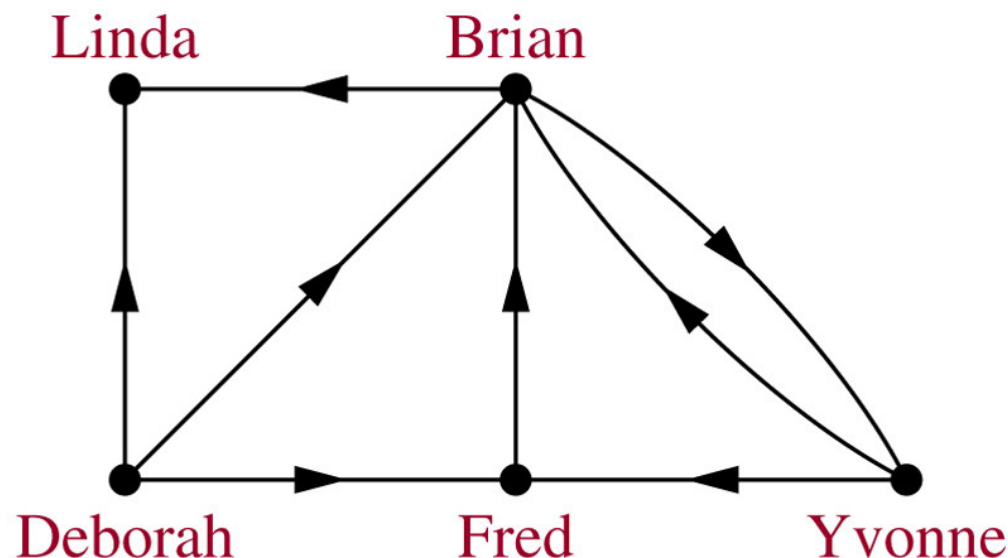


FIGURE 8 (10.1)

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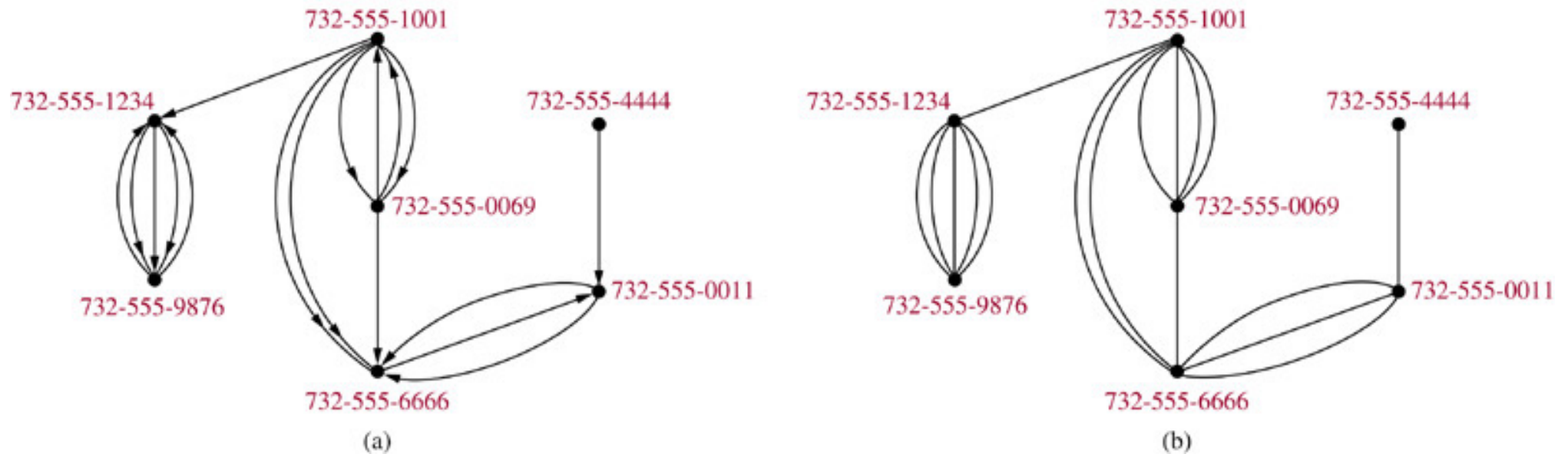


FIGURE 8 A Call Graph.

Example 9. (Precedence graphs and concurrent processing)

Computer programs can be executed more rapidly by executing certain statements concurrently. It is important not to execute a statement that requires results of statements not yet executed.

Simple digraph $\Rightarrow$ Each **statement** is represented by a **vertex**, and there is an **edge** from a to b if the statement of b **cannot be executed before** the statement of a .

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S_1 $a := 0$
 S_2 $b := 1$
 S_3 $c := a + 1$
 S_4 $d := b + a$
 S_5 $e := d + 1$
 S_6 $e := c + d$

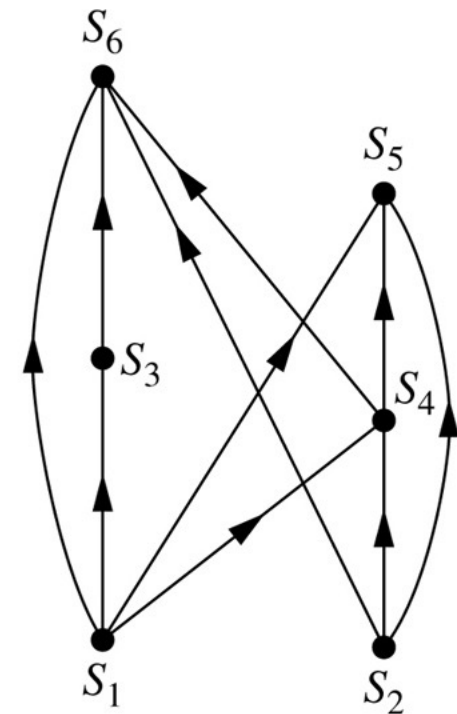


FIGURE 10 A Precedence Graph.

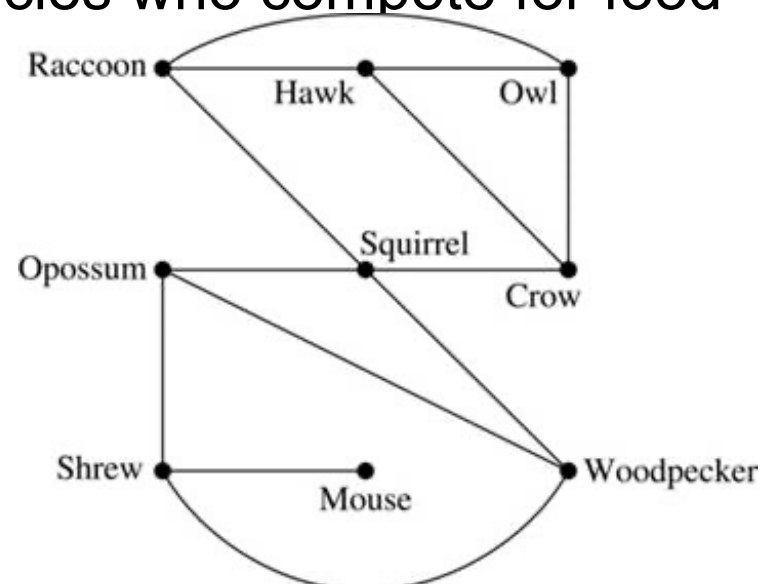
Biological Applications

Graph models are used extensively in many areas of the biological science. We will describe two such models, one to ecology and the other to molecular biology.

Niche overlap graphs model competition between species in an ecosystem

- Vertices represent species and an edge connects two vertices when they represent species who compete for food resources.

Example: This is the niche overlap graph for a forest ecosystem with nine species.



Biological Applications

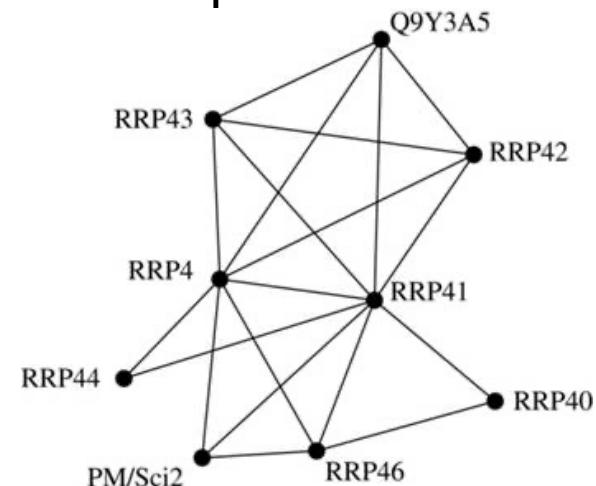
We can model the interaction of proteins in a cell using a ***protein interaction network***.

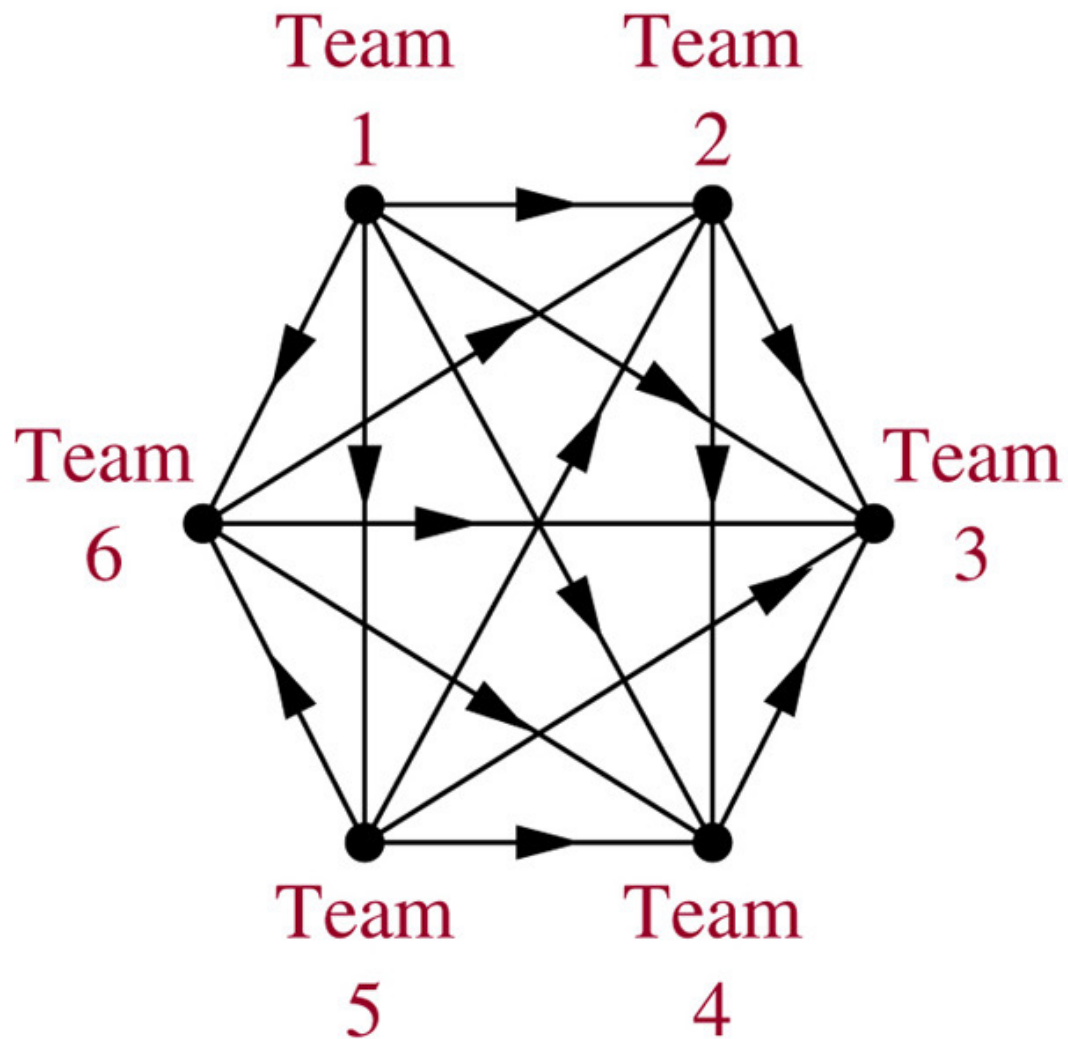
In a ***protein interaction graph***, vertices represent proteins and vertices are connected by an edge if the proteins they represent interact.

Protein interaction graphs can be huge and can contain more than 100,000 vertices, each representing a different protein, and more than 1,000,000 edges, each representing an interaction between proteins

Protein interaction graphs are often split into smaller graphs, called ***modules***, which represent the interactions between proteins involved in a particular function.

Example: This is a module of the protein interaction graph of proteins that degrade RNA in a human cell.





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FIGURE 13 A Graph Model of a **Round-Robin Tournament**.

§10.2 Graph Terminology and Special Types of Graphs

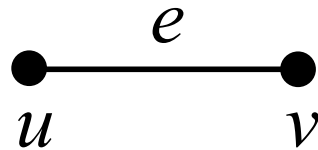
Definition 1. Two vertices u, v in an undirected graph G are called **adjacent** (or **neighbors**) in G if there is an edge e between u and v . Such an edge e is called *incident with* the vertices u and v and e is said to *connect* u and v . $N(A) = \bigcup_{v \in A} N(v)$.

Definition 2. The set of all neighbors of a vertex v of $G = (V, E)$, denoted by $N(v)$, is called the **neighborhood** of v . If A is a subset of V , we denote by $N(A)$ the set of all vertices in G that are adjacent to at least one vertex in A . So,

Definition 3. The **degree of a vertex** in a undirected graph is the number of edges incident with it, except that a loop at a vertex contributes two to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

§10.2 Graph Terminology and Special Types of Graphs

Def 1. G : undirected graph



u and v are **adjacent** (or **neighbors**)

e is **incident** with u and v

u and v are **endpoints** of e

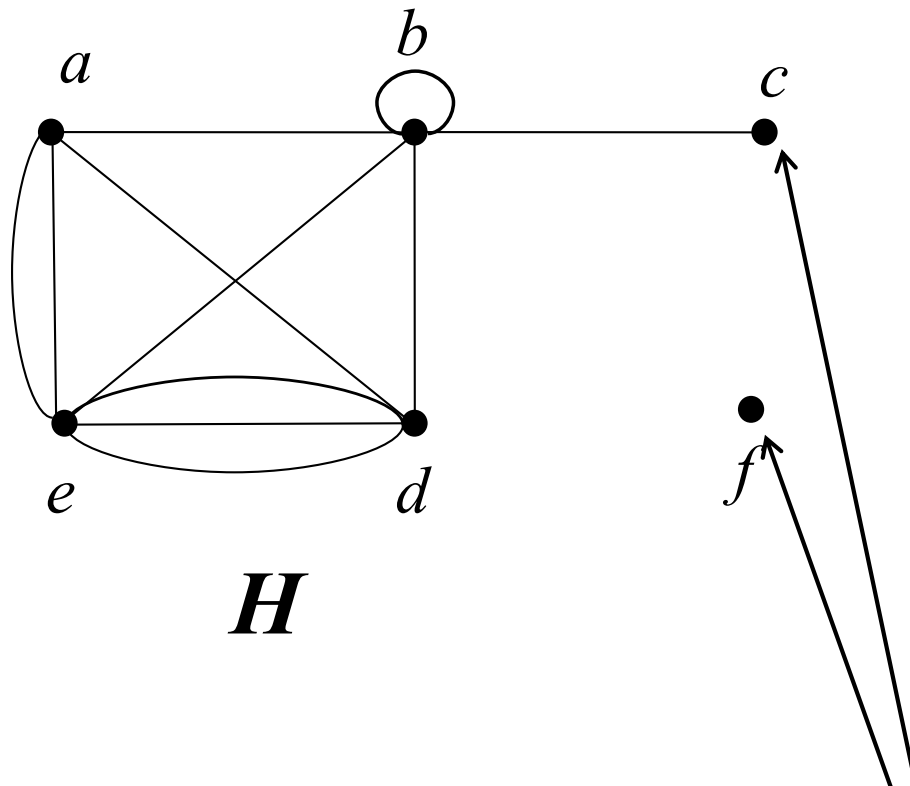
Def 3. The **degree** of a vertex v , denoted by **$\deg(v)$** , in an undirected graph is the number of edges incident with it.

(Note : A loop adds 2 to the degree.)

(a loop: )

Example 1. What are the degrees of the vertices in the graph H ?

Sol :



$$\deg(a)=4$$

$$\deg(b)=6$$

$$\deg(c)=1$$

$$\deg(d)=5$$

$$\deg(e)=6$$

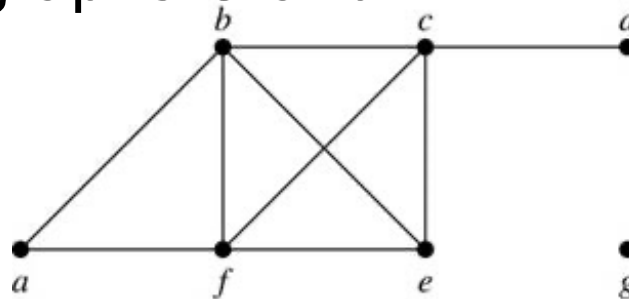
$$\deg(f)=0$$

Def. A vertex of degree 0 is called **isolated**.

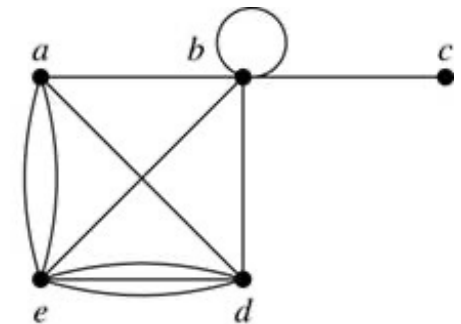
A vertex of degree 1 is called **pendant**.

Degrees and Neighborhoods of Vertices

Example: What are the degrees and neighborhoods of the vertices in the graphs G and H ?



G



H

Solution:

G : $\deg(a) = 2$, $\deg(b) = \deg(c) = \deg(f) = 4$, $\deg(d) = 1$,
 $\deg(e) = 3$, $\deg(g) = 0$.

$N(a) = \{b, f\}$, $N(b) = \{a, c, e, f\}$, $N(c) = \{b, d, e, f\}$, $N(d) = \{c\}$,

$N(e) = \{b, c, f\}$, $N(f) = \{a, b, c, e\}$, $N(g) = \emptyset$.

H : $\deg(a) = 4$, $\deg(b) = \deg(e) = 6$, $\deg(c) = 1$, $\deg(d) = 5$.

$N(a) = \{b, d, e\}$, $N(b) = \{a, b, c, d, e\}$, $N(c) = \{b\}$,

$N(d) = \{a, b, e\}$, $N(e) = \{a, b, d\}$.

Example 2.

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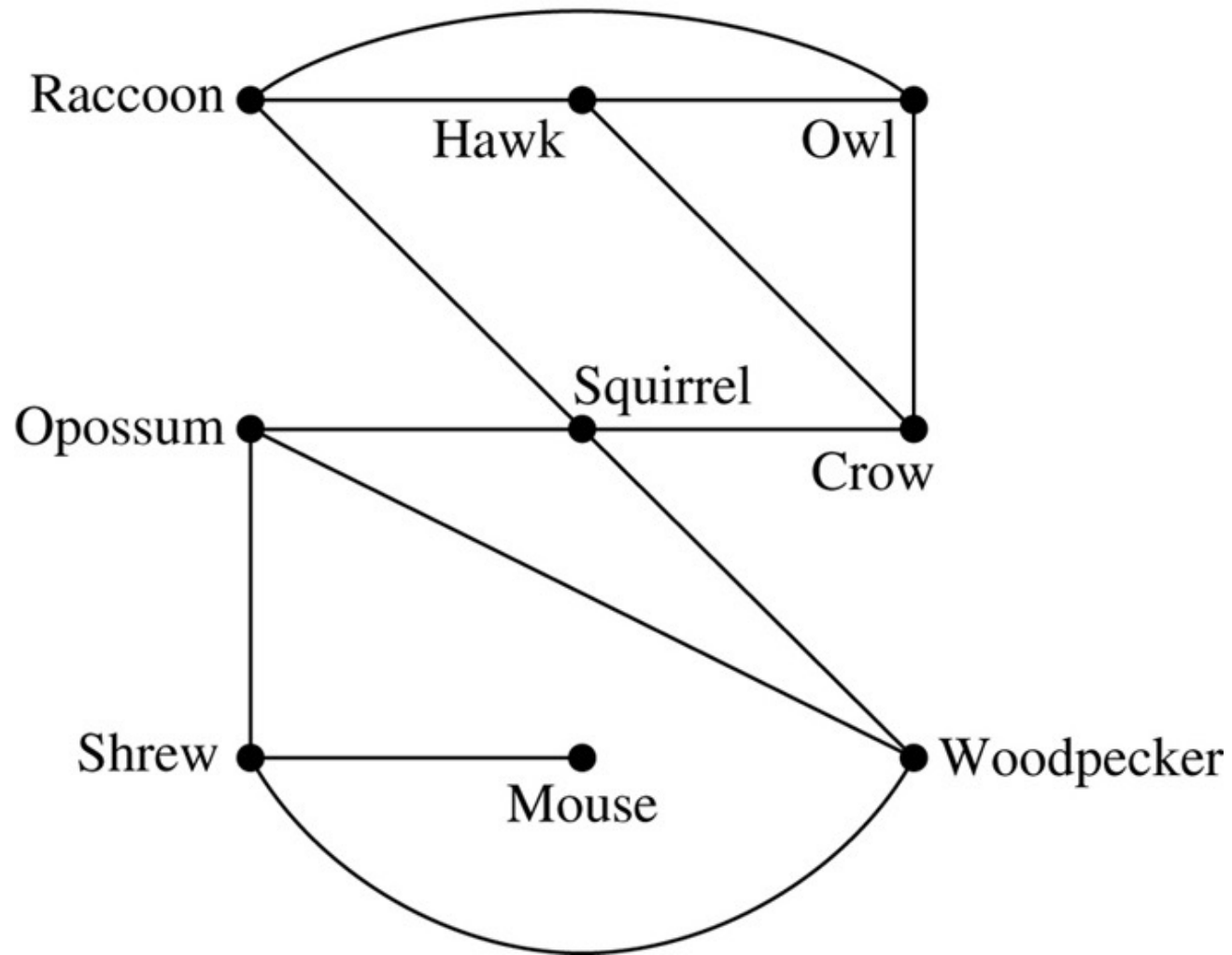


FIGURE 11 A Niche Overlap Graph.

Degrees of Vertices

Theorem 1 (*Handshaking Theorem*): If $G = (V, E)$ is an undirected graph with m edges, then

$$2m = \sum_{v \in V} \deg(v)$$

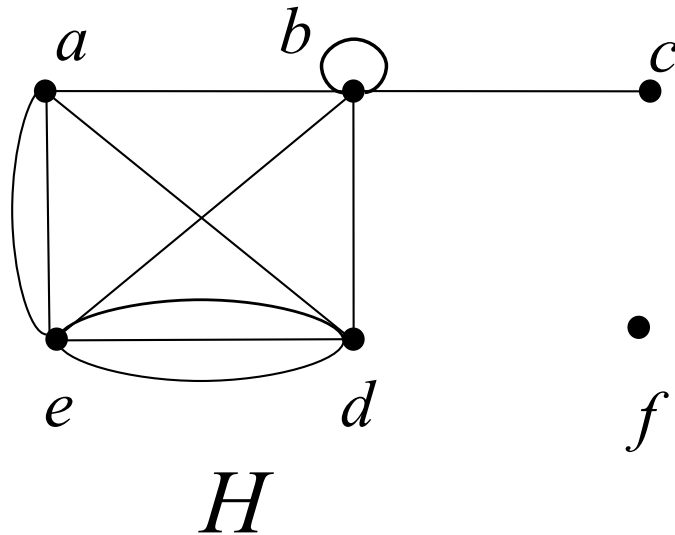
Proof:

Each edge contributes twice to the degree count of all vertices. Hence, both the left-hand and right-hand sides of this equation equal twice the number of edges.

Think about the graph where vertices represent the people at a party and an edge connects two people who have shaken hands.

每加一條 **edge** $\{u, v\}$ 會同時使 u 跟 v 各增加一個 **degree**

eg.



The graph H has 11 edges, and

$$\sum_{v \in V} \deg(v) = 22$$

Example 3. How many edges are there in a graph with 10 vertices each of degree six?

Sol :

$$10 \cdot 6 = 2e \quad \Rightarrow \quad e=30$$

Handshaking Theorem

Example: If a graph has 5 vertices, can each vertex have degree 3?

Solution: This is not possible by the handshaking theorem, because the sum of the degrees of the vertices $3 \cdot 5 = 15$ is odd.

Degree of Vertices

Theorem 2: An undirected graph has an even number of vertices of odd degree.

Proof: Let V_1 be the vertices of even degree and V_2 be the vertices of odd degree in an undirected graph $G = (V, E)$ with m edges. Then

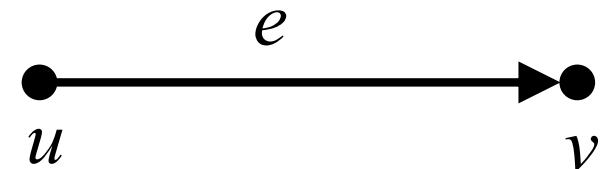
even $\rightarrow 2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$

must be even
since $\deg(v)$
is even for
each $v \in V_1$

This sum must be even because $2m$ is even and the sum of the degrees of the vertices of even degrees is also even. Because this is the sum of the degrees of all vertices of odd degree in the graph, there must be an even number of such vertices.

- Recall the definition of a directed graph.
- Definition:** An *directed graph* $G = (V, E)$ consists of V , a nonempty set of *vertices* (or *nodes*), and E , a set of *directed edges* or *arcs*. Each edge is an ordered pair of vertices. The directed edge (u, v) is said to start at u and end at v .
- Definition:** Let (u, v) be an edge in G . Then u is the *initial vertex* of this edge and is *adjacent to* v and v is the *terminal* (or *end*) *vertex* of this edge and is *adjacent from* u . The initial and terminal vertices of a loop are the same.

Def 4. $G = (V, E)$: directed graph,
 $e = (u, v) \in E : u$ is adjacent to v
 v is adjacent from u
 u : initial vertex of e
 v : terminal (end) vertex of e



Def 5.

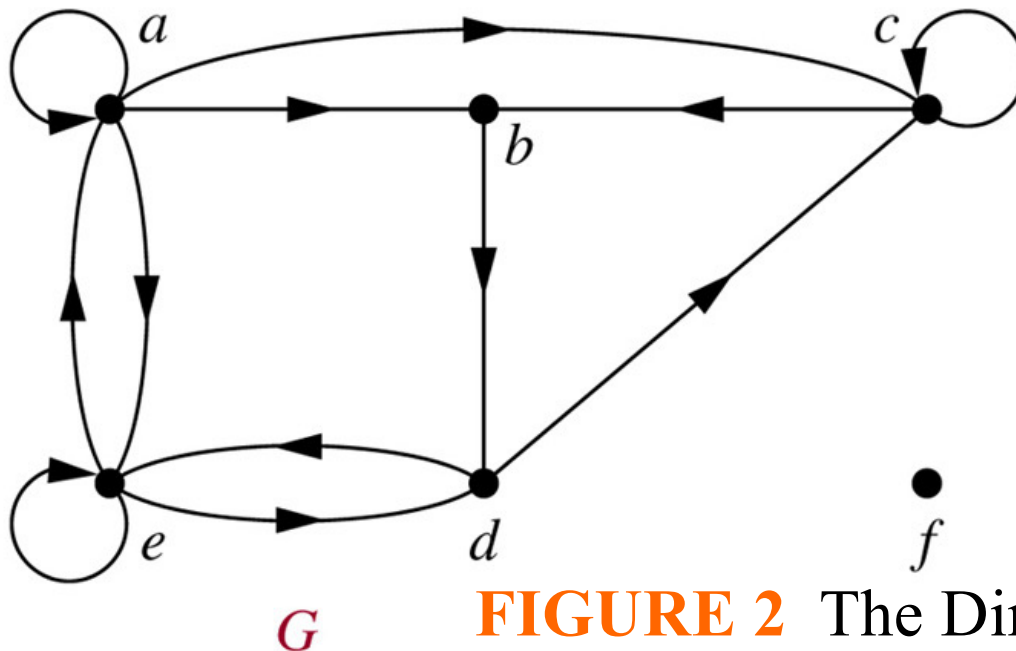
$G = (V, E)$: directed graph, $v \in V$

$\deg^-(v)$: # of edges with v as a terminal.
(in-degree)

$\deg^+(v)$: # of edges with v as a initial vertex
(out-degree)

Example 4.

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$\deg^-(a)=2, \deg^+(a)=4$
 $\deg^-(b)=2, \deg^+(b)=1$
 $\deg^-(c)=3, \deg^+(c)=2$
 $\deg^-(d)=2, \deg^+(d)=2$
 $\deg^-(e)=3, \deg^+(e)=3$
 $\deg^-(f)=0, \deg^+(f)=0$

FIGURE 2 The Directed Graph G .

Thm 3. Let $G = (V, E)$ be a digraph. Then

$$\sum_{v \in V} \deg^{-}(v) = \sum_{v \in V} \deg^{+}(v) = |E|$$

- **Proof:** The first sum counts the number of outgoing edges over all vertices and the second sum counts the number of incoming edges over all vertices. It follows that both sums equal the number of edges in the graph.

每增加一條 edge



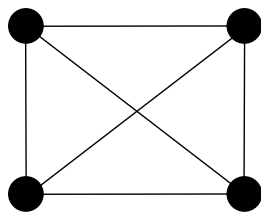
$\deg^{+}(u)$ 增加 1

$\deg^{-}(v)$ 增加 1

(ex39上面) A simple graph $G=(V, E)$ is called **regular** if every vertex of this graph has the same degree. A regular graph is called **n -regular** if $\deg(v)=n$, $\forall v \in V$.

eg.

K_4 :



is 3-regular.

Some Special Simple Graphs

Example 5.

The **complete graph on n vertices**, denoted by K_n , is the simple graph that contains **exactly one edge between each pair of distinct vertices.**

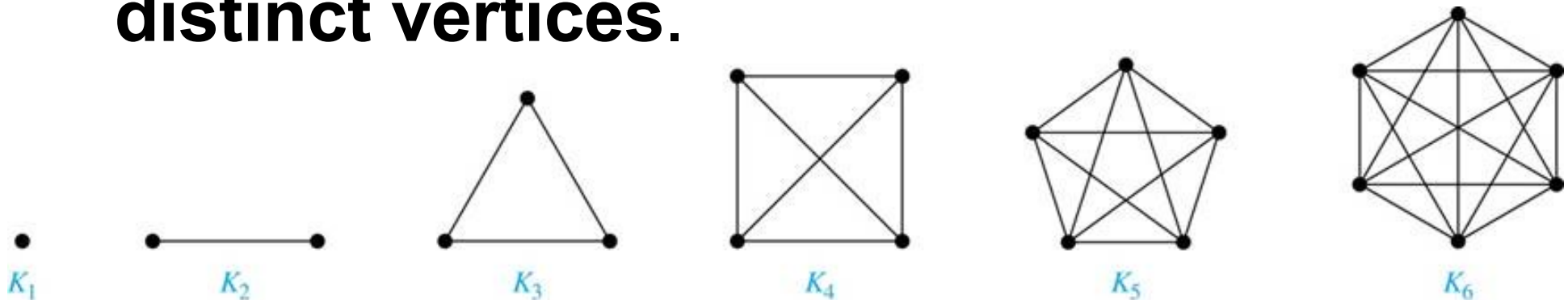


FIGURE 3 The Graphs K_n for $1 \leq n \leq 6$.

Note. K_n is $(n-1)$ -regular, $|V(K_n)|=n$,

$$|E(K_n)| = \binom{n}{2}$$

Example 6. The cycle C_n , $n \geq 3$, consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.

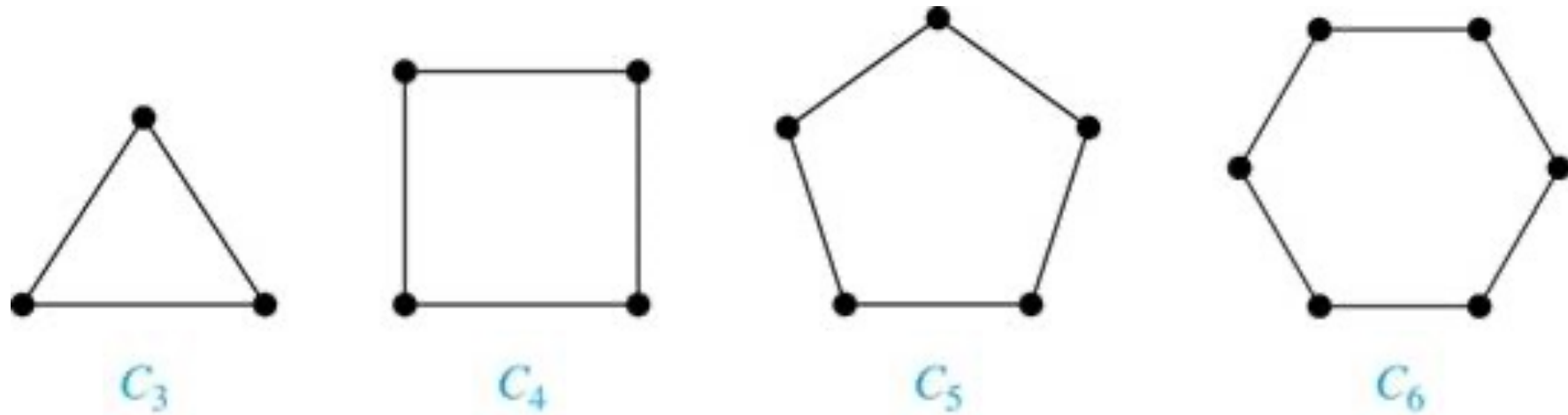


FIGURE 4 The Cycles C_3 , C_4 , C_5 , and C_6 .

Note C_n is 2-regular, $|V(C_n)| = n$, $|E(C_n)| = n$

Example 7. We obtained the wheel W_n when we add an additional vertex to the cycle C_n , for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n , by new edges.

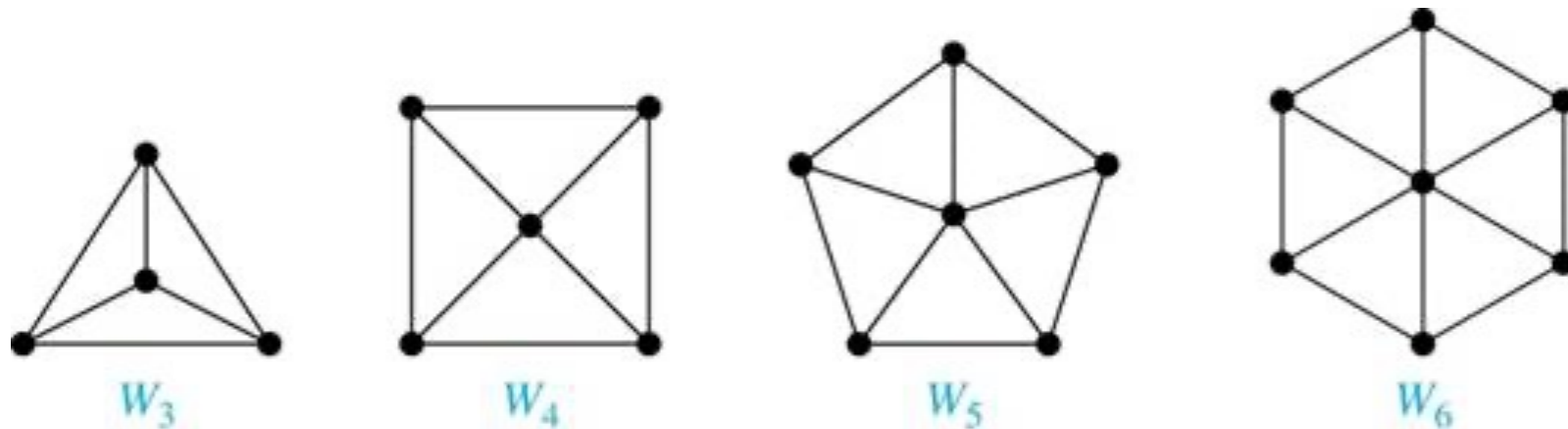
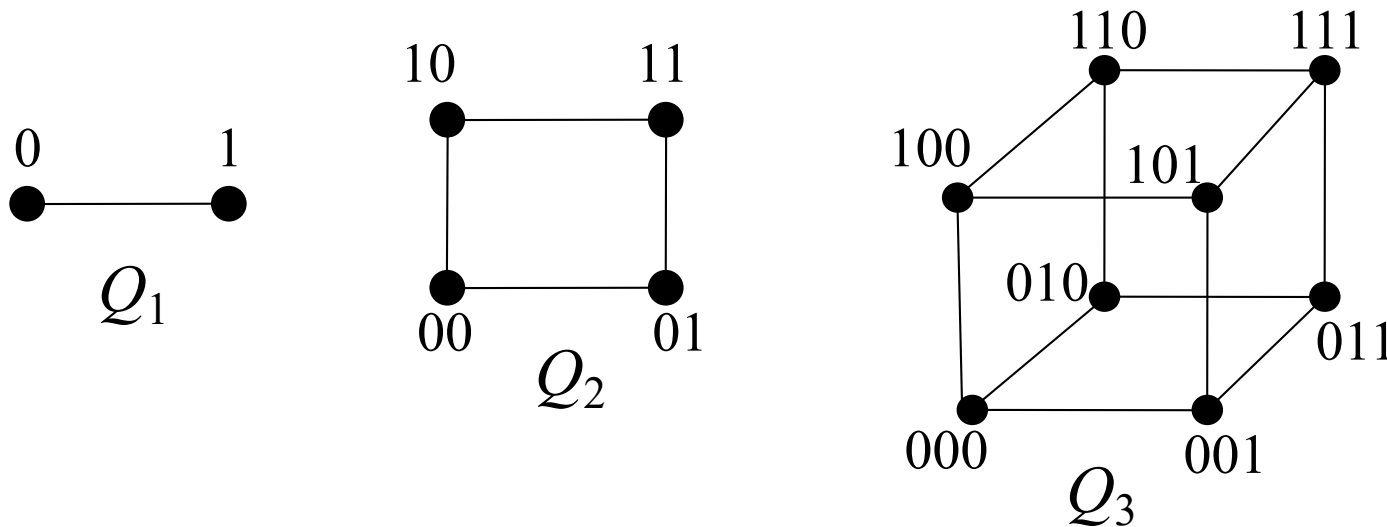


FIGURE 5 The Wheels W_3 , W_4 , W_5 , and W_6 .

Note. $|V(W_n)| = n + 1$, $|E(W_n)| = 2n$,
 W_n is not a regular graph if $n \neq 3$.

Example 8. The n -dimensional hypercube, or n -cube, denoted by Q_n , is the graph that has vertices representing the 2^n bit strings of length n . Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position.



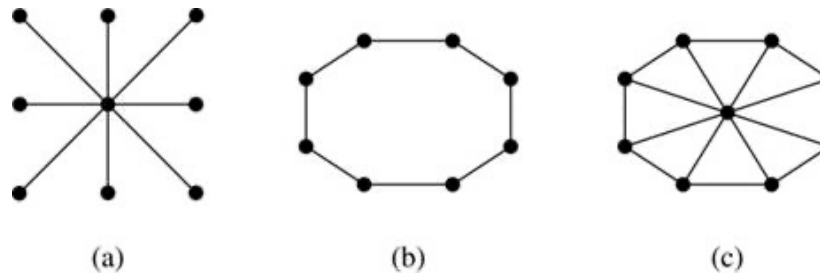
另法： Q_{n-1} 點數
即每個dim邊數，
共 n -dim

Note. Q_n is n -regular, $|V(Q_n)| = 2^n$, $|E(Q_n)| = (2^n n)/2 = 2^{n-1} n$

deg總和 / 2

Special Types of Graphs and Computer Network Architecture

Various special graphs play an important role in the design of computer networks.



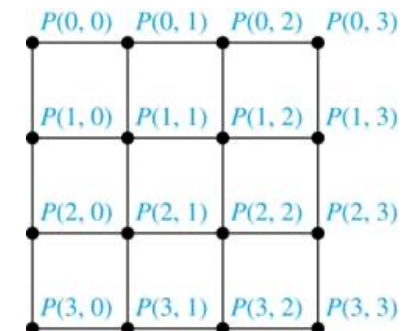
Some local area networks use a **star topology**, which is a complete bipartite graph $K_{1,n}$, as shown in (a). All devices are connected to a central control device.

Other local networks are based on a **ring topology**, where each device is connected to exactly two others using C_n , as illustrated in (b). Messages may be sent around the ring.

Others, as illustrated in (c), use a W_n – based topology, combining the features of a star topology and a ring topology.

Various special graphs also play a role in parallel processing where processors need to be interconnected as one processor may need the output generated by another.

- The n -dimensional hypercube, or n -cube, Q_n , is a common way to connect processors in parallel, e.g., Intel Hypercube.
- Another common method is the *mesh* network, illustrated here for 16 processors.



Bipartite Graphs

Def 6. A simple graph $G=(V,E)$ is called **bipartite** if V can be partitioned into V_1 and V_2 , $V_1 \cap V_2 = \emptyset$, such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 .

Example 9.

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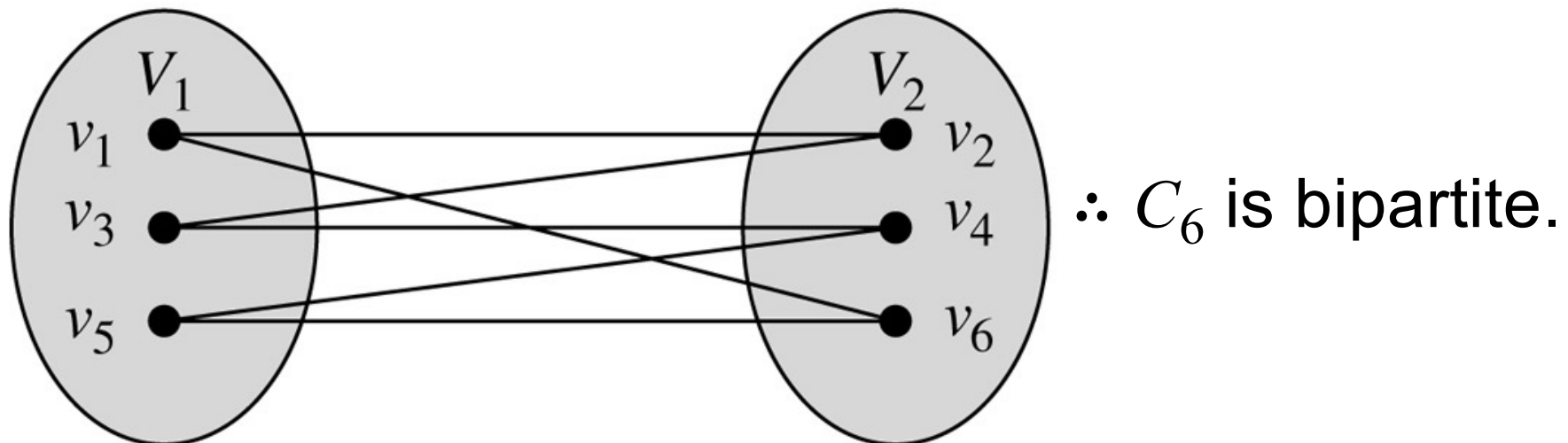


FIGURE 7 Showing That C_6 Is Bipartite.

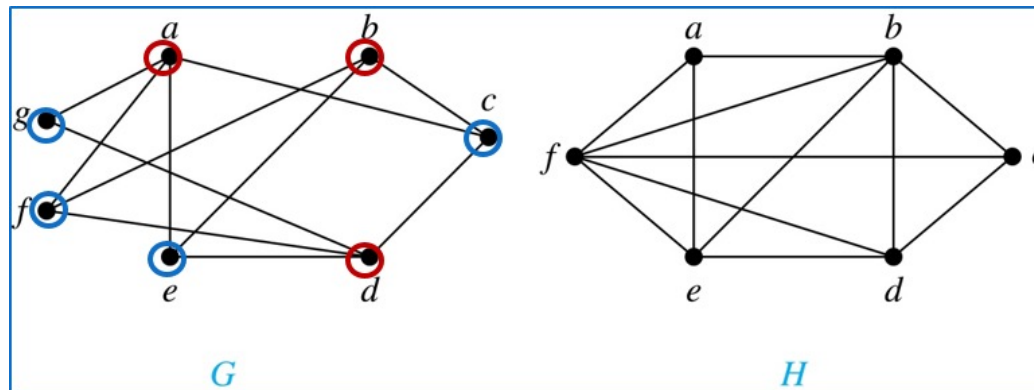
Bipartite Graphs₁

Definition: A simple graph G is *bipartite* if V can be partitioned into two disjoint subsets V_1 and V_2 such that every edge connects a vertex in V_1 and a vertex in V_2 . In other words, there are no edges which connect two vertices in V_1 or in V_2 .

It is not hard to show that an equivalent definition of a bipartite graph is a graph where it is possible to color the vertices red or blue so that no two adjacent vertices are the same color.

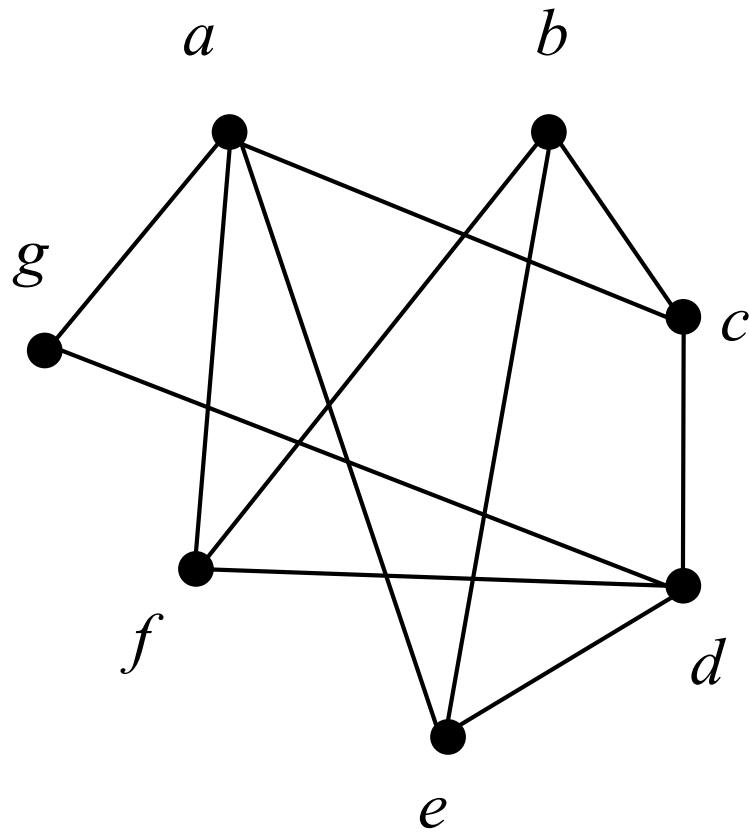
Example 11. Is the graph G bipartite?

G is
bipartite

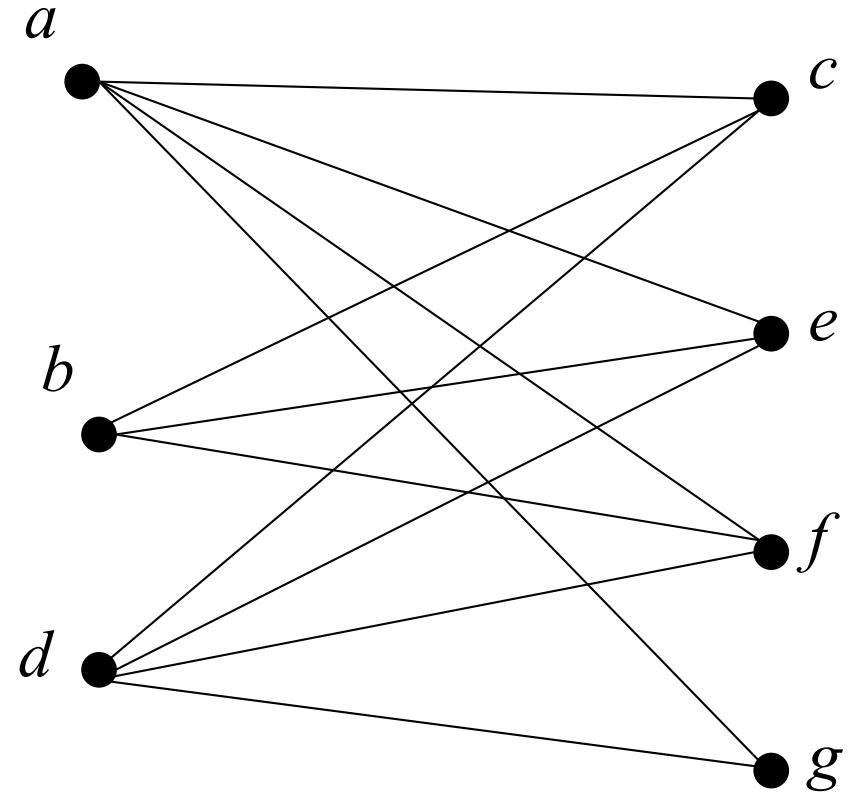
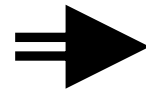


H is not bipartite since if we color a red, then the adjacent vertices f and b must both be blue.

Example 11. Is the graph G bipartite?



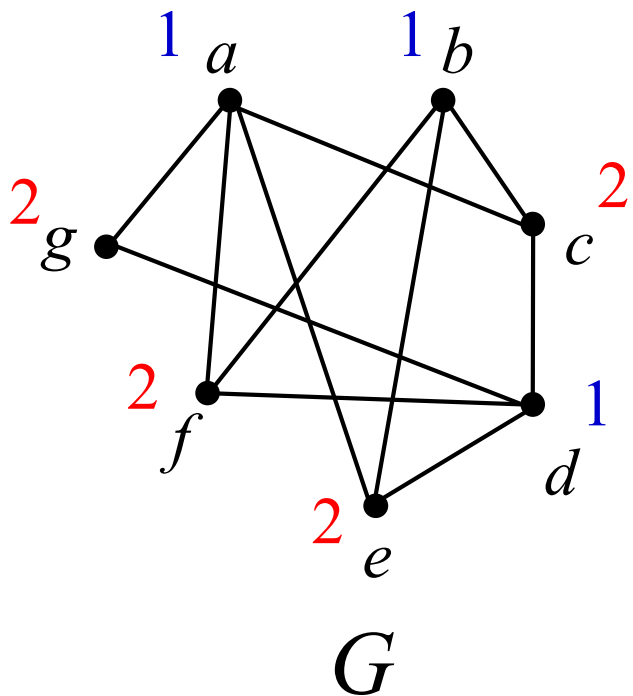
G



Yes !

Thm 4. A simple graph is bipartite if and only if it is possible to **assign one of two different colors to each vertex** of the graph so that **no two adjacent vertices** are assigned the **same color**.

Example 12. Use Thm 4 to show that G is bipartite.



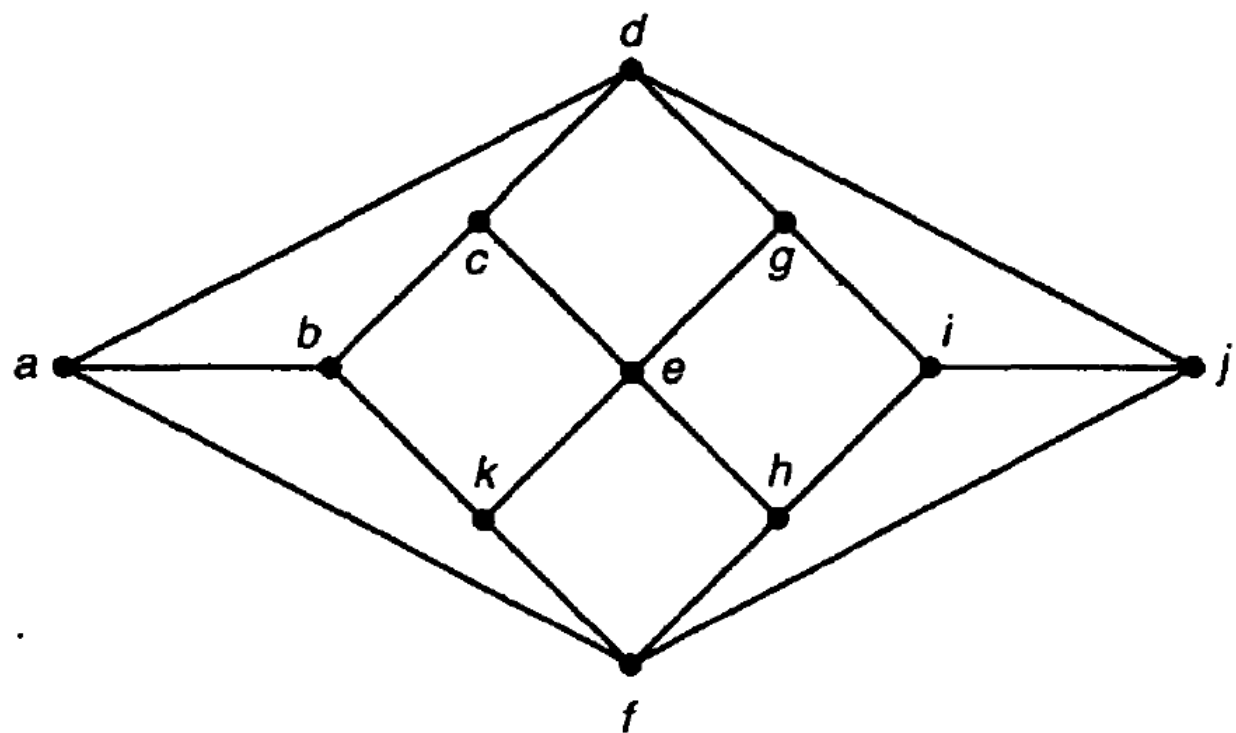


Figure 11.21

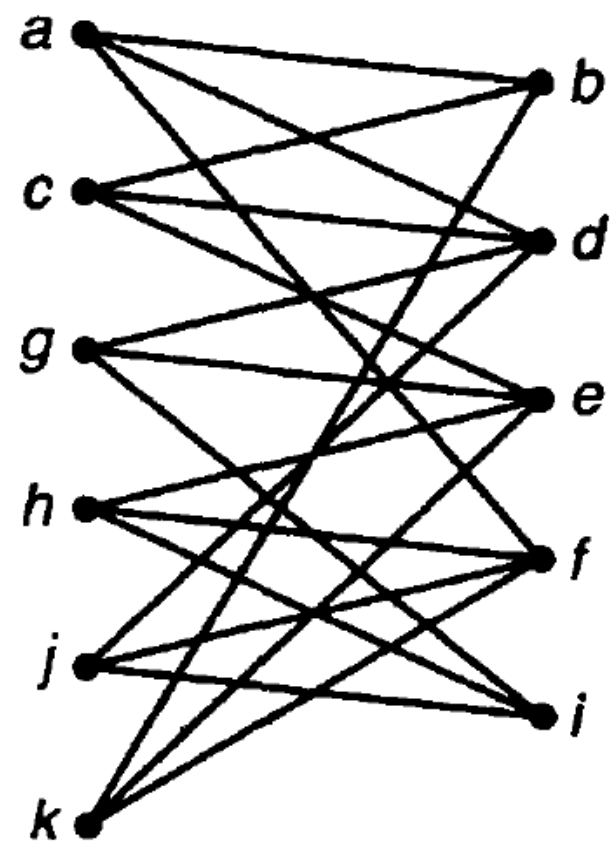
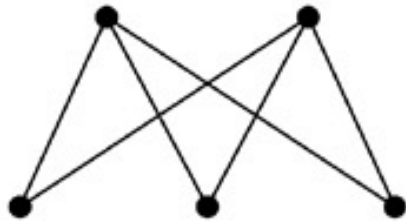


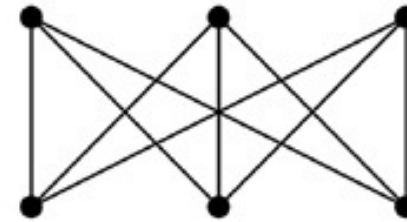
Figure 11.22

- Example 13. Complete Bipartite graphs ($K_{m,n}$)

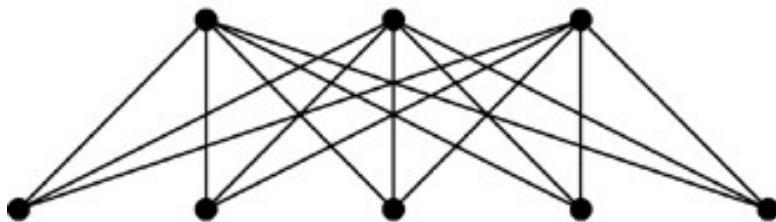
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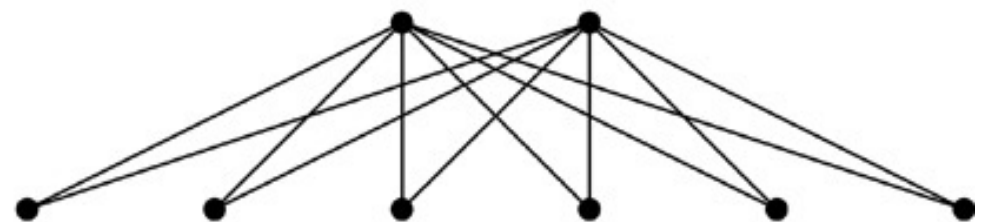
$K_{2,3}$



$K_{3,3}$



$K_{3,5}$



$K_{2,6}$

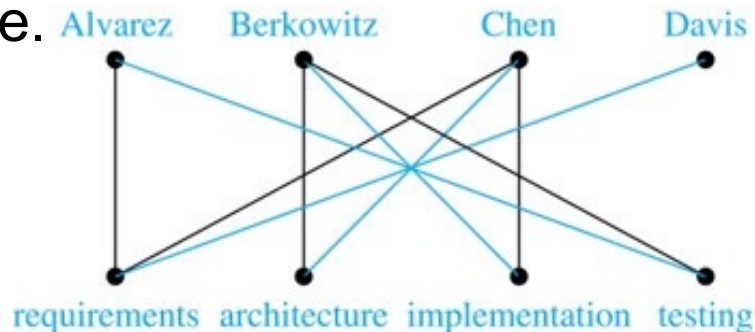
FIGURE 9 Some Complete Bipartite Graphs.

Note. $|V(K_{m,n})| = m+n$, $|E(K_{m,n})| = mn$,
 $K_{m,n}$ is regular if and only if $m=n$.

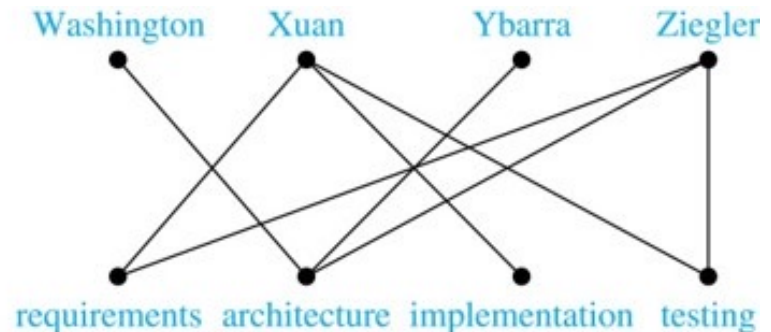
Bipartite Graphs and Matchings

Bipartite graphs are used to model applications that *involve matching the elements of one set to elements in another*, for example:

Job assignments - vertices represent the jobs and the employees, edges link employees with those jobs they have been trained to do. A common goal is to match jobs to employees so that the most jobs are done.



(a)



(b)

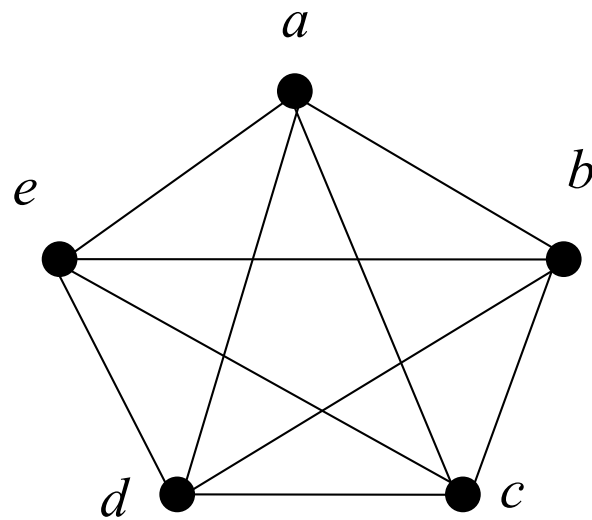
Marriage - vertices represent the men and the women and edges link a man and a woman if they are an acceptable spouse. We may wish to find the largest number of possible marriages.

See the text for more about matchings in bipartite graphs.

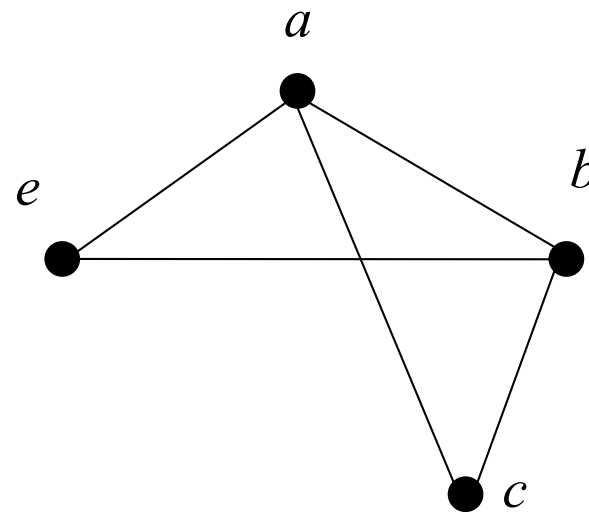
New Graphs from Old

Def 6. A **subgraph** of a graph $G=(V, E)$ is a graph $H=(W, F)$ where $W \subseteq V$ and $F \subseteq E$.
(注意 F 要連接 W 裡的點)

Example 18. A subgraph of K_5



K_5



subgraph of K_5

FIGURE 15 A Subgraph of K_5 .

Def 9. The **union** of two simple graphs

$G_1=(V_1, E_1)$ and $G_2=(V_2, E_2)$ is the simple graph

$$G_1 \cup G_2 = (V_1 \cup V_2, E_1 \cup E_2)$$

Example 19.

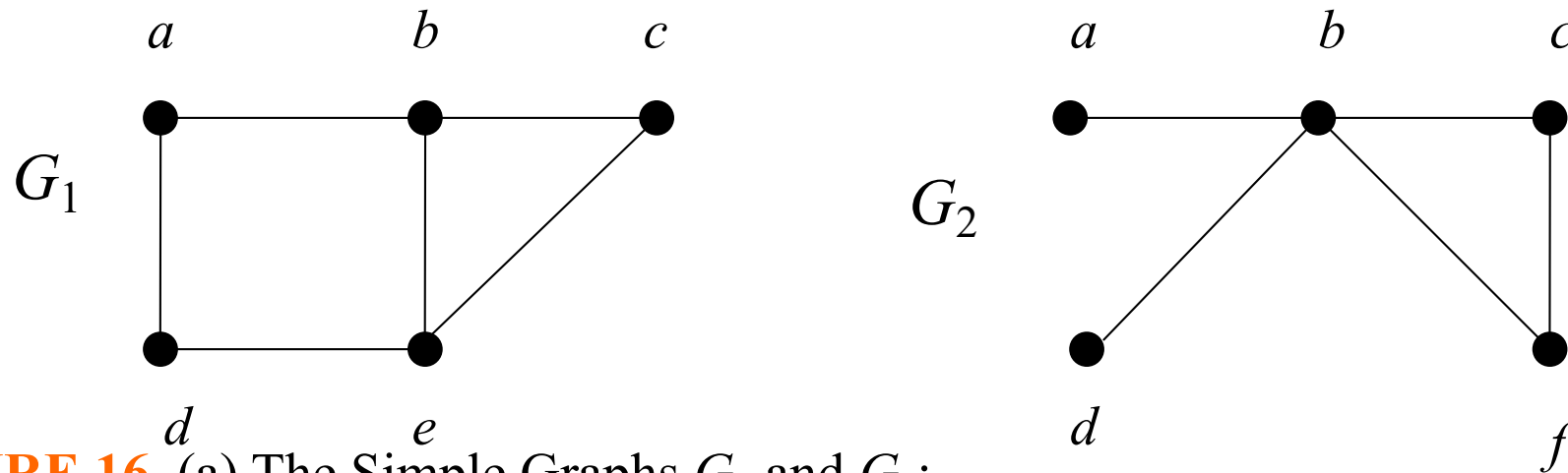
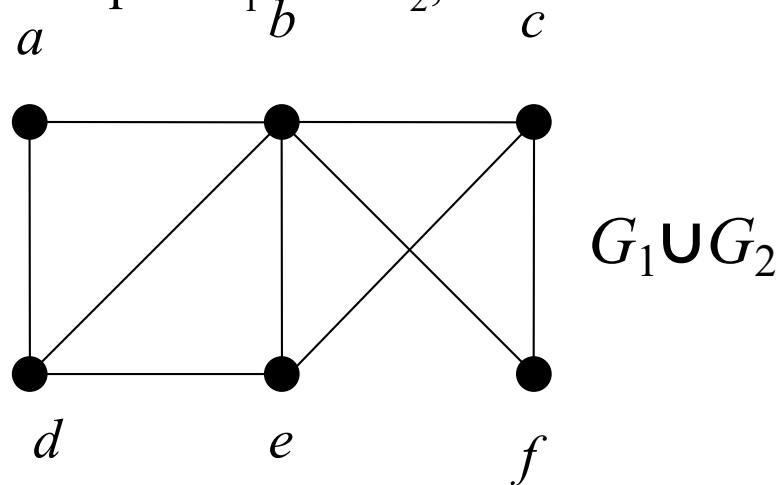
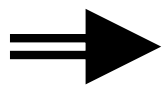
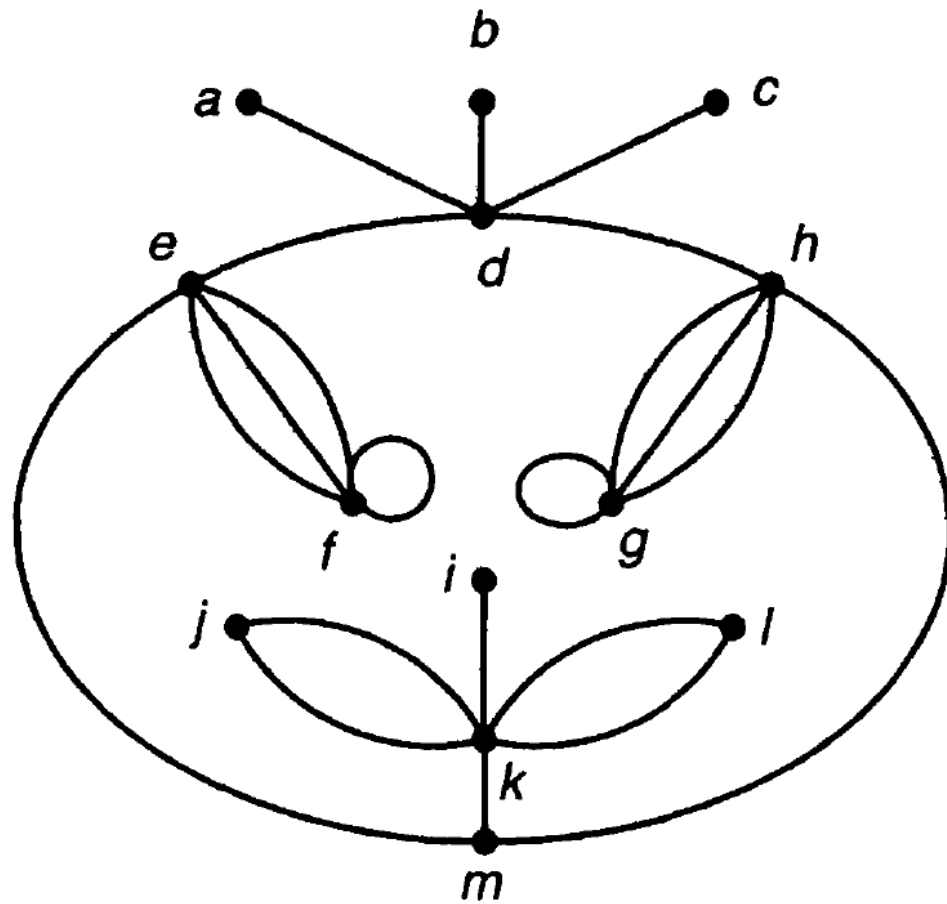


FIGURE 16 (a) The Simple Graphs G_1 and G_2 ;

(b) Their Union $G_1 \cup G_2$.





GraphBuster (Ghostbuster)

Figure 11.3

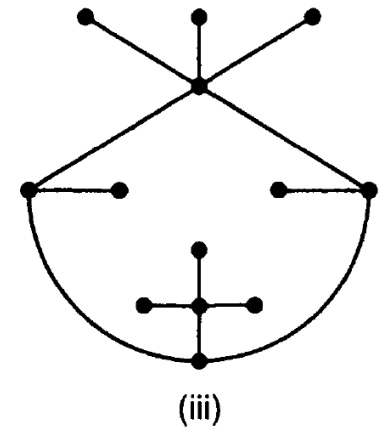
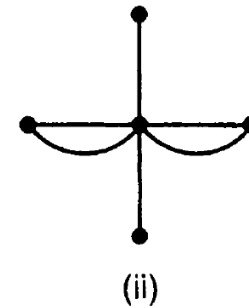
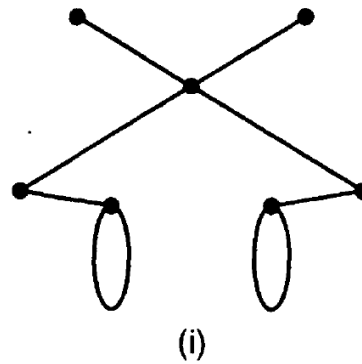
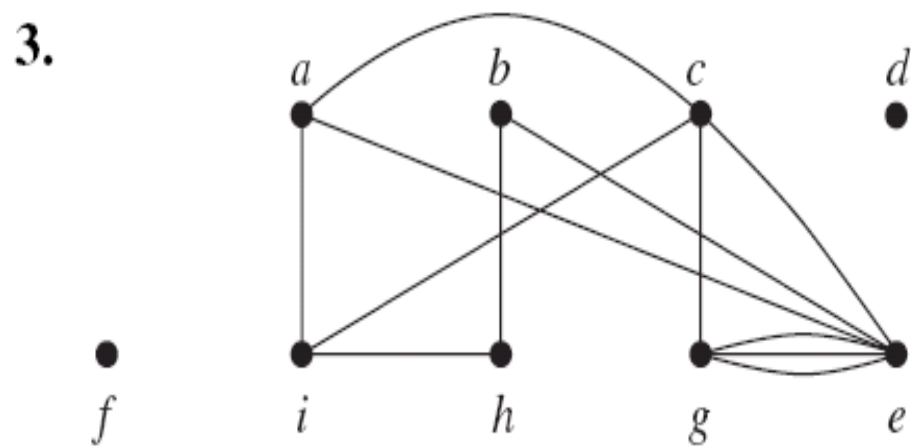
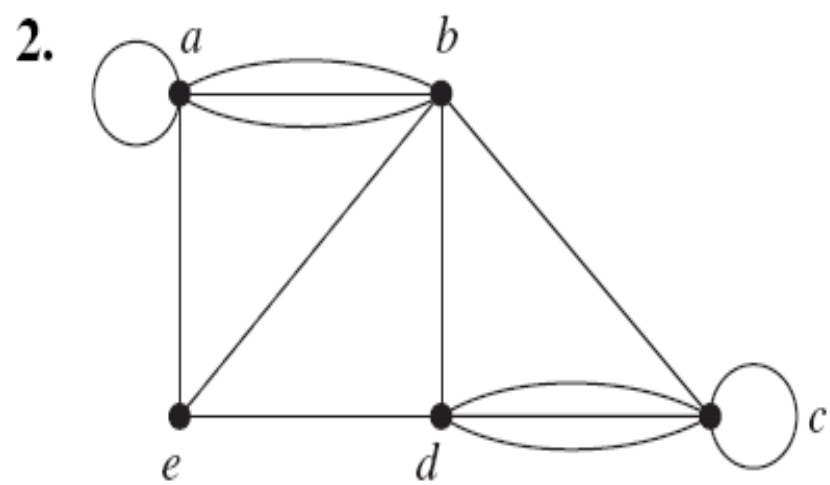
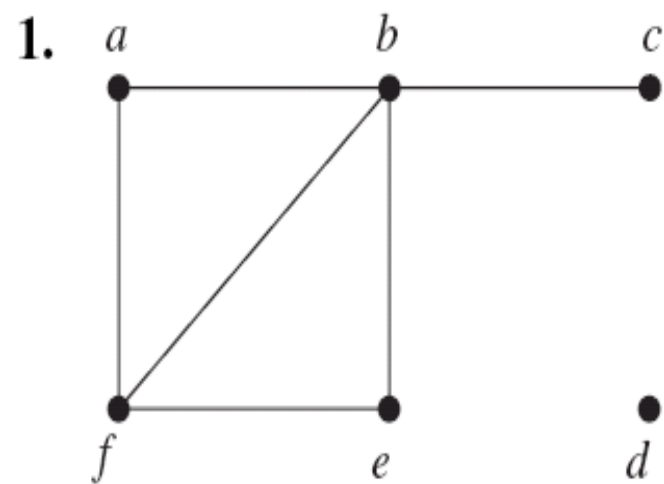
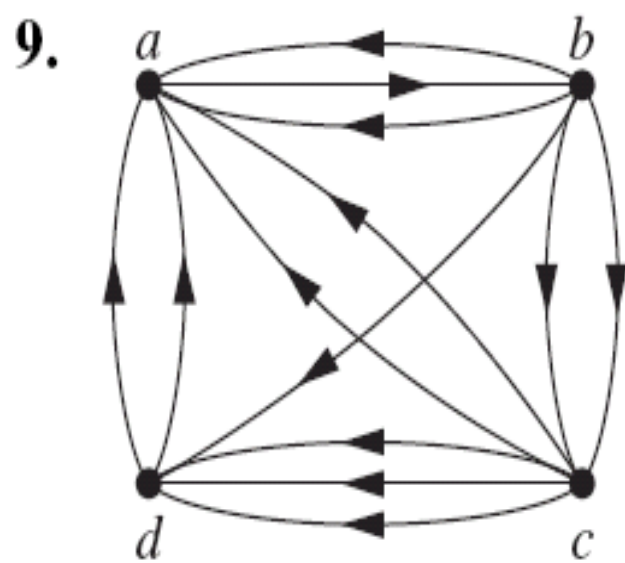
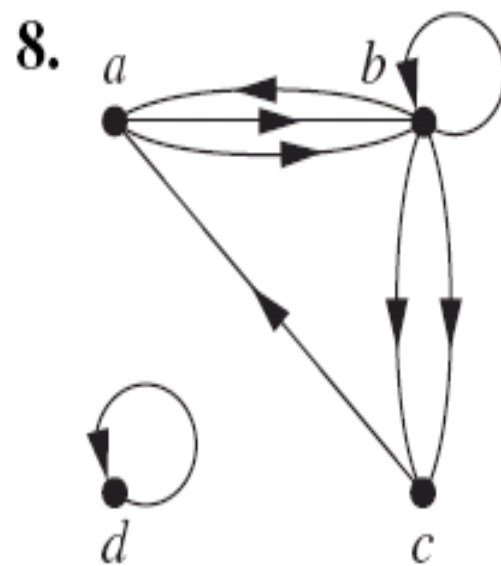
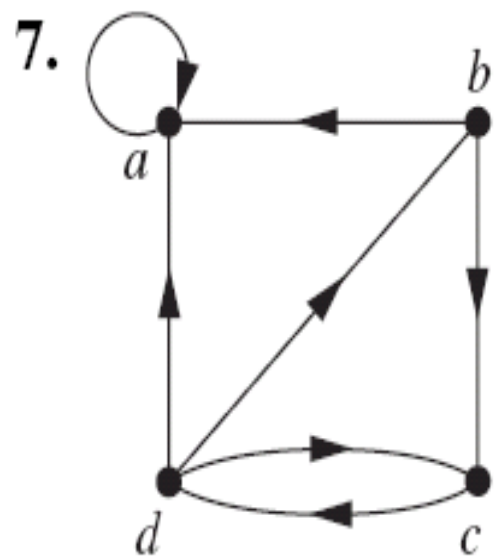
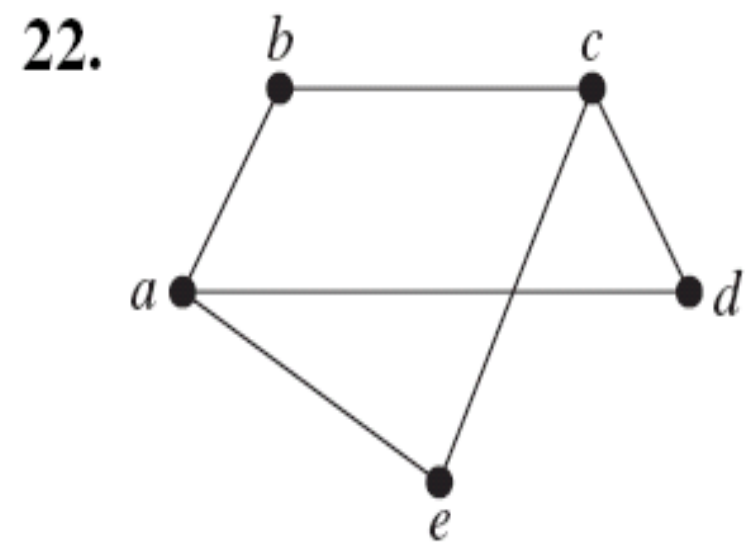
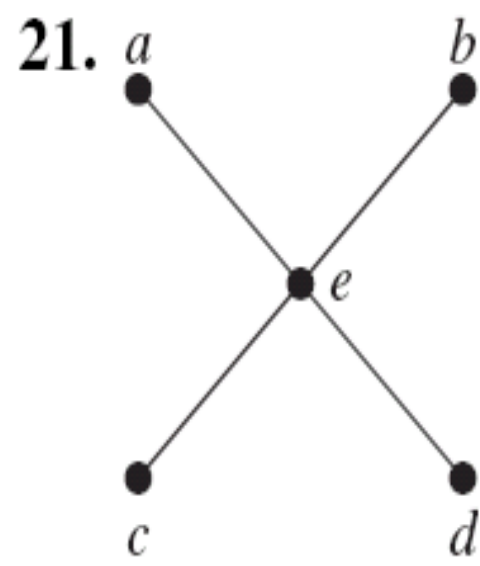


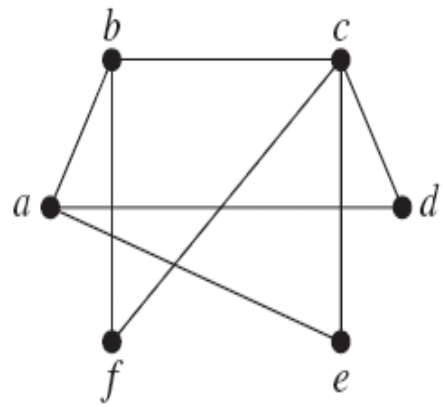
Figure 11.10



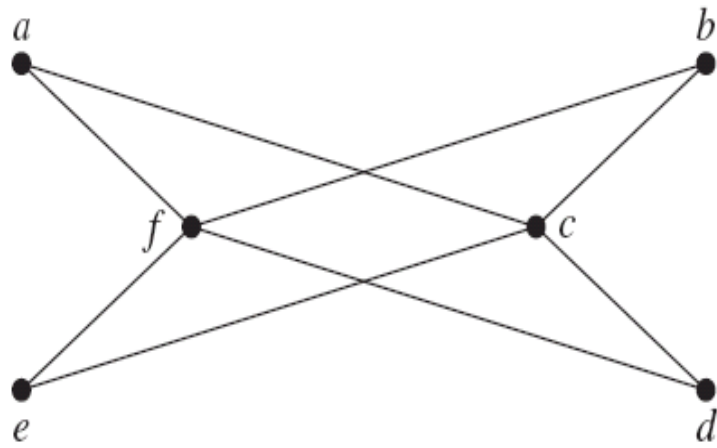




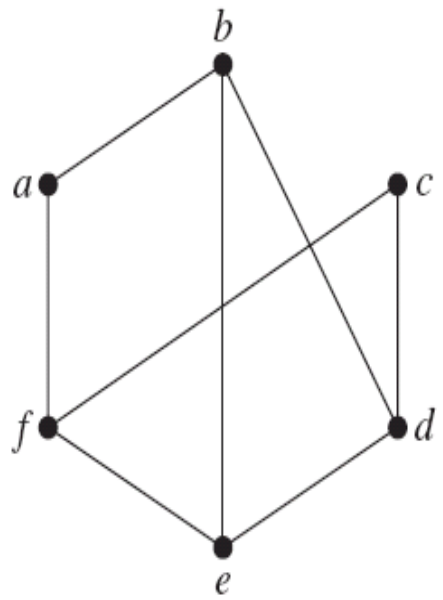
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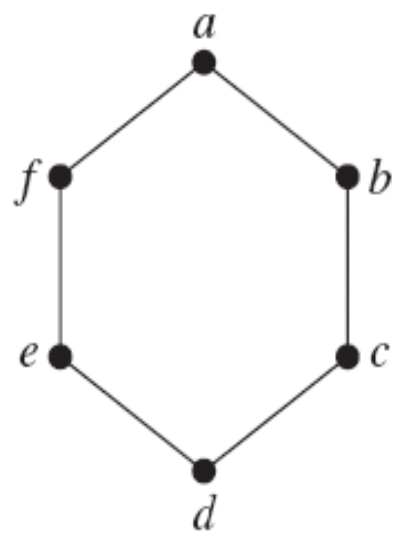
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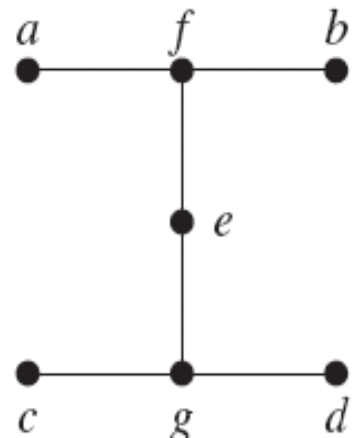
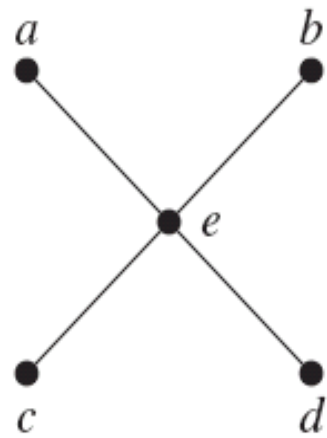
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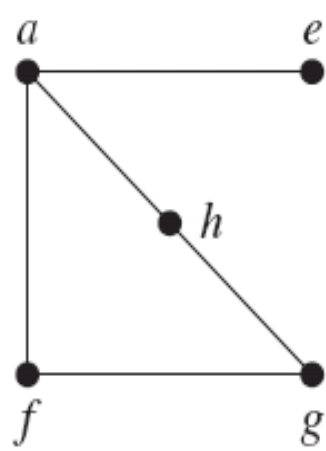
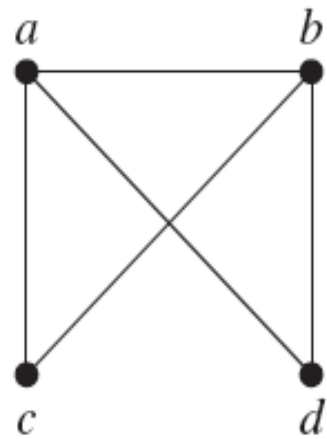
56.



57.



58.



§10.3 Representing Graphs and Graph Isomorphism

✂Adjacency list

Example 1. Use adjacency lists to describe the simple graph given below.

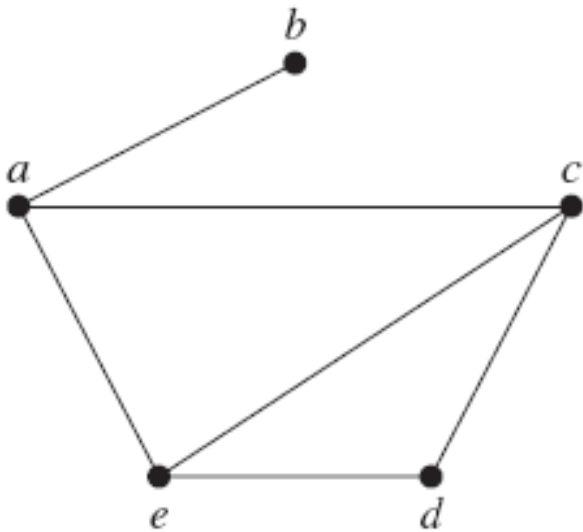
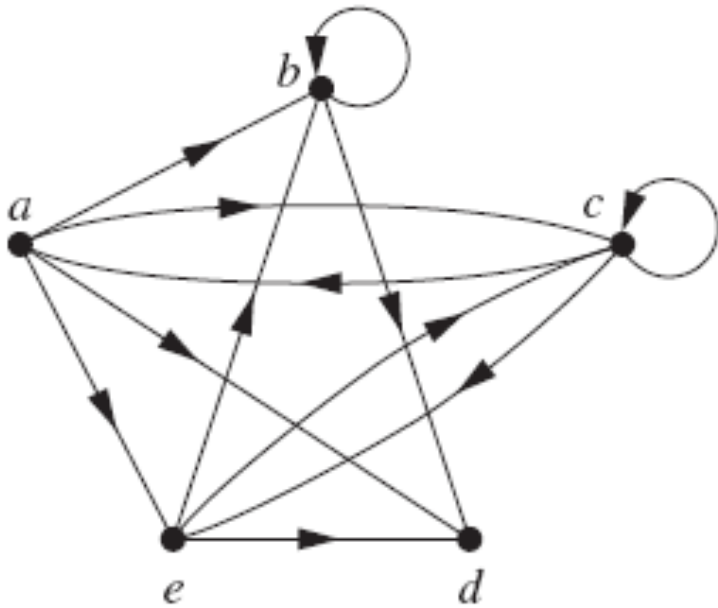


FIGURE 1 A Simple Graph.

Sol :

Vertex	Adjacent Vertices
<i>a</i>	<i>b, c, e</i>
<i>b</i>	<i>a</i>
<i>c</i>	<i>a, d, e</i>
<i>d</i>	<i>c, e</i>
<i>e</i>	<i>a, c, d</i>

Example 2. (digraph)



Initial vertex	Terminal vertices
a	b, c, d, e
b	b, d
c	a, c, e
d	
e	b, c, d

FIGURE 2 A Directed Graph.

※Adjacency Matrices

Def. $G=(V, E)$: simple graph, $V=\{v_1, v_2, \dots, v_n\}$. (順序沒關係)

A matrix A is called the **adjacency matrix** of G

if $A=[a_{ij}]_{n \times n}$, where $a_{ij} = \begin{cases} 1, & \text{if } \{v_i, v_j\} \in E, \\ 0, & \text{otherwise.} \end{cases}$

Example 3.

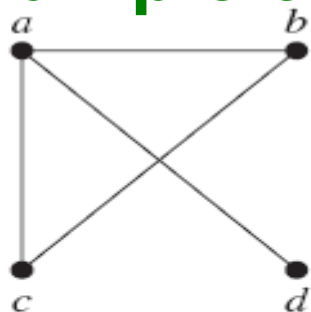


FIGURE 3
Simple Graph.

$$A_1 = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A_2 = \begin{matrix} & \begin{matrix} b & d & c & a \end{matrix} \\ \begin{matrix} b \\ d \\ c \\ a \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

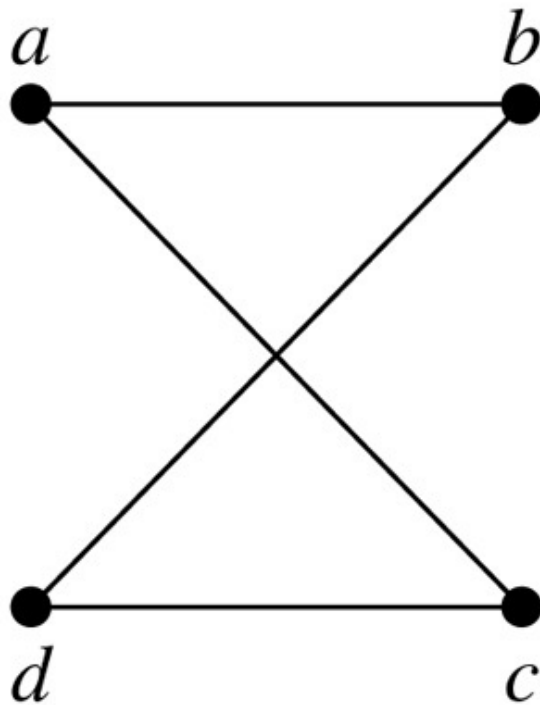
Note:

1. There are $n!$ different adjacency matrices for a graph with n vertices.
2. The adjacency matrix of an undirected graph is **symmetric**.

Draw a graph with the adjacency matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

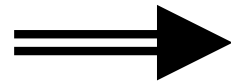
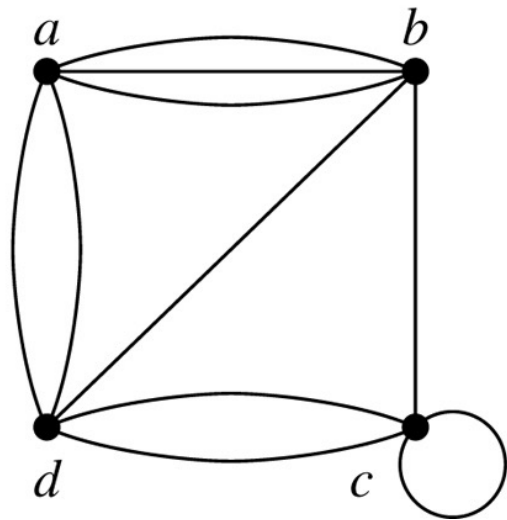
Example 4.



with respect to the ordering of vertices a, b, c, d .

FIGURE 4 A Graph with the Given Adjacency Matrix.

Example 5. (Pseudograph) (矩陣未必是0,1矩陣.)



$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix} \end{matrix}$$

FIGURE 5 A Pseudograph.

Def. If $A=[a_{ij}]$ is the adjacency matrix for the directed graph, then

$$a_{ij} = \begin{cases} 1 & , \text{ if } \begin{matrix} \bullet & \longrightarrow & \bullet \\ v_i & & v_j \end{matrix} \\ 0 & , \text{ otherwise} \end{cases}$$

(故矩陣未必對稱)

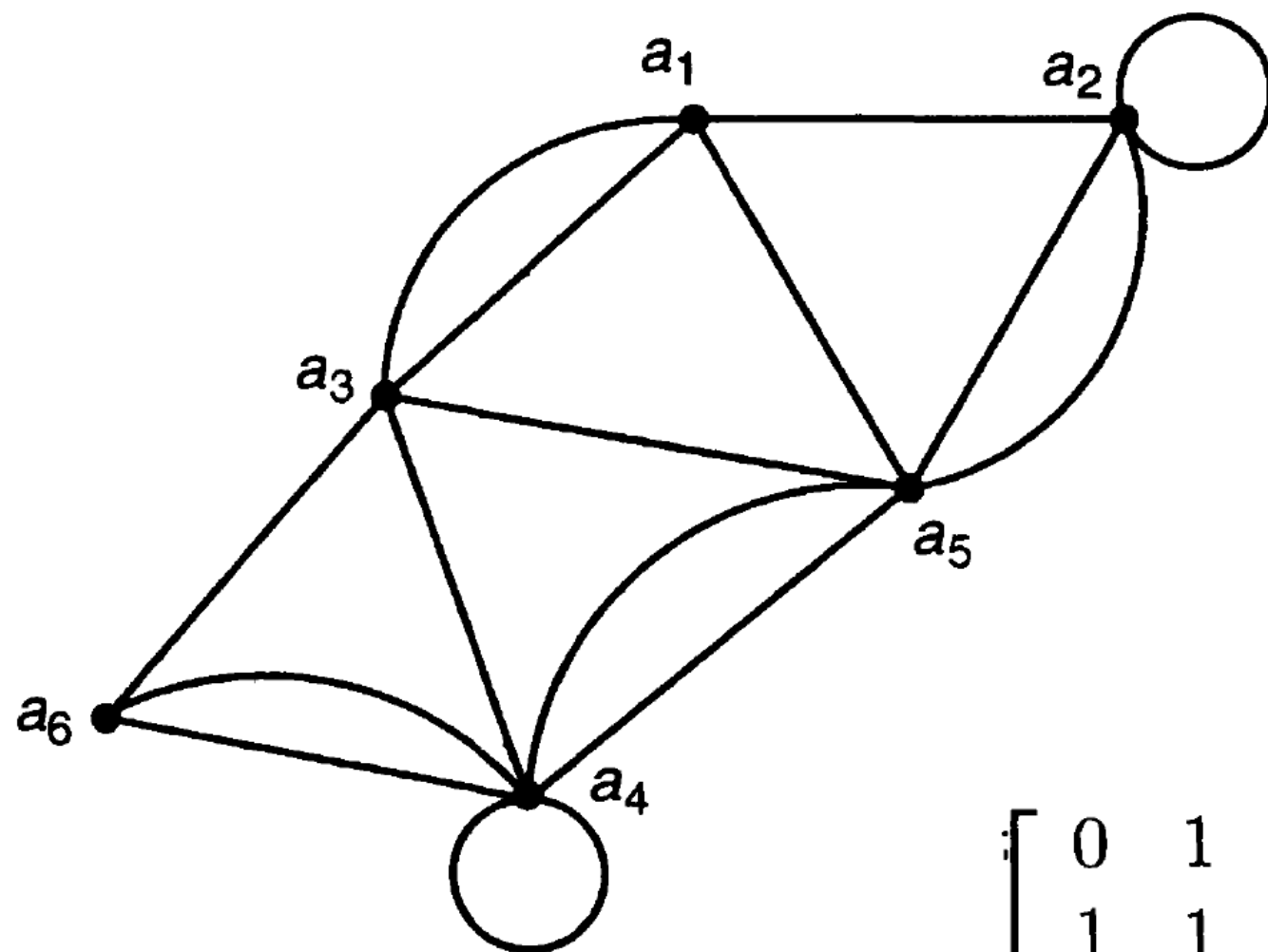


Figure 11.11

$$\begin{bmatrix}
 0 & 1 & 2 & 0 & 1 & 0 \\
 1 & 1 & 0 & 0 & 2 & 0 \\
 2 & 0 & 0 & 1 & 1 & 1 \\
 0 & 0 & 1 & 1 & 2 & 2 \\
 1 & 2 & 1 & 2 & 0 & 0 \\
 0 & 0 & 1 & 2 & 0 & 0
 \end{bmatrix}$$

Representation of Graphs: Incidence Matrices₁

Definition: Let $G = (V, E)$ be an undirected graph with vertices where $v_1, v_2, \dots, v_n$ and edges $e_1, e_2, \dots, e_m$. The incidence matrix with respect to the ordering of V and E is the $n \times m$ matrix $\mathbf{M} = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$$

※Simple Graph and Incidence Matrices

Example 6.

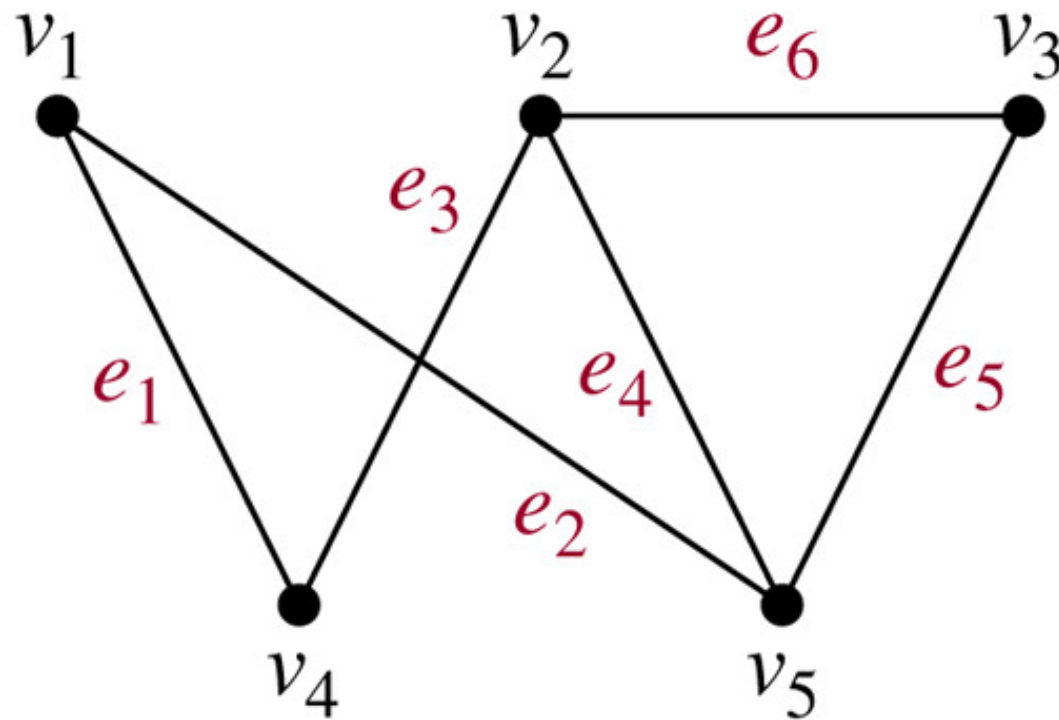


FIGURE 6 An Undirected Graph.

$$\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{bmatrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{bmatrix}.$$

※Pseudograph and Incidence Matrices

Example 7.

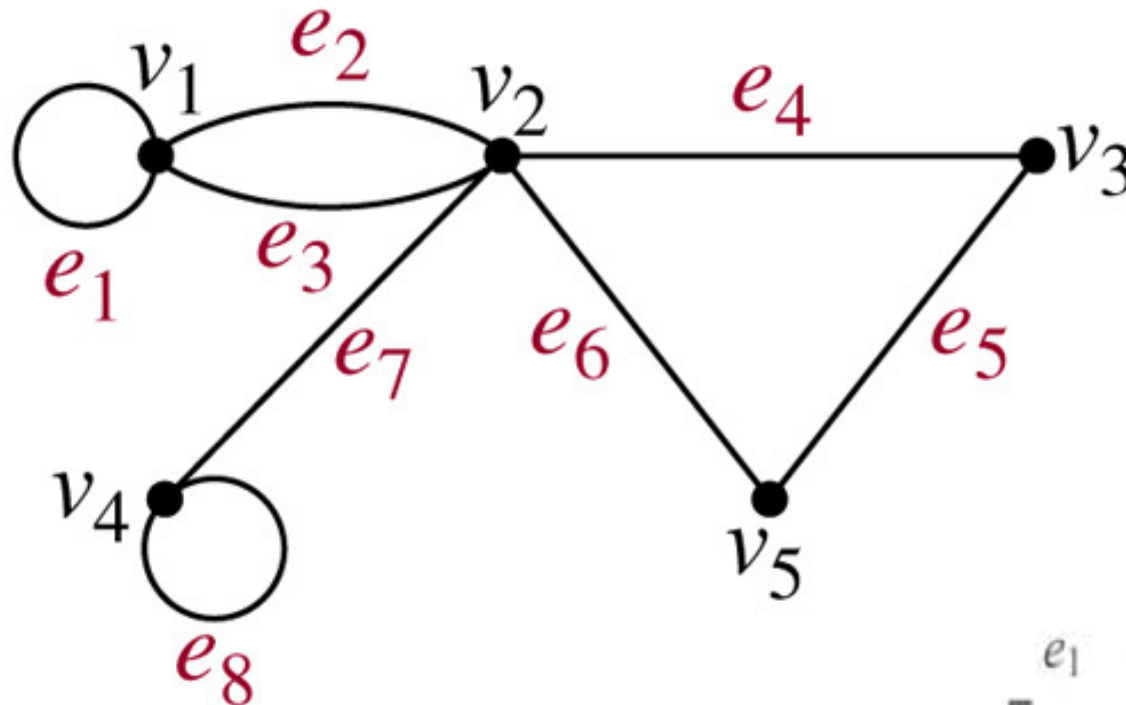
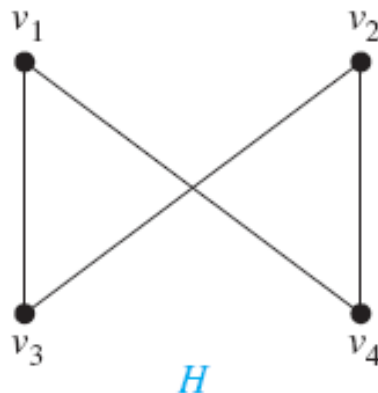
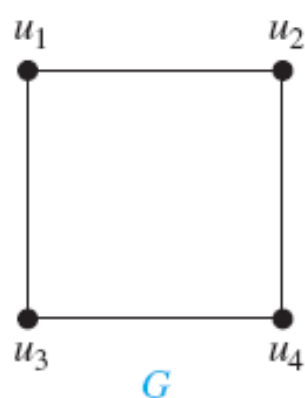


FIGURE 7 A Pseudograph.

$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}.$$

✂ Isomorphism of Graphs



G is isomorphic to H

FIGURE 8 The Graphs G and H .

Def 1.

The simple graphs $G_1=(V_1, E_1)$ and $G_2=(V_2, E_2)$ are **isomorphic** if there is an one-to-one and onto function f from V_1 to V_2 with the property that $a \sim b$ in G_1 iff $f(a) \sim f(b)$ in G_2 , $\forall a, b \in V_1$
 f is called an isomorphism.

Isomorphism of Graphs₃

It is difficult to determine whether two simple graphs are isomorphic using brute force because there are $n!$ possible one-to-one correspondences between the vertex sets of two simple graphs with n vertices.

The best algorithms for determining whether two graphs are isomorphic have exponential worst case complexity in terms of the number of vertices of the graphs.

Sometimes it is not hard to show that two graphs are not isomorphic. We can do so by finding a property, preserved by isomorphism, that only one of the two graphs has. Such a property is called ***graph invariant***.

There are many different useful graph invariants that can be used to distinguish nonisomorphic graphs, such as the **number of vertices**, **number of edges**, and **degree sequence** (list of the degrees of the vertices in nonincreasing order). We will encounter others in later sections of this chapter.

Example 8. Show that G and H are isomorphic.

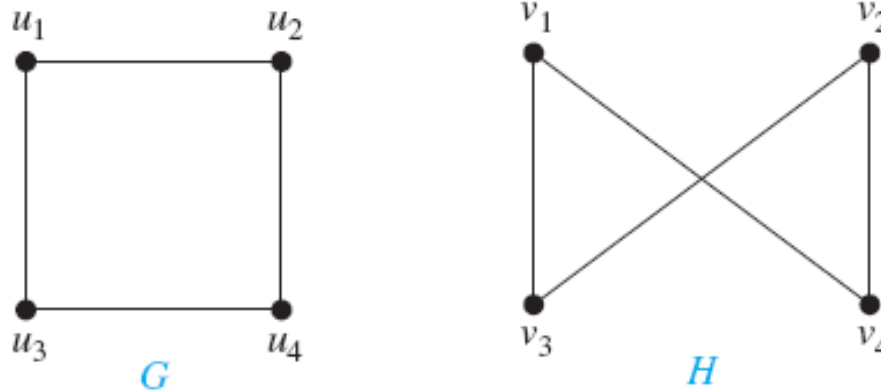


FIGURE 8 The Graphs G and H .

Sol.

The function f with $f(u_1) = v_1, f(u_2) = v_4, f(u_3) = v_3,$ and $f(u_4) = v_2$ is a one-to-one correspondence between $V(G)$ and $V(H)$.

※ graph invariant –

Isomorphism graphs 必有：(1) 相同的點數
(2) 相同的邊數 (3) 相同的degree分佈。

※給定二圖，判斷它們是否isomorphic的問題一般來說不易解，而且答案常是否定的。

Example 9. Show that G and H are not isomorphic.

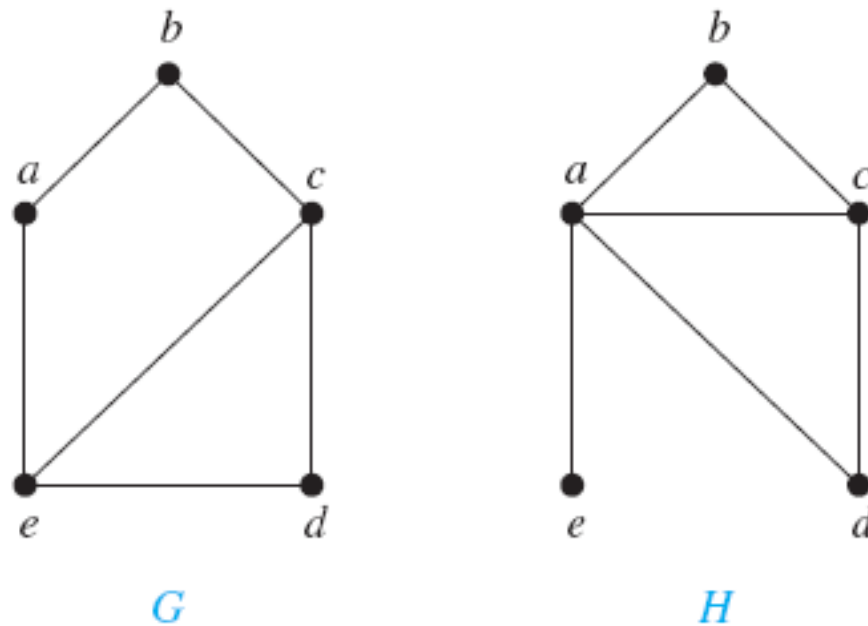


FIGURE 9 The Graphs G and H .

Sol :

G 有 degree = 1 的點， H 沒有

Not isomorphic !

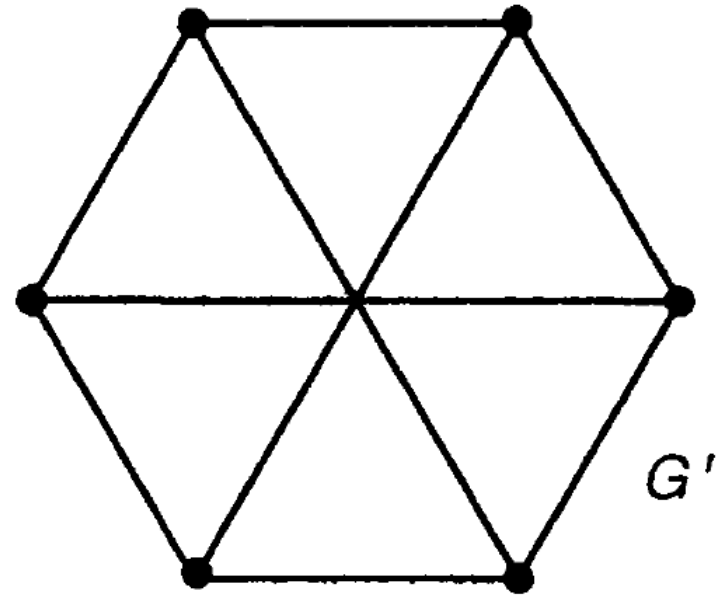
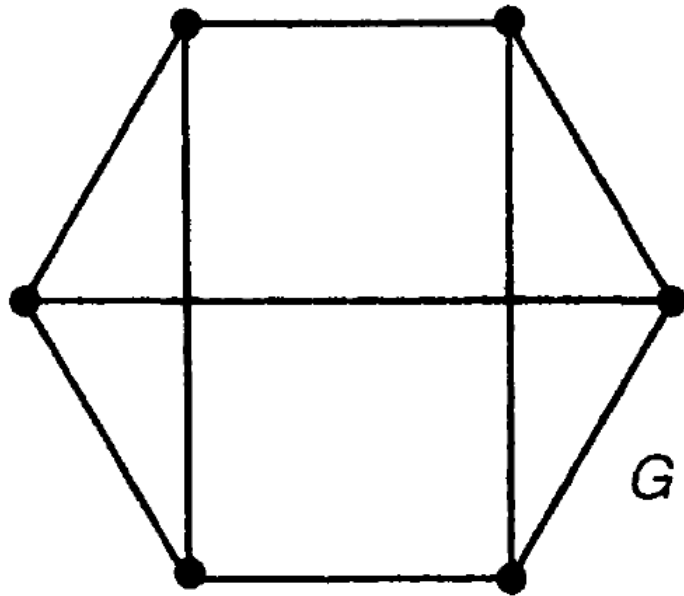
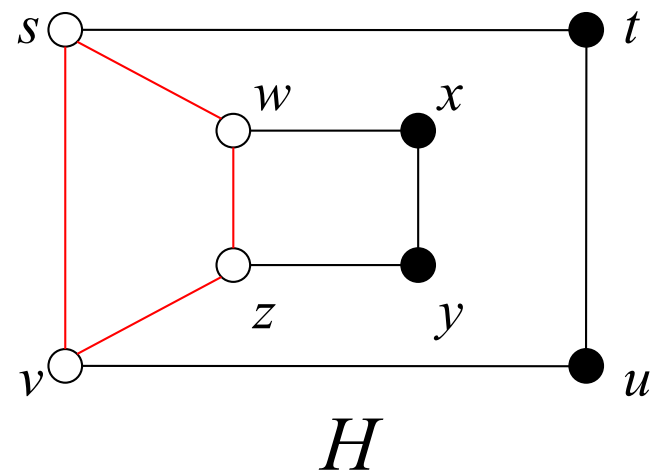
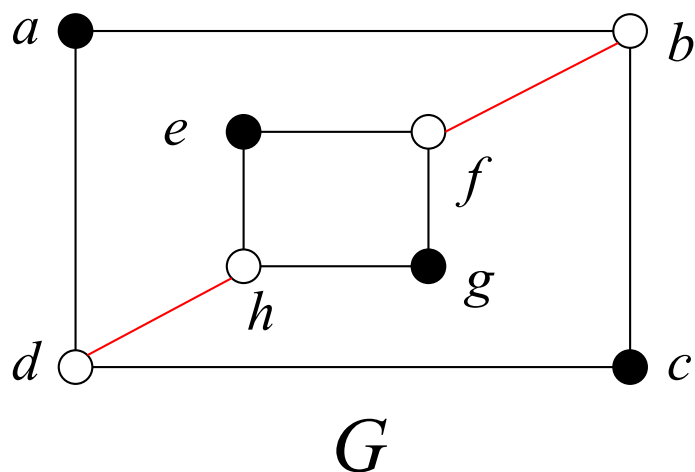


Figure 11.7

Example 10.

Determine whether G and H are isomorphic.



Sol : $\because$ G 中 degree 為 3 的點有 d, h, f, b

它們不能接成 4-cycle

但 H 中 degree 為 3 的點 s, w, z, v 可接成 4-cycle

$\therefore$ 不是 isomorphic.

另法 : G 中 $\deg = 3$ 的點，旁邊都只連了另一個 $\deg = 3$ 的點

但 H 中 $\deg = 3$ 的點旁邊都連了 2 個 $\deg = 3$ 的點

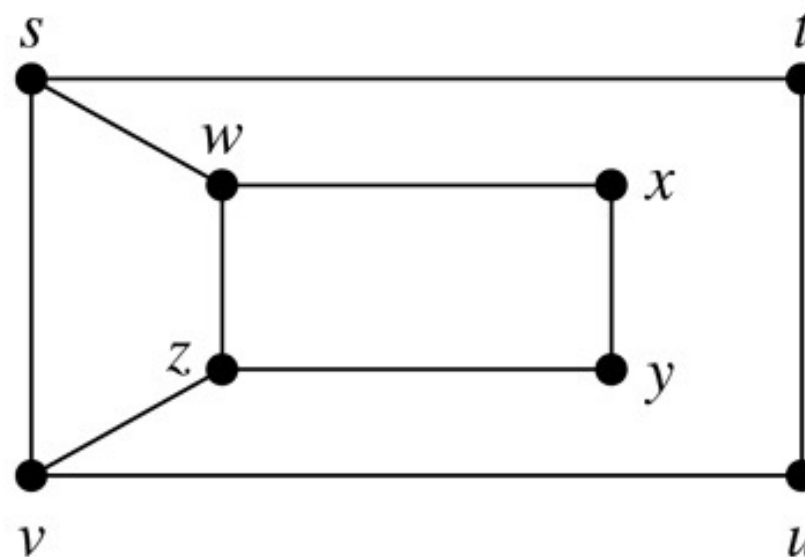
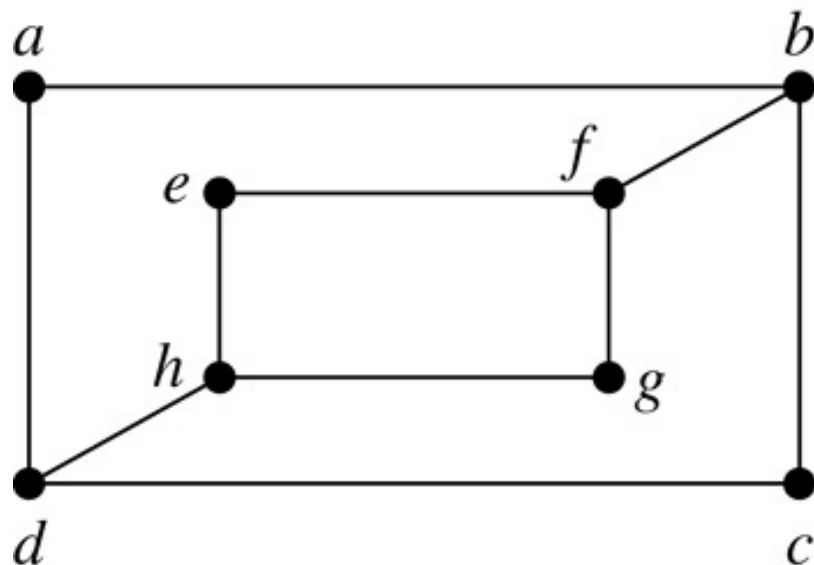


FIGURE 10 The Graphs G and H .

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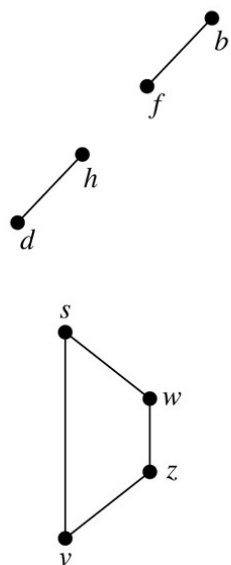
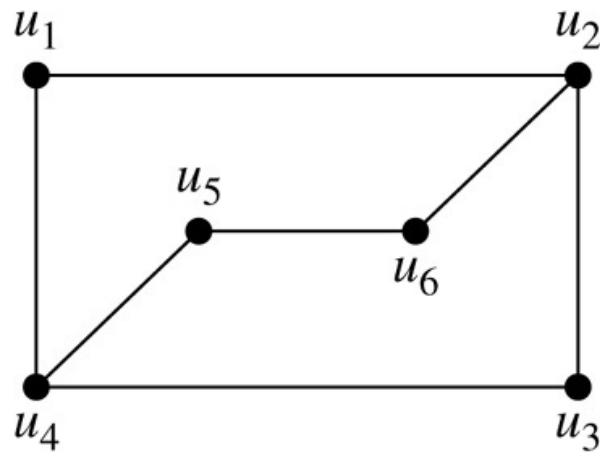
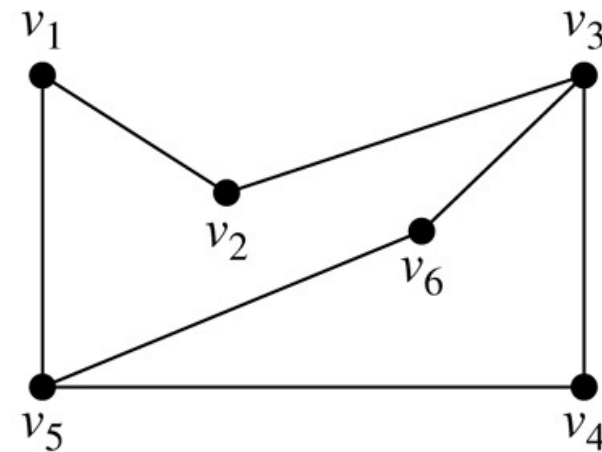


FIGURE 11 The Subgraphs of G and H Made Up of Vertices of Degree Three and the Edges Connecting Them.

Example 11. Determine whether the graphs G and H are isomorphic.



G

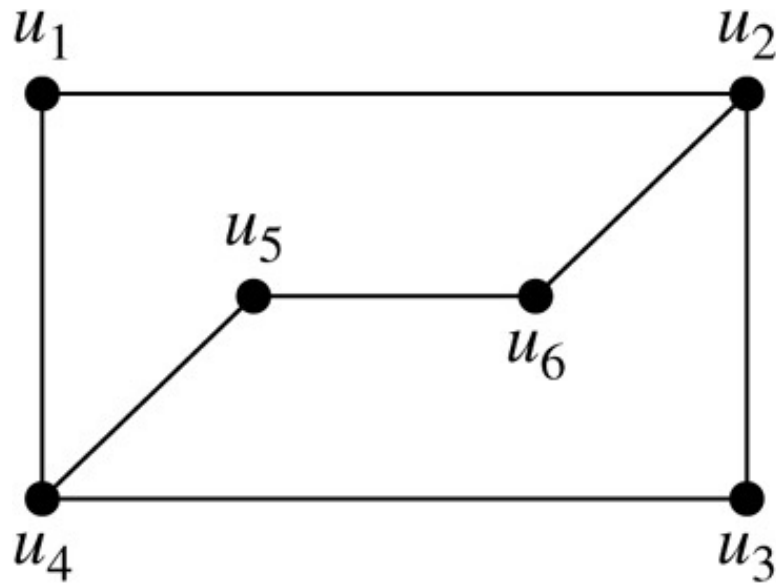


H

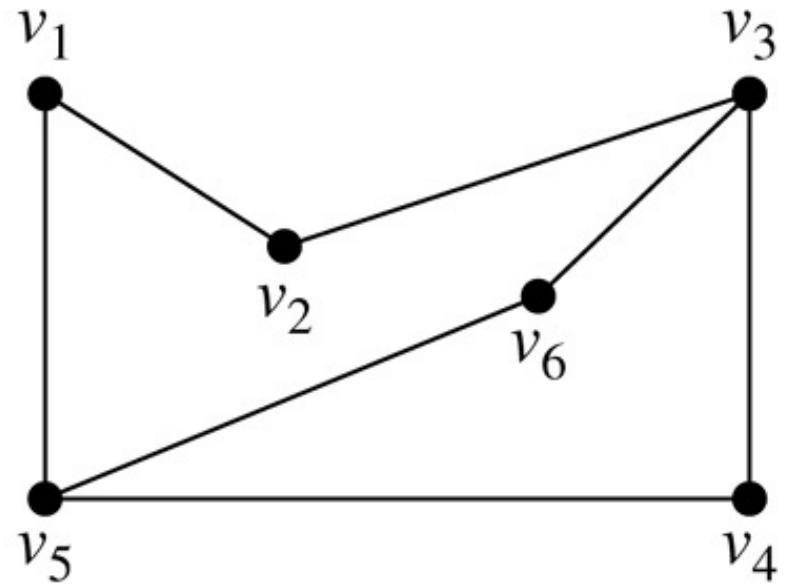
$$\mathbf{A}_G = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix},$$

$$\mathbf{A}_H = \begin{matrix} & \begin{matrix} v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \end{matrix} \\ \begin{matrix} v_6 \\ v_3 \\ v_4 \\ v_5 \\ v_1 \\ v_2 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}.$$

Example 11. Determine whether the graphs G and H are isomorphic.



G



H

Sol:

$$f(u_1)=v_6, f(u_2)=v_3, f(u_3)=v_4, f(u_4)=v_5, f(u_5)=v_1, f(u_6)=v_2$$

$\Rightarrow$ Yes

Algorithms for Graph Isomorphism

The best algorithms known for determining whether two graphs are isomorphic have exponential worst-case time complexity (in the number of vertices of the graphs).

However, there are algorithms with linear average-case time complexity.

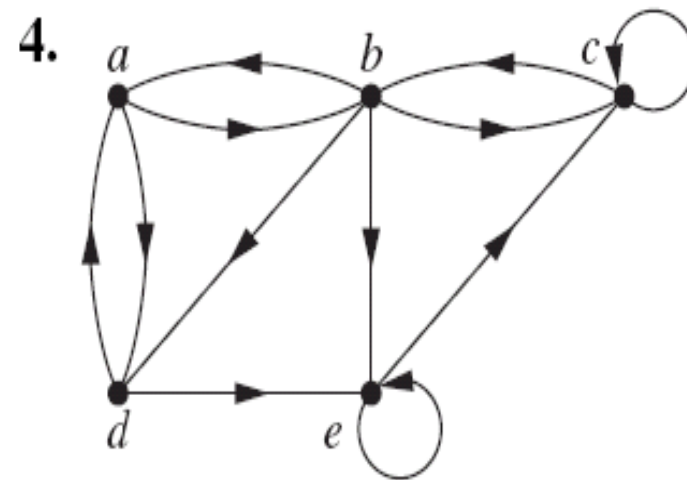
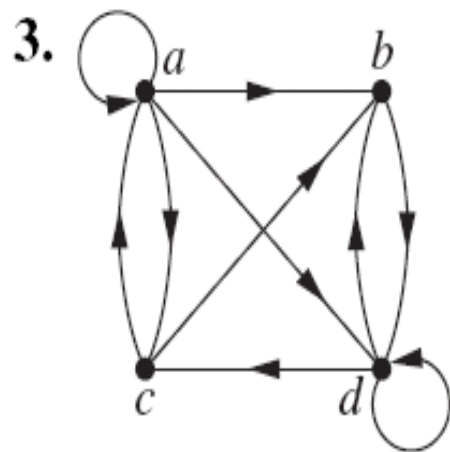
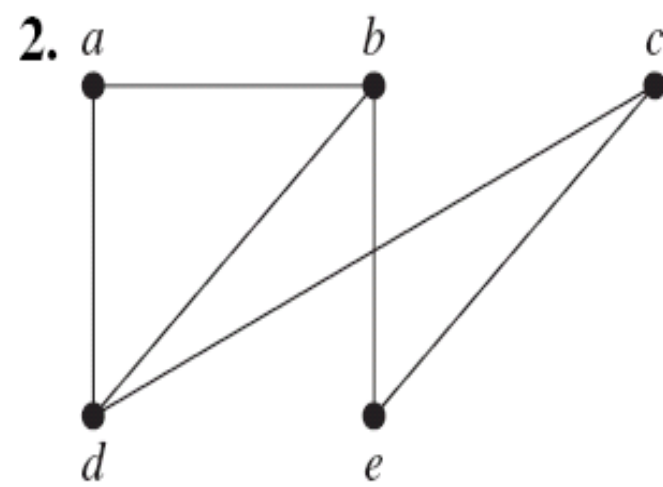
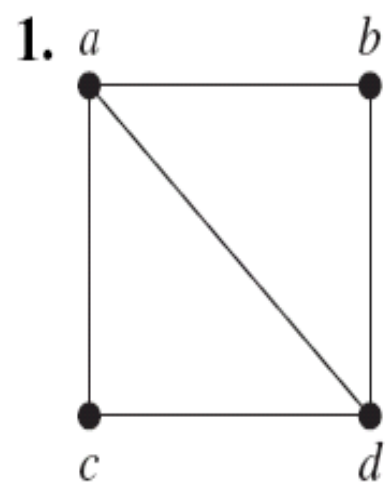
You can use a public domain program called NAUTY to determine in less than a second whether two graphs with as many as 100 vertices are isomorphic.

Graph isomorphism is a problem of special interest because it is one of a few NP problems not known to be either tractable or **NP-complete** (see Section 3.3).

Applications of Graph Isomorphism

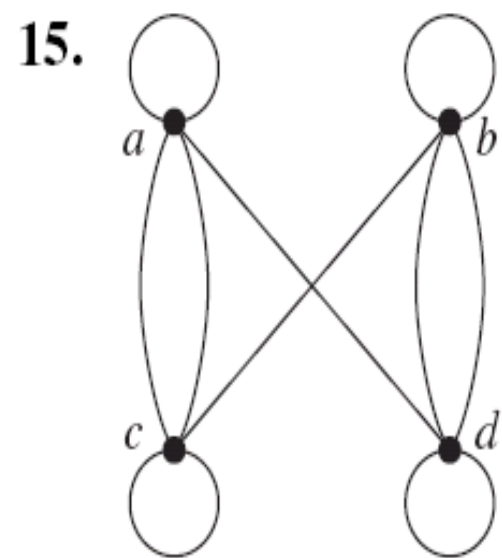
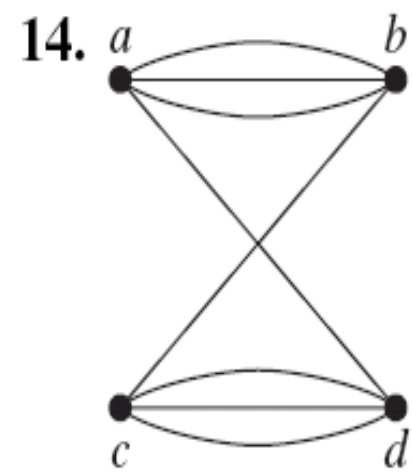
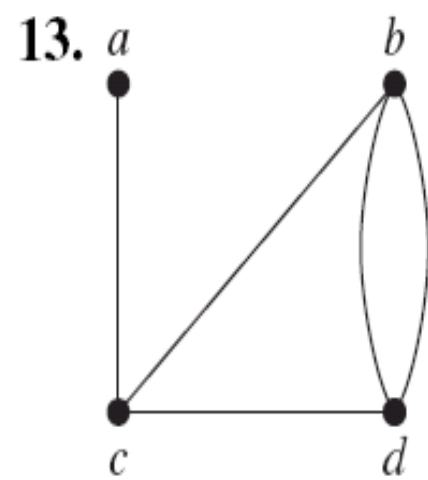
The question whether graphs are isomorphic plays an important role in applications of graph theory. For example,

- chemists use molecular graphs to model **chemical compounds**. Vertices represent atoms and edges represent chemical bonds. When a new compound is synthesized, a database of molecular graphs is checked to determine whether the graph representing the new compound is isomorphic to the graph of a compound that this already known.
- **Electronic circuits** are modeled as graphs in which the vertices represent components and the edges represent connections between them. Graph isomorphism is the basis for
 - the verification that a particular layout of a circuit corresponds to the design's original schematics.
 - determining whether a chip from one vendor includes the intellectual property of another vendor.



$$\mathbf{10.} \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{11.} \quad \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

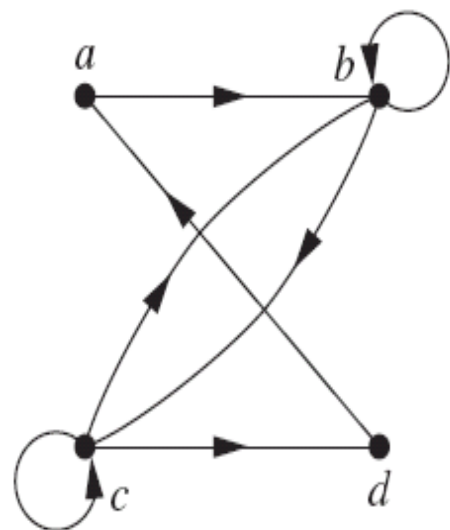


16. $\begin{bmatrix} 1 & 3 & 2 \\ 3 & 0 & 4 \\ 2 & 4 & 0 \end{bmatrix}$

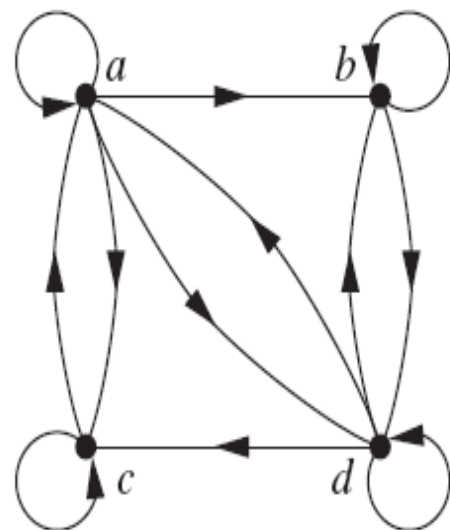
17. $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$

18. $\begin{bmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{bmatrix}$

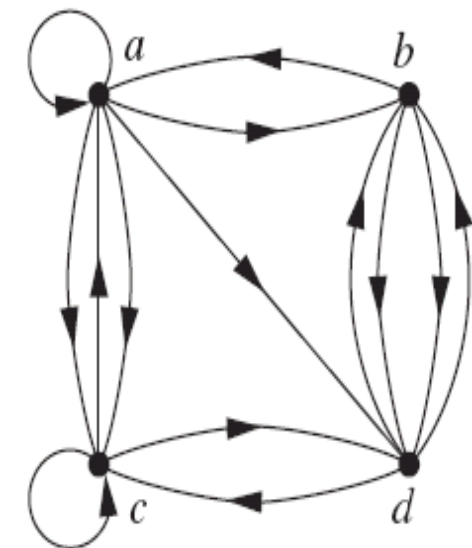
19.



20.



21.

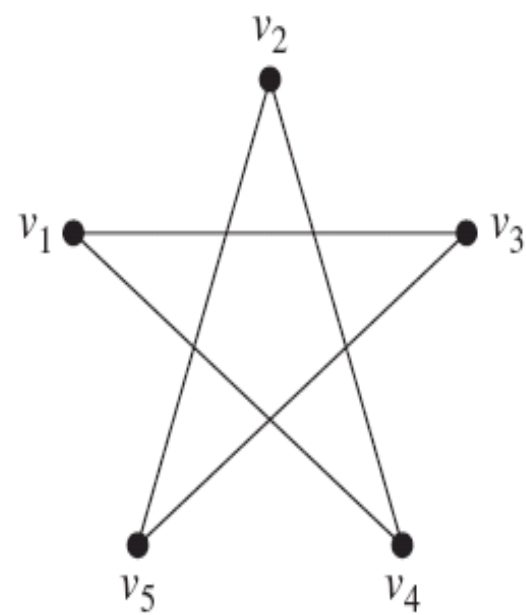
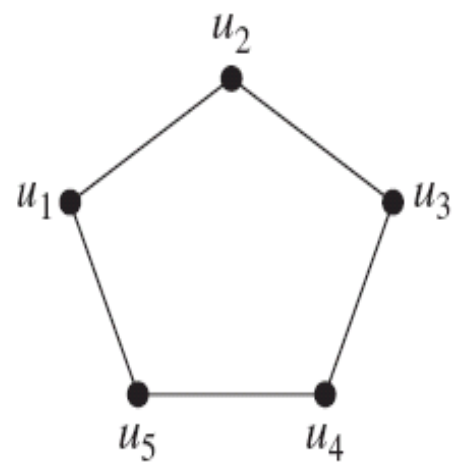


22. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

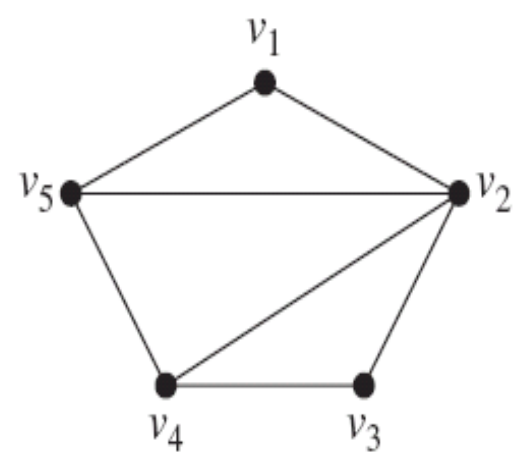
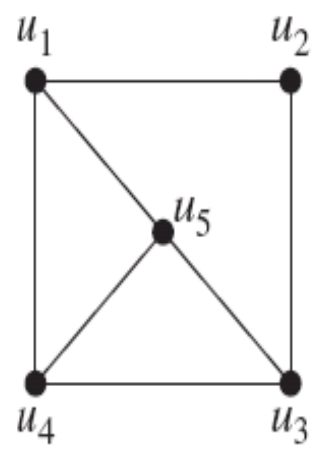
23. $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 0 & 0 \\ 0 & 2 & 2 \end{bmatrix}$

24. $\begin{bmatrix} 0 & 2 & 3 & 0 \\ 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$

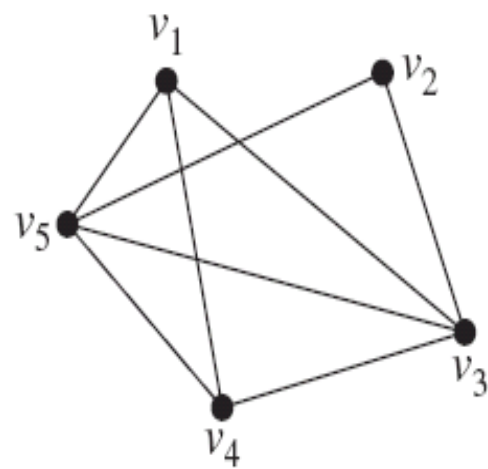
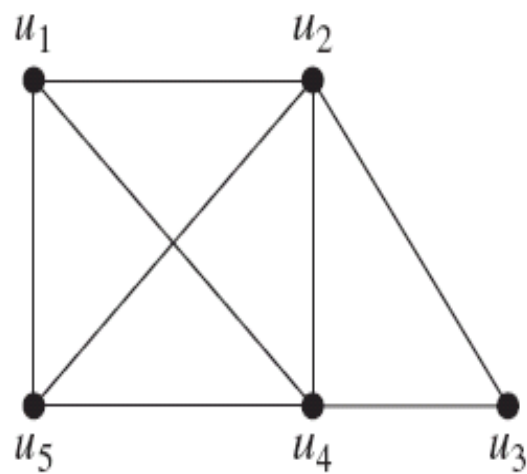
35.



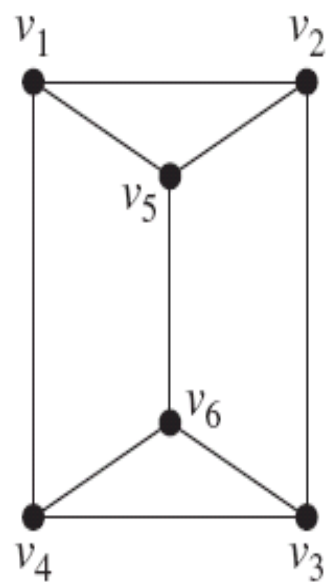
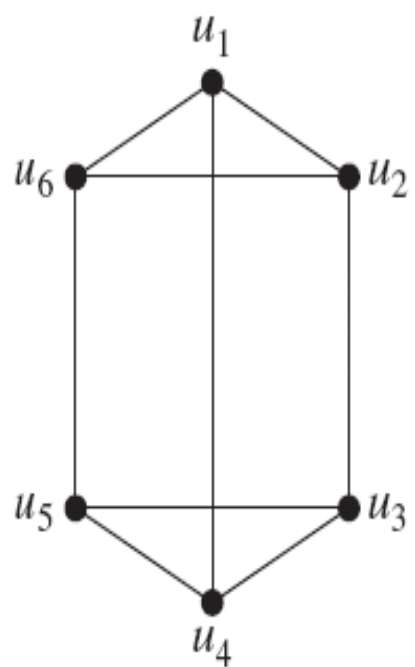
36.



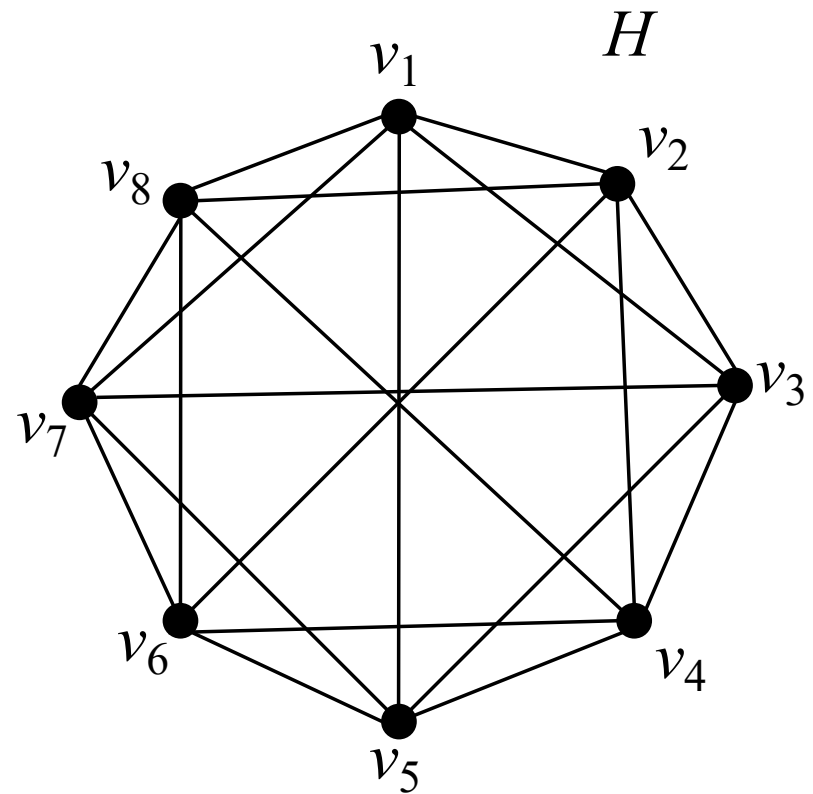
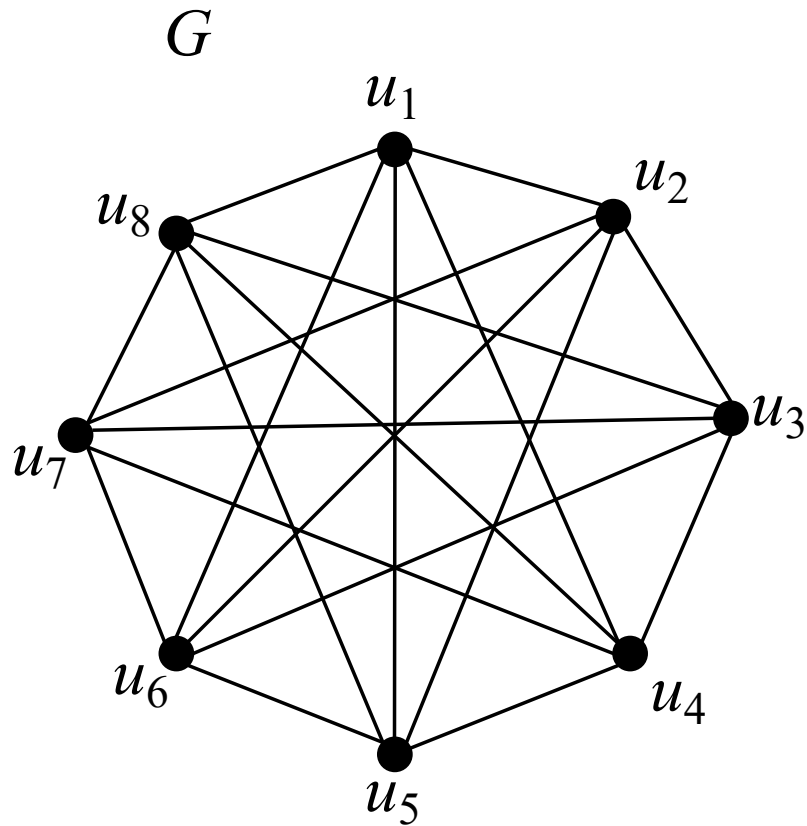
38.



39.



28. Determine whether the graphs G and H are isomorphic.



此二圖都是**vertex-transitive**，每個點的角色是一樣的，故考慮 u_1 及 v_1 即可
 v_1 的5個neighbor可以連接成**5-cycle**，但 u_1 的不行，故不是**isomorphic**

§10.4: Connectivity

Def. 1,2 :

In an undirected graph, a **path of length n** from u to v is a sequence of $n+1$ adjacent vertices going from vertex u to vertex v .

(e.g., $P: u=x_0, x_1, x_2, \dots, x_n=v$.) (P has n edges.)

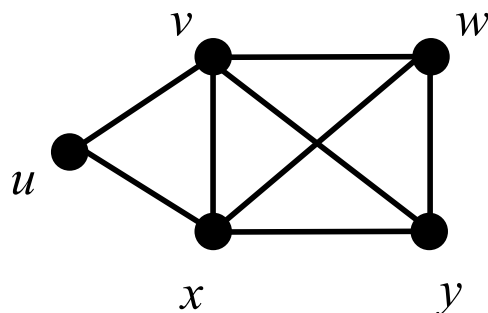
Def:

path中的點及邊不可重複

trail: 允許點重複的路徑 (邊不可重複)

walk: 允許點及邊重複的路徑

Example



path: u, v, y

trail: u, v, w, y, v, x, y

walk: u, v, w, v, x, v, y

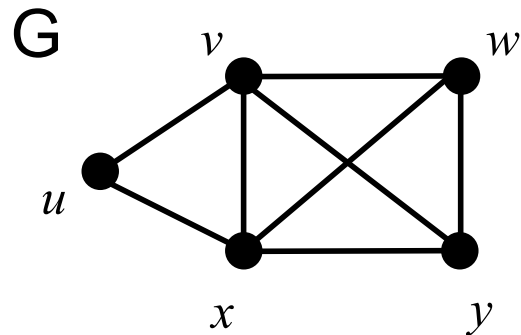
Def:

cycle: path with $u=v$

circuit: trail with $u=v$

closed walk: walk with $u=v$

Example

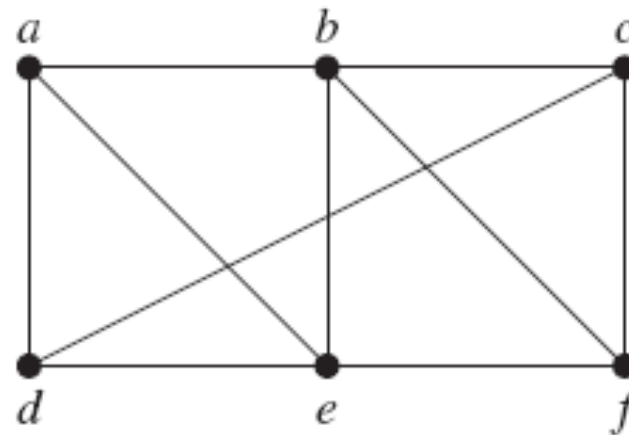


cycle: u, v, y, x, u

trail: u, v, w, y, v, x, u

walk: $u, v, w, v, x, v, y, x, u$

- **Example:** In the simple graph here:
 - a, d, c, f, e is a simple path of length 4.
 - d, e, c, a is not a path because e is not connected to c .
 - b, c, f, e, b is a circuit of length 4.
 - a, b, e, d, a, b is a path of length 5, but it is not a simple path.

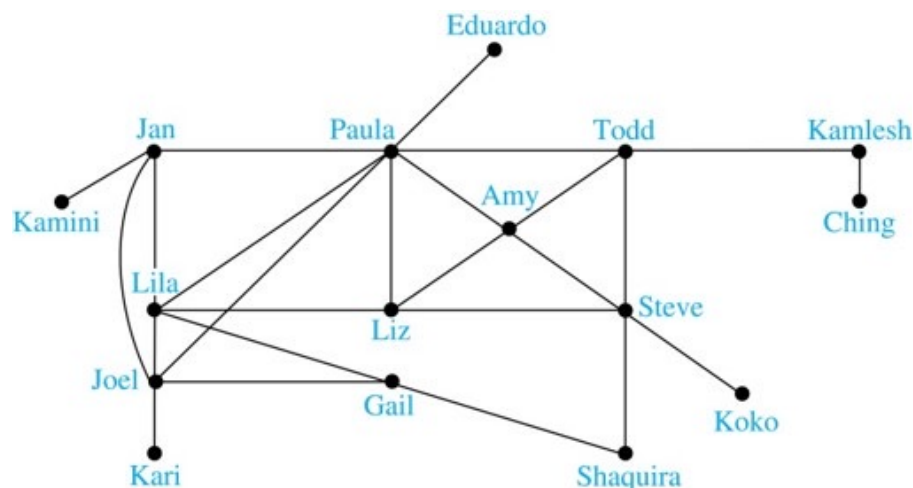


Example 1.

FIGURE 1 A Simple Graph.

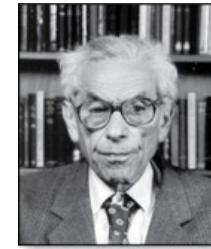
Degrees of Separation

Example: *Paths in Acquaintanceship Graphs.* In an acquaintanceship graph there is a path between two people if there is a chain of people linking these people, where two people adjacent in the chain know one another. In this graph there is a chain of six people linking Kamini and Ching.



Some have speculated that almost every pair of people in the world are linked by a small chain of no more than six, or maybe even, five people. The play *Six Degrees of Separation* by John Guare is based on this notion.

Erdős numbers



Paul
Erdős

Example: *Erdős numbers*.

In a collaboration graph, two people a and b are connected by a path when there is a sequence of people starting with a and ending with b such that the endpoints of each edge in the path are people who have collaborated.

- In the academic collaboration graph of people who have written papers in mathematics, the *Erdős number* of a person m is the length of the shortest path between m and the prolific mathematician Paul Erdős.
- To learn more about Erdős numbers, visit

<http://www.ams.org/mathscinet/collaborationDistance.html>

TABLE 1 The Number of Mathematicians with a Given Erdős Number (as of early 2006).

<i>Erdős Number</i>	<i>Number of People</i>
0	1
1	504
2	6,593
3	33,605
4	83,642
5	87,760
6	40,014
7	11,591
8	3,146
9	819
10	244
11	68
12	23
13	5

Bacon Numbers

In the Hollywood graph, two actors a and b are linked when there is a chain of actors linking a and b , where every two actors adjacent in the chain have acted in the same movie.

The *Bacon number* of an actor c is defined to be the length of the shortest path connecting c and the well-known actor Kevin Bacon. (Note that we can define a similar number by replacing Kevin Bacon by a different actor.)

The *oracle of Bacon* web site <http://oracleofbacon.org/how.php> provides a tool for finding Bacon numbers.

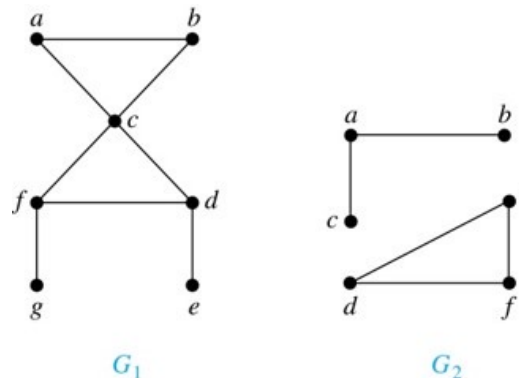
TABLE 2 The Number of Actors with a Given Bacon Number (as of August 2017).

<i>Bacon Number</i>	<i>Number of People</i>
0	1
1	3,452
2	401,636
3	1,496,104
4	390,878
5	4,388
6	631
7	131
8	9
9	1

Connectedness in Undirected Graphs

Definition: An undirected graph is called *connected* if there is a path between every pair of vertices. An undirected graph that is not *connected* is called *disconnected*. We say that we *disconnect* a graph when we remove vertices or edges, or both, to produce a disconnected subgraph.

Example: G_1 is connected because there is a path between any pair of its vertices, as can be easily seen. However G_2 is not connected because there is no path between vertices a and f , for example.



[Jump to long description](#)

Example 4.

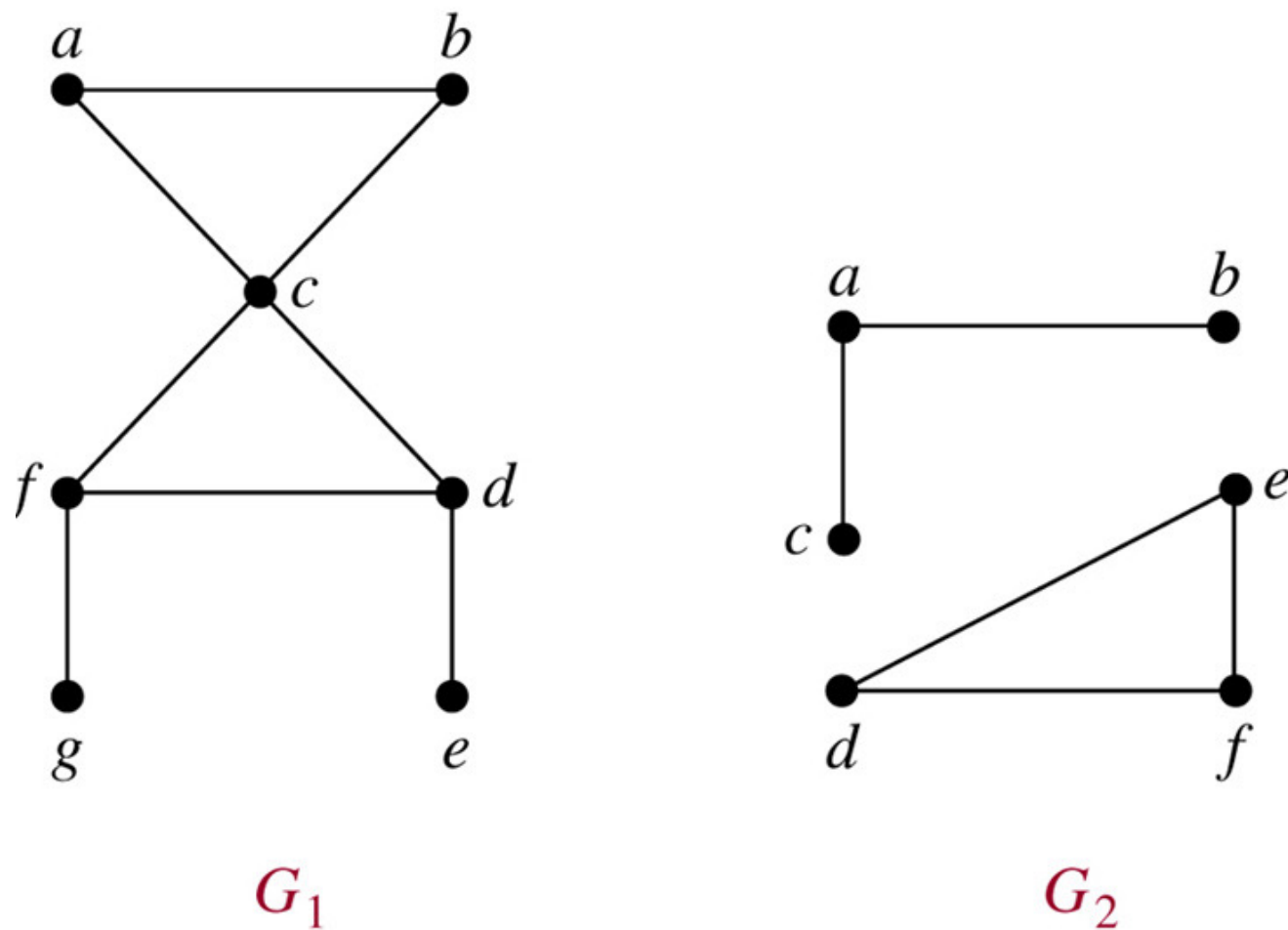


FIGURE 2 The Graphs G_1 and G_2 .

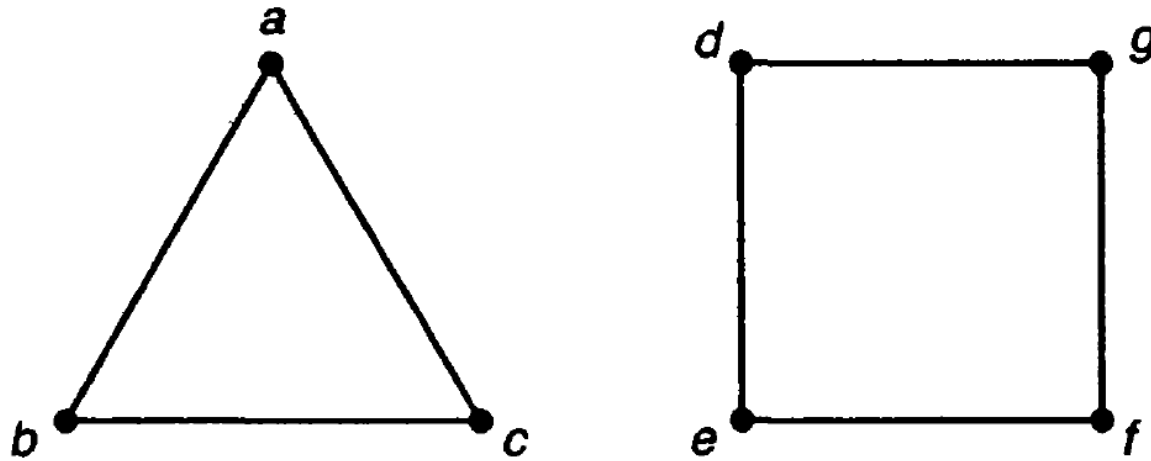


Figure 11.8

disconnected

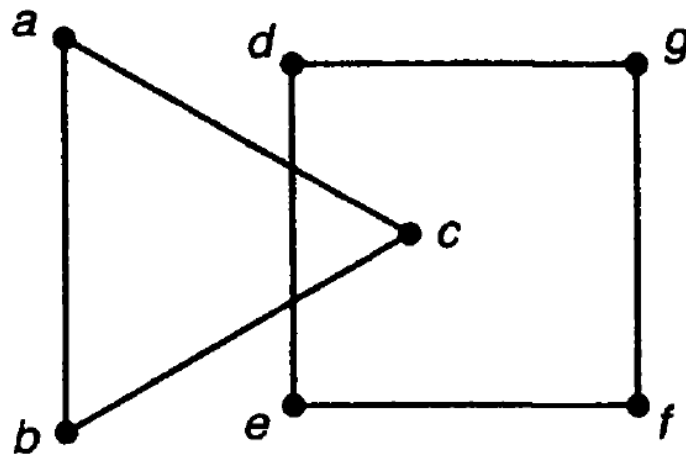
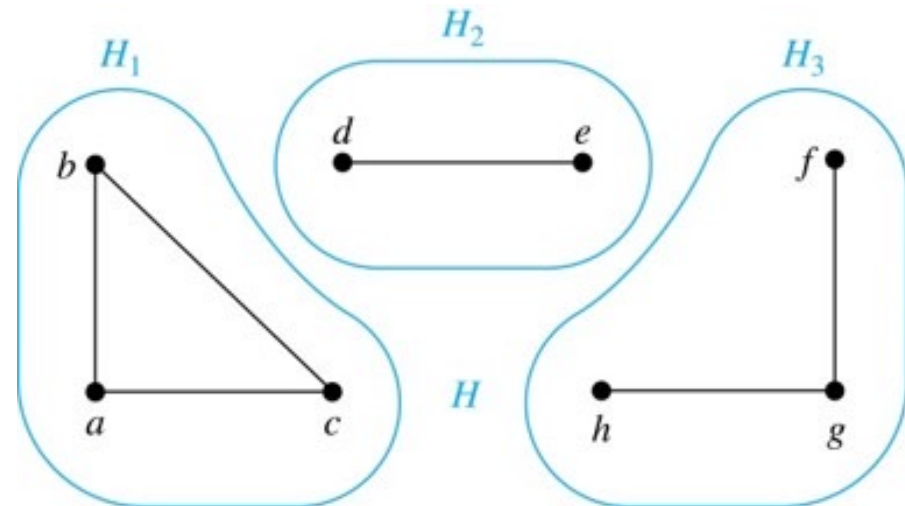


Figure 11.9

Connected Components

Definition: A *connected component* of a graph G is a connected subgraph of G that is not a proper subgraph of another connected subgraph of G . A graph G that is not connected has two or more connected components that are disjoint and have G as their union.

Example: The graph H is the union of three disjoint subgraphs H_1 , H_2 , and H_3 , none of which are proper subgraphs of a larger connected subgraph of G . These three subgraphs are the connected components of H .



THEOREM 1

There is a simple path between every pair of distinct vertices of a connected undirected graph.

Example 5 What are the connected components of the graph H ?

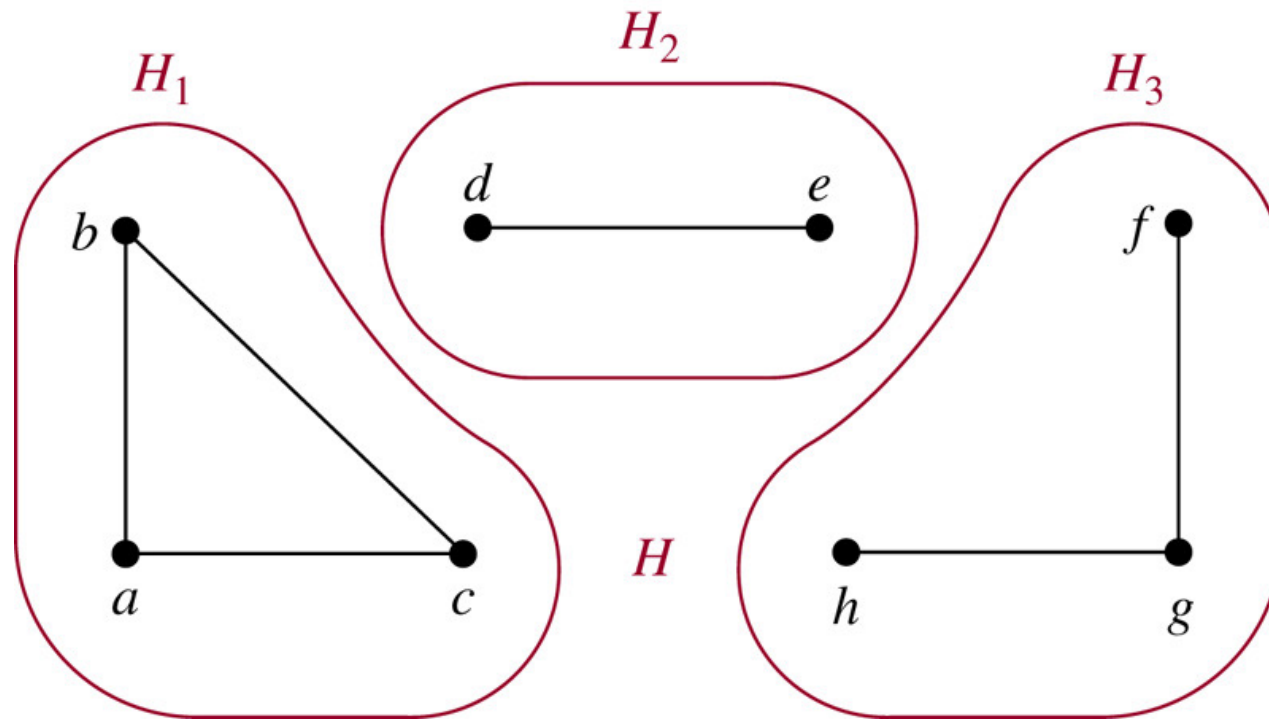


FIGURE 3 The Graph H and Its Connected Components H_1 , H_2 , and H_3 .

Def:

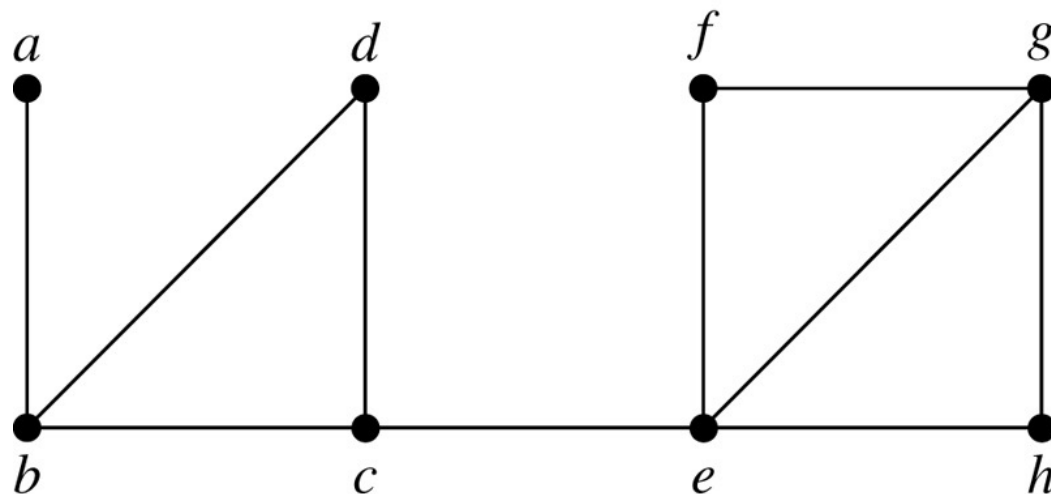
A *cut vertex* separates one connected component into several components if it is removed.

A *cut edge* separates one connected component into two components if it is removed.

Example 7. Find the cut vertices and cut edges in the graph G .

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Sol:

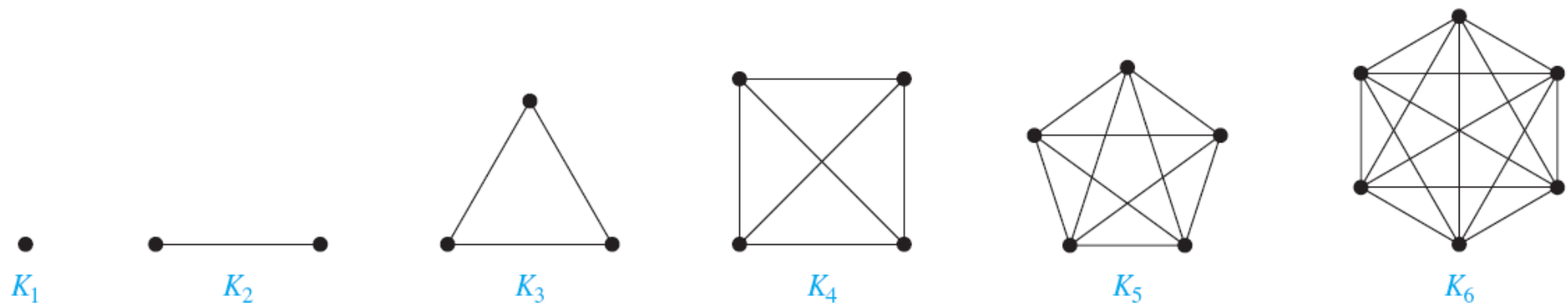


cut vertices: b, c, e

cut edges:

$\{a, b\}, \{c, e\}$

FIGURE 4 The Graph G .



VERTEX CONNECTIVITY Not all graphs have cut vertices. For example, the complete graph K_n , where $n \geq 3$, has no cut vertices. When you remove a vertex from K_n and all edges incident to it, the resulting subgraph is the complete graph K_{n-1} , a connected graph. Connected graphs without cut vertices are called **nonseparable graphs**, and can be thought of as more connected than those with a cut vertex. We can extend this notion by defining a more granulated measure of graph connectivity based on the minimum number of vertices that can be removed to disconnect a graph.

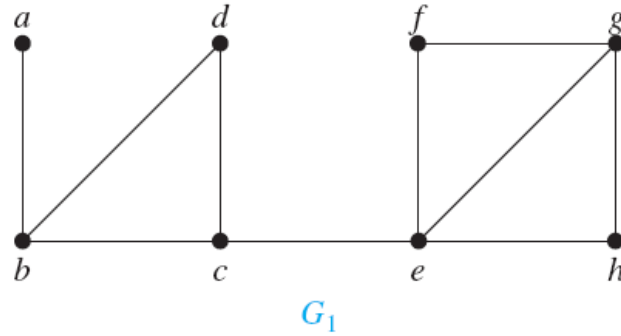
connected graph, except a complete graph, has a vertex cut. We define the **vertex connectivity** of a noncomplete graph G , denoted by $\kappa(G)$, as the minimum number of vertices in a vertex cut.

Consequently, for every graph G , $\kappa(G)$ is minimum number of vertices that can be removed from G to either disconnect G or produce a graph with a single vertex. We have $0 \leq \kappa(G) \leq n - 1$ if G has n vertices, $\kappa(G) = 0$ if and only if G is disconnected or $G = K_1$, and $\kappa(G) = n - 1$ if and only if G is complete [see Exercise 52(a)].

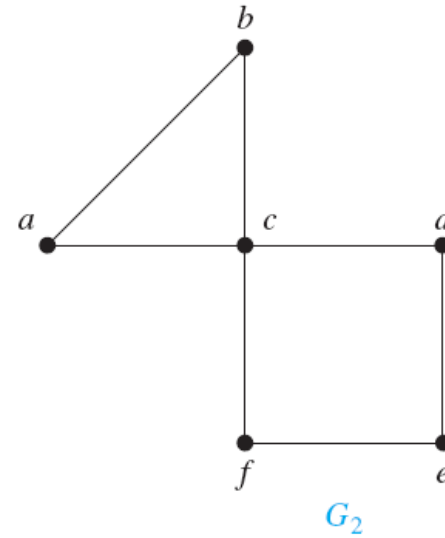
The larger $\kappa(G)$ is, the more connected we consider G to be. Disconnected graphs and K_1 have $\kappa(G) = 0$, connected graphs with cut vertices and K_2 have $\kappa(G) = 1$, graphs without cut vertices that can be disconnected by removing two vertices and K_3 have $\kappa(G) = 2$, and so on. We say that a graph is **k -connected** (or **k -vertex-connected**), if $\kappa(G) \geq k$. A graph G is 1-connected if it is connected and not a graph containing a single vertex; a graph is 2-connected, or

Example 8.

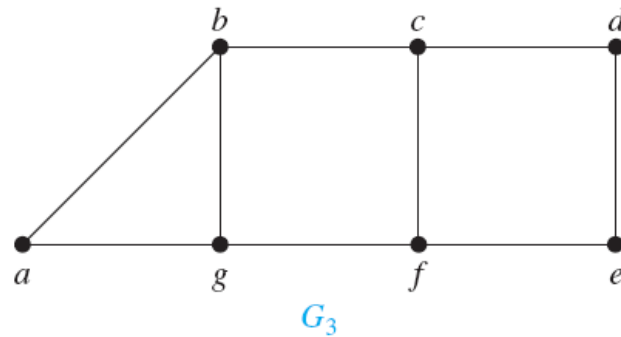
$$\kappa(G1) = 1$$



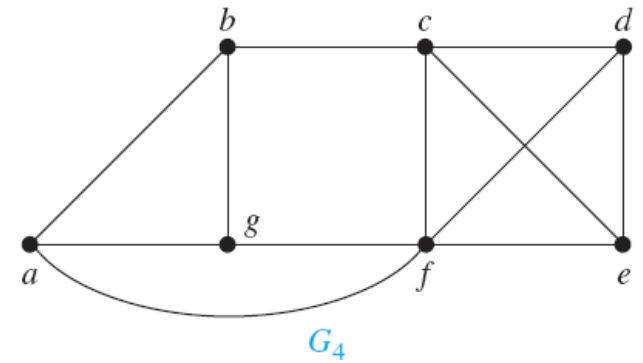
$$\kappa(G2) = 1$$



$$\kappa(G3) = 2$$



$$\kappa(G4) = 2$$



$$\kappa(G5) = 3$$

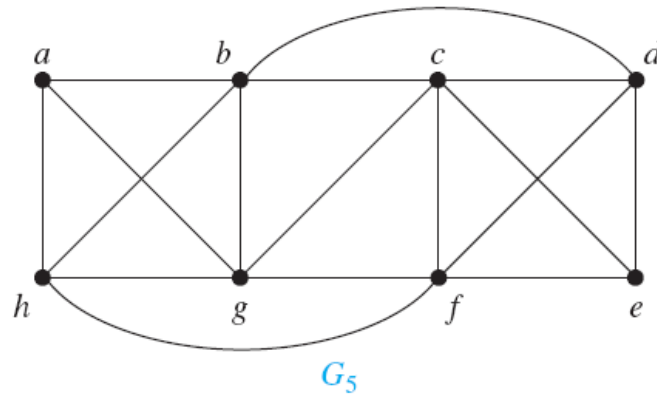


FIGURE 4 Some Connected Graphs

Connectedness in Directed Graphs

Def. 4: A directed graph is *strongly connected* if there is a path from a to b for any two vertices a, b . A directed graph is *weakly connected* if there is a path between every two vertices in the underlying undirected graphs.

Example 9 Are the directed graphs G and H strongly connected or weakly connected?

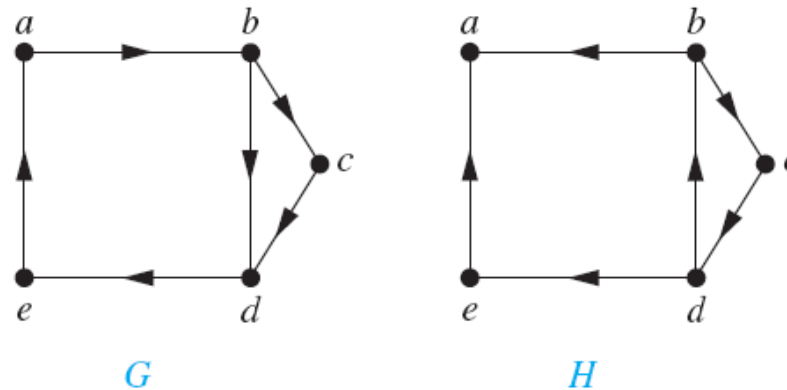


FIGURE 5 The Directed Graphs G and H .

Paths and Isomorphism

Note that connectedness, and the existence of a circuit or simple circuit of length k are graph invariants with respect to isomorphism.

Example 12. Determine whether the graphs G and H are isomorphic.

Sol: No,
因 G 沒有三角形，
 H 有

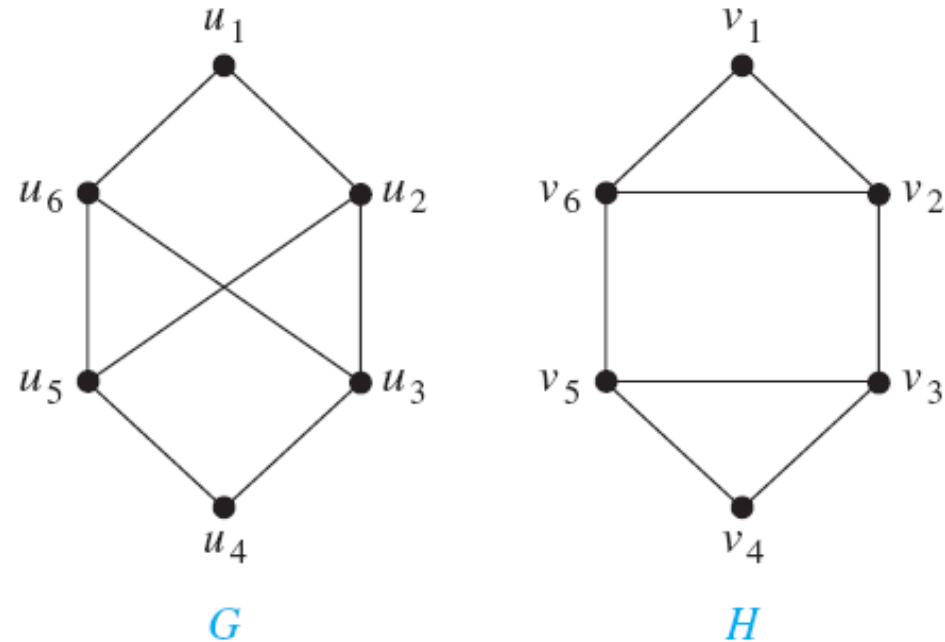
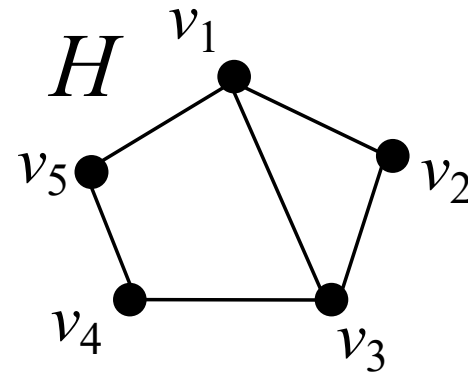
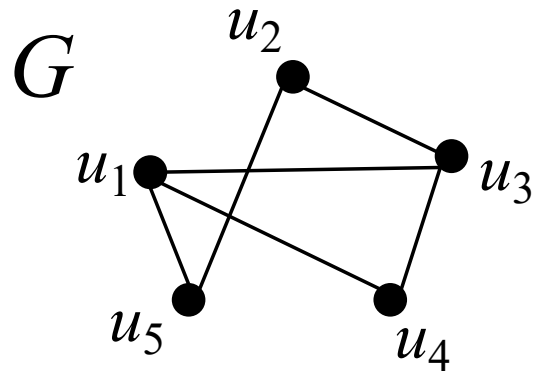


FIGURE 6 The Graphs G and H .

Example 13. Determine whether the graphs G and H are isomorphic.



Sol.

Both G and H have 5 vertices, 6 edges, two vertices of deg 3, three vertices of deg 2, a 3-cycle, a 4-cycle, and a 5-cycle. $\Rightarrow G$ and H may be isomorphic.

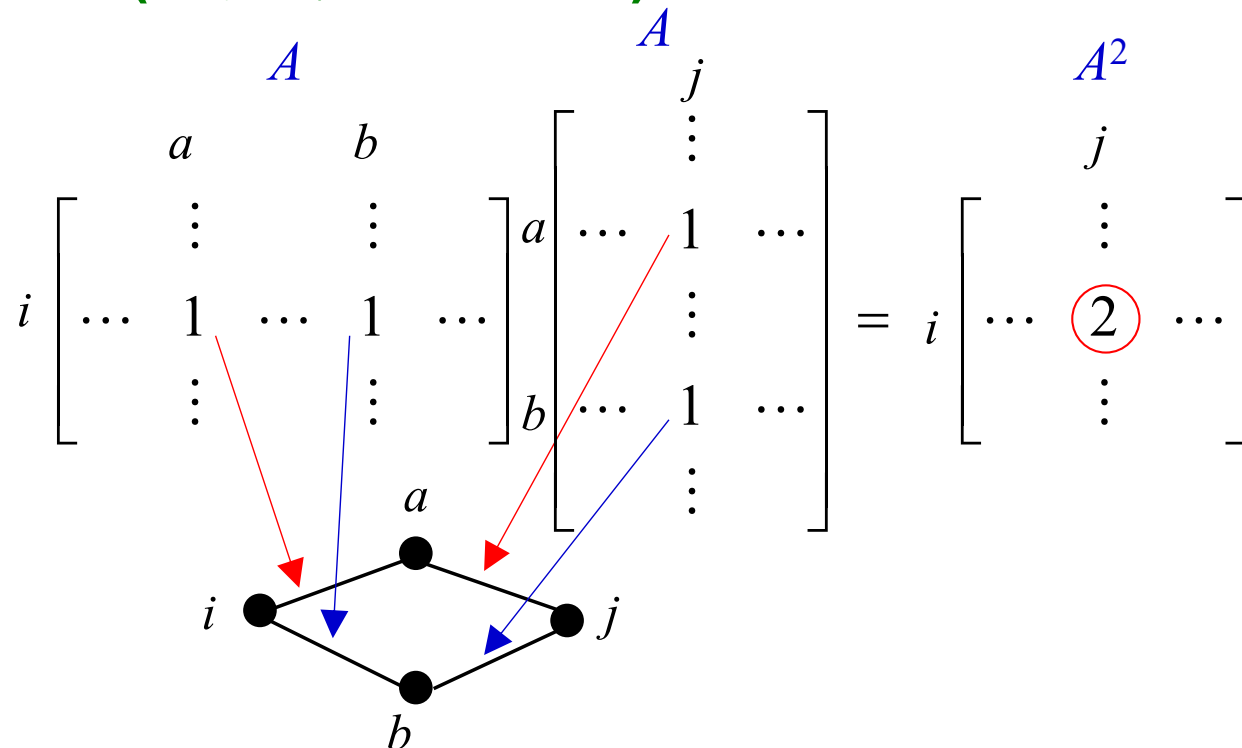
The function f with $f(u_1) = v_1, f(u_2) = v_4, f(u_3) = v_3, f(u_4) = v_2$ and $f(u_5) = v_5$ is a one-to-one correspondence between $V(G)$ and $V(H)$. $\Rightarrow G$ and H maybe be isomorphic.

Counting Paths between Vertices

Theorem 2:

Let G be a graph with adjacency matrix A with respect to the ordering $v_1, v_2, \dots, v_n$. The number of **walks** of length r from v_i to v_j is equal to $(A^r)_{i,j}$.

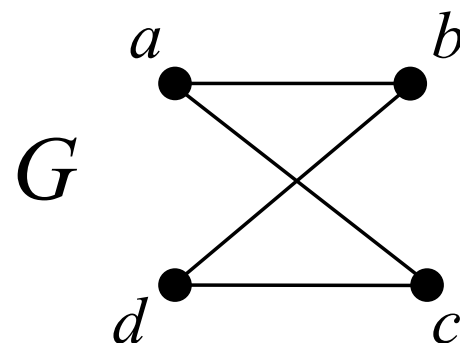
Proof (僅簡單舉例說明)



Example 15. How many **paths** of length 4 are there from a to d in the graph G ?

Sol.

The adjacency matrix of G (ordering as a, b, c, d) is



$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix} \Rightarrow A^4 = \begin{bmatrix} 8 & 0 & 0 & 8 \\ 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 \\ 8 & 0 & 0 & 8 \end{bmatrix} \Rightarrow 8$$

Q: 哪8條？

$a-b-a-b-d, a-b-a-c-d, a-c-a-b-d, a-c-a-c-d,$
 $a-b-d-b-d, a-b-d-c-d, a-c-d-b-d, a-c-d-c-d$

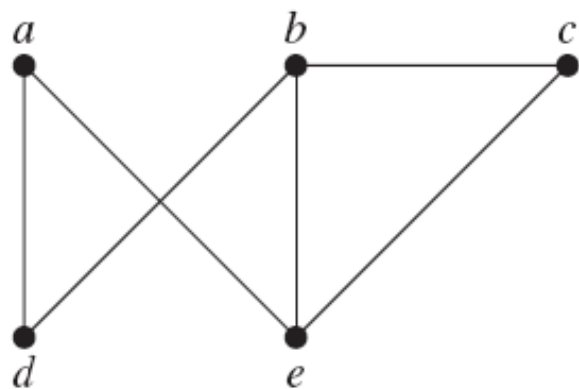
1. Does each of these lists of vertices form a path in the following graph? Which paths are simple? Which are circuits? What are the lengths of those that are paths?

a) a, e, b, c, b

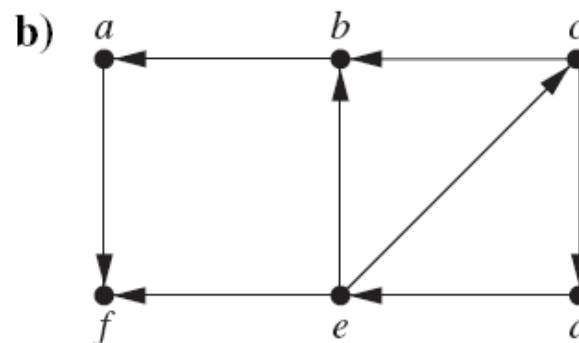
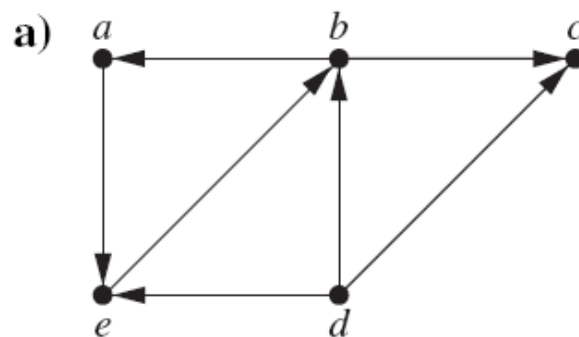
b) a, e, a, d, b, c, a

c) e, b, a, d, b, e

d) c, b, d, a, e, c

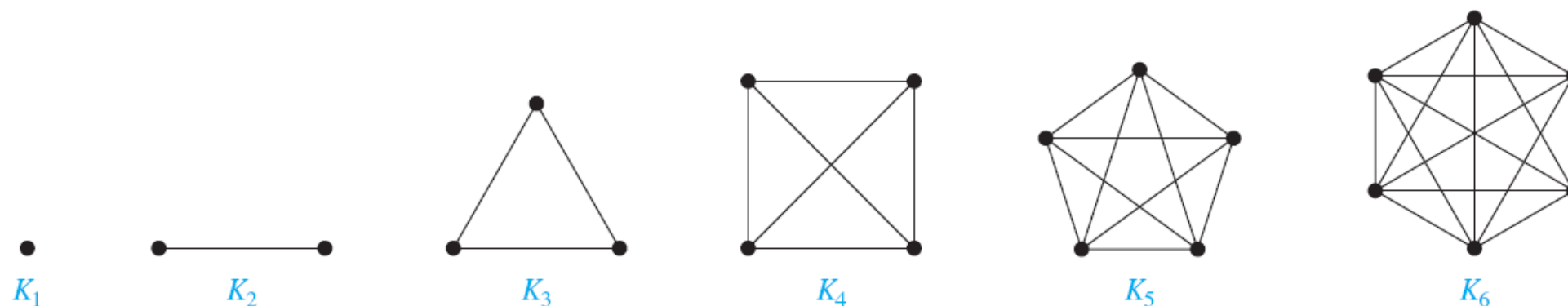


14. Find the strongly connected components of each of these graphs.



19. Find the number of paths of length n between two different vertices in K_4 if n is

- a) 2. b) 3. c) 4. d) 5.



25. Find the number of paths of length n between any two nonadjacent vertices in $K_{3,3}$ for the values of n in Exercise 19.

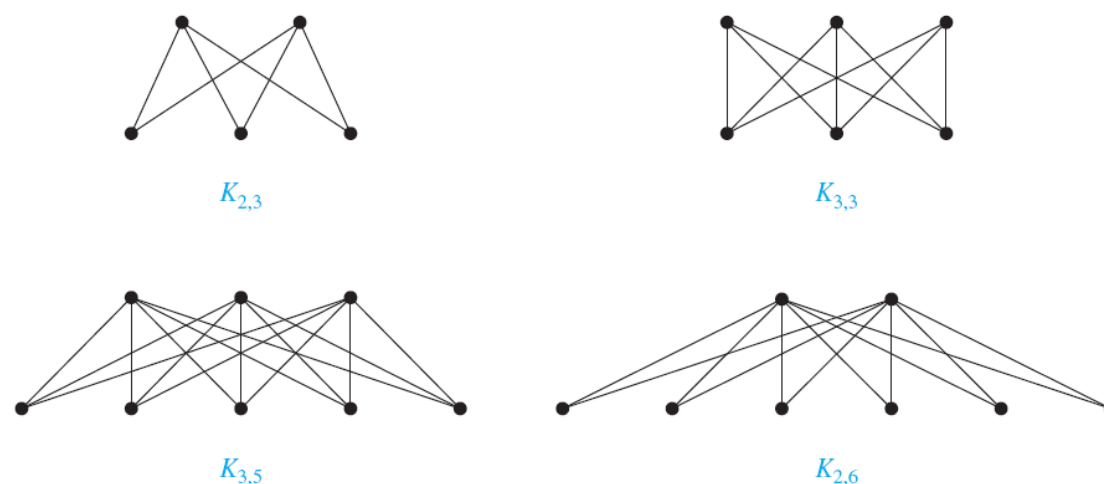
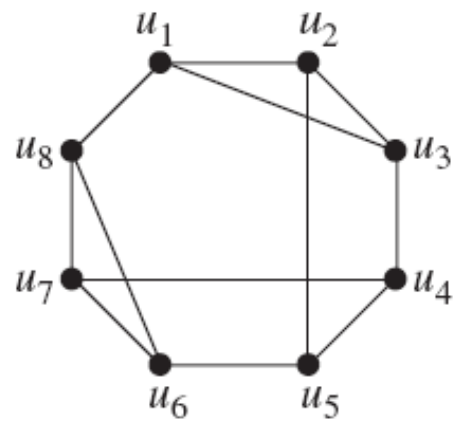
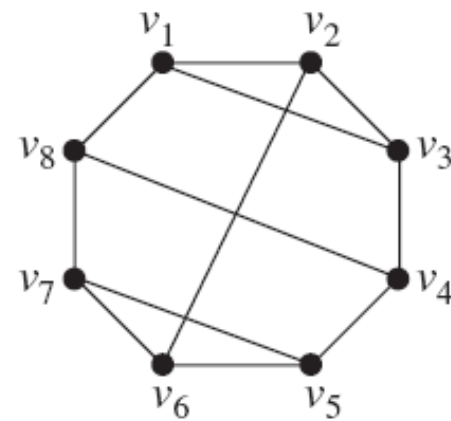


FIGURE 9 Some Complete Bipartite Graphs.

22. Use paths either to show that these graphs are not isomorphic or to find an isomorphism between them.



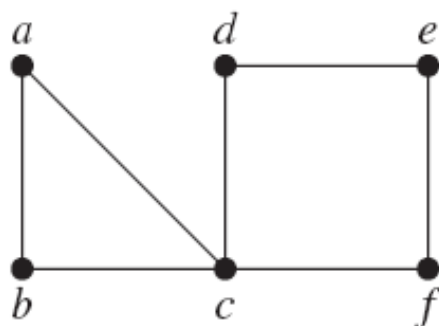
G



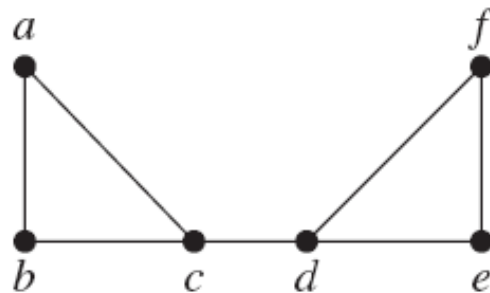
H

In Exercises 31–33 find all the cut vertices of the given graph.

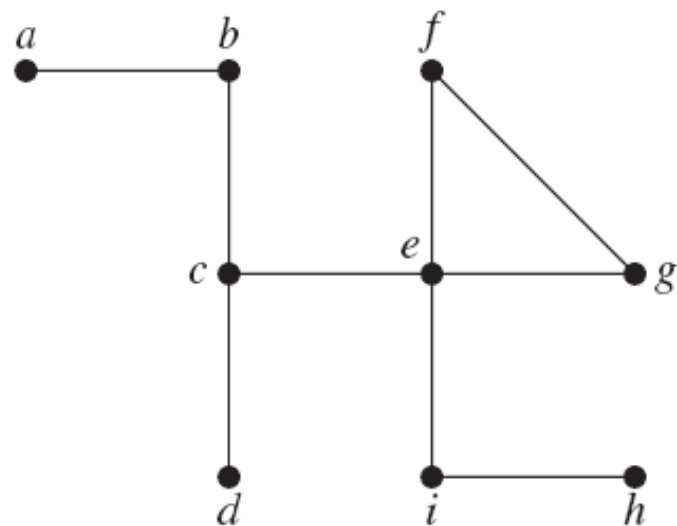
31.



32.



33.



§10.5: Euler & Hamilton Paths

- The town of Königsberg, Prussia (now Kalingrad, Russia) was divided into four sections by the branches of the Pregel river. In the 18th century seven bridges connected these regions.
- People wondered whether whether it was possible to follow a path that crosses each bridge exactly once and returns to the starting point.
- The Swiss mathematician Leonard Euler proved that no such path exists. This result is often considered to be the first theorem ever proved in graph theory.



Leonard Euler
(1707-1783)

- 1736, Euler solved the Königsberg Bridge Problem (七橋問題)

Königsberg Bridge Problem

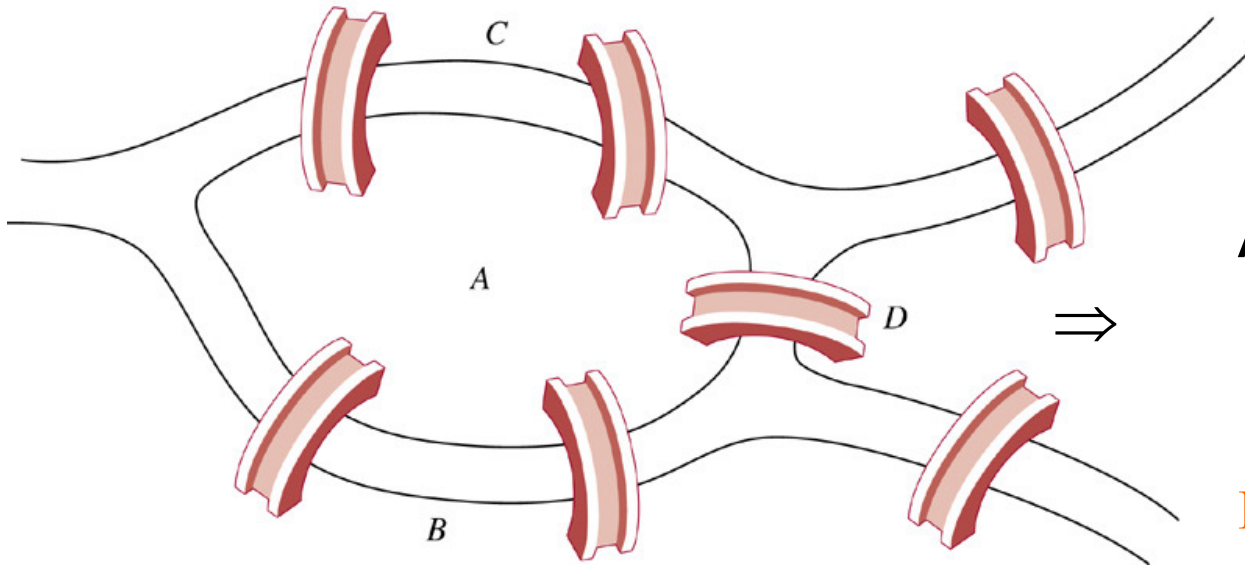


FIGURE 1 The Seven Bridges of Königsberg.

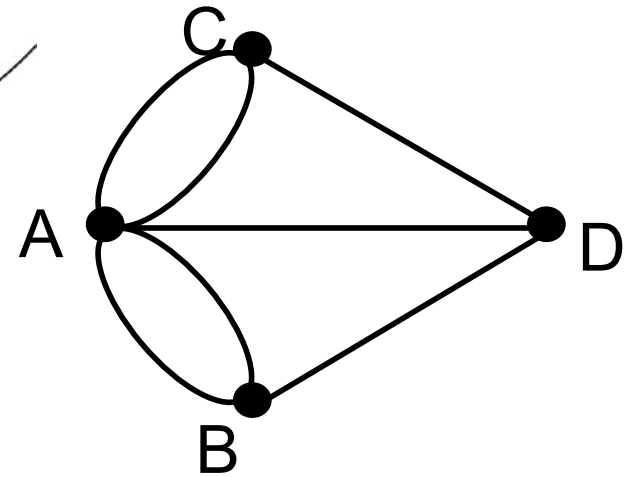


FIGURE 2 Multigraph Model of the Town of Königsberg.

Q: 是否存在一種走法，可以走過每條邊一次，並回到起點？

Ans: 因為每次經過一個點，都需要從一條邊進入該點，再用另一條邊離開，所以經過每個點一次要使用掉一對邊。

⇒ 每個點上連接的邊數必須是偶數才行

⇒ 此種走法不存在

Def 1:

An ***Euler circuit*** in a graph G is a simple circuit containing every edge of G .

An ***Euler path*** in G is a simple path containing every edge of G .

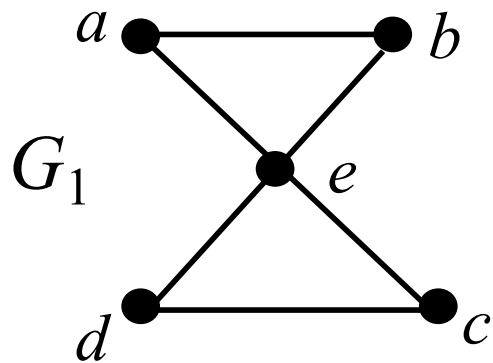
Thm. 1:

A connected multigraph with at least two vertices has an Euler circuit if and only if each of its vertices has even degree.

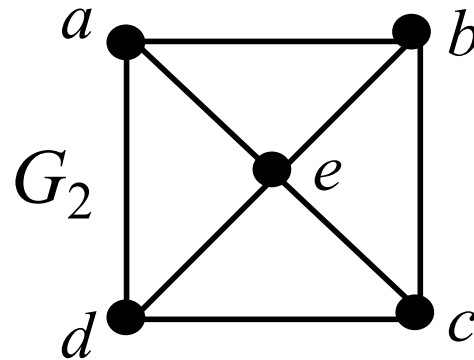
Thm. 2:

A connected multigraph has an Euler path (**but not an Euler circuit**) if and only if it has exactly 2 vertices of odd degree.

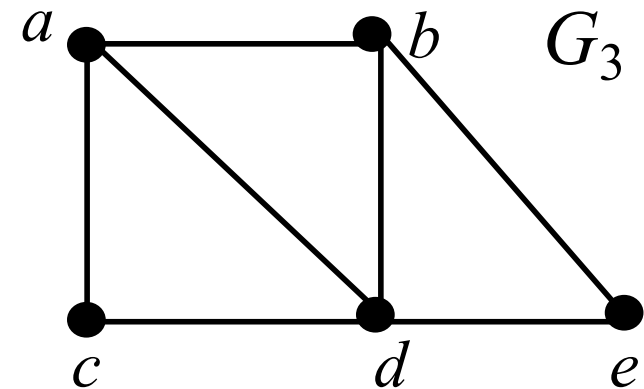
Example 1. Which of the following graphs have an Euler circuit or an Euler path?



Euler circuit



none



Euler path

- **Solution:** The graph G_1 has an Euler circuit (e.g., a, e, c, d, e, b, a). But, as can easily be verified by inspection, neither G_2 nor G_3 has an Euler circuit. Note that G_3 has an Euler path (e.g., a, c, d, e, b, d, a, b), but there is no Euler path in G_2 , which can be verified by inspection.

Example 2.

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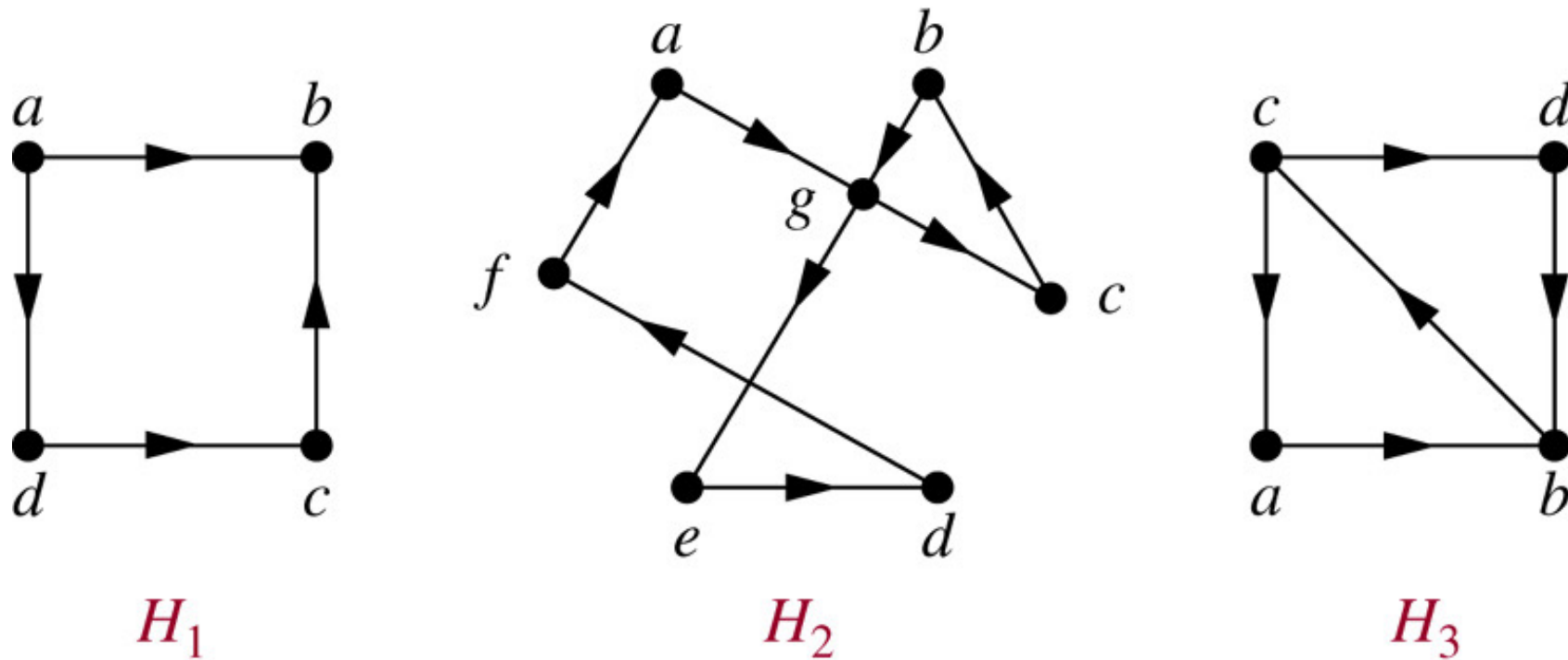
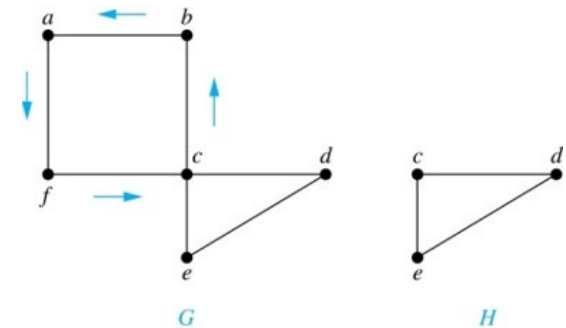


FIGURE 4 The Directed Graphs H_1 , H_2 , and H_3 .

Sufficient Conditions for Euler Circuits and Paths₁

Suppose that G is a connected multigraph with ≥ 2 vertices, all of even degree. Let $x_0 = a$ be a vertex of even degree. Choose an edge $\{x_0, x_1\}$ incident with a and proceed to build a simple path $\{x_0, x_1\}, \{x_1, x_2\}, \dots, \{x_{n-1}, x_n\}$ by adding edges one by one until another edge can not be added.

We illustrate this idea in the graph G here. We begin at a and choose the edges $\{a, f\}$, $\{f, c\}$, $\{c, b\}$, and $\{b, a\}$ in succession.



The path begins at a with an edge of the form $\{a, x\}$; we show that it must terminate at a with an edge of the form $\{y, a\}$. Since each vertex has an even degree, there must be an even number of edges incident with this vertex. Hence, every time we enter a vertex other than a , we can leave it. Therefore, the path can only end at a .

If all of the edges have been used, an Euler circuit has been constructed.

Otherwise, consider the subgraph H obtained from G by deleting the edges already used.

In the example H consists of the vertices c, d, e .

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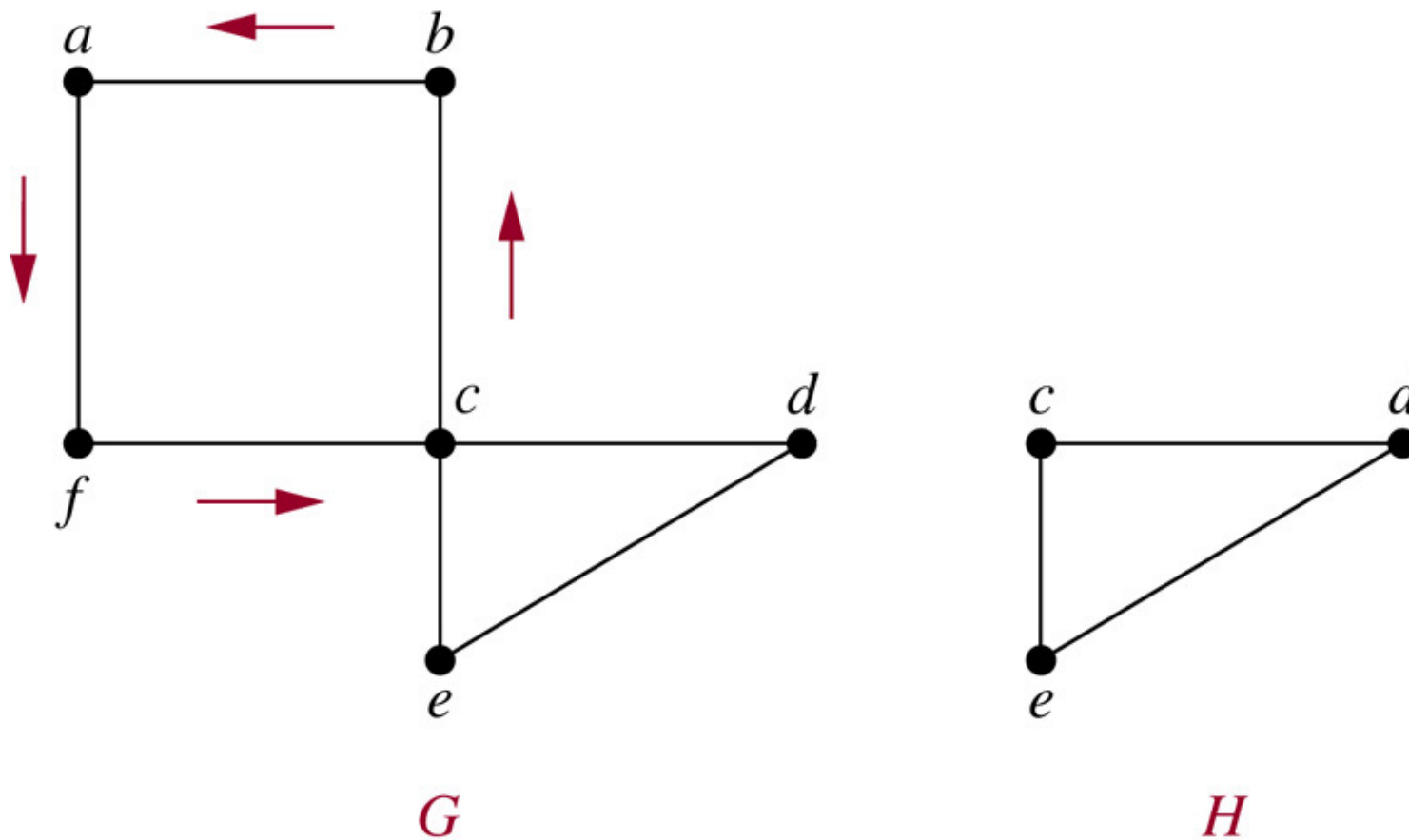
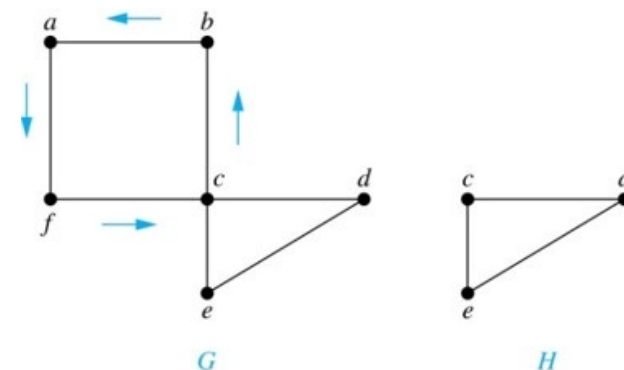


FIGURE 5 Constructing and Euler Circuit in G .

Sufficient Conditions for Euler Circuits and Paths₂

Because G is connected, H must have at least one vertex in common with the circuit that has been deleted.

In the example, the vertex is c .



Every vertex in H must have even degree because all the vertices in G have even degree and for each vertex, pairs of edges incident with this vertex have been deleted. Beginning with the shared vertex construct a path ending in the same vertex (as was done before). Then splice this new circuit into the original circuit.

In the example, we end up with the circuit a, f, c, d, e, c, b, a .

Continue this process until all edges have been used. This produces an Euler circuit. Since every edge is included and no edge is included more than once.

Similar reasoning can be used to show that a graph with exactly two vertices of odd degree must have an Euler path connecting these two vertices of odd degree

Def 1:

An *Euler circuit* in a graph G is a simple circuit containing every edge of G .

An *Euler path* in G is a simple path containing every edge of G .

Thm. 1:

A **connected multigraph** with at least two vertices **has an Euler circuit** if and only if **each** of its **vertices has even degree**.

Thm. 2:

A connected multigraph has an Euler path (but not an Euler circuit) if and only if it has exactly 2 vertices of odd degree.

Algorithm 1 (Constructing Euler Circuits.)

Procedure *Euler*(G : connected multigraph with all vertices of even degree)

circuit := a circuit in G beginning at an arbitrary chosen vertex with edges successively added to form a path that returns to this vertex

H := G with the edges of this circuit removed

while H has edges

begin

subcircuit := a circuit in H beginning at a vertex in H that also is an endpoint of an edge of *circuit*

H := H with edges of *subcircuit* and all isolated vertices removed

circuit := *circuit* with *subcircuit* inserted at appropriate vertex

end {*circuit* is an Euler circuit}

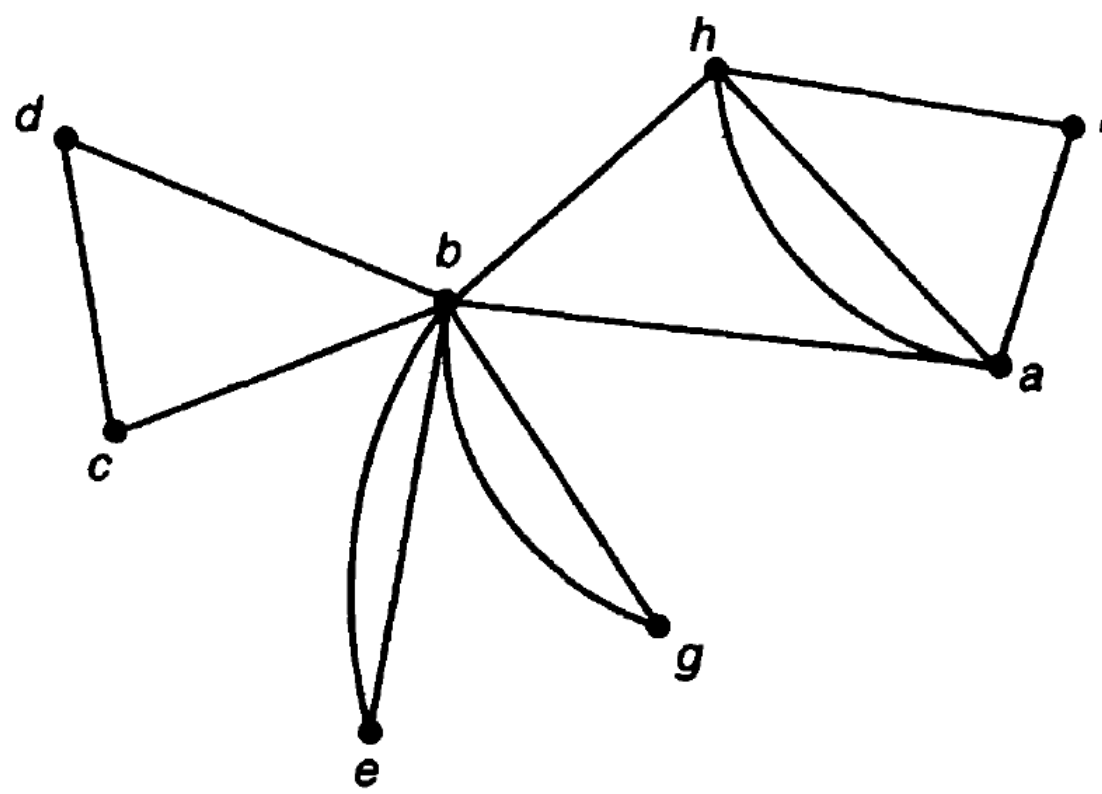


Figure 11.14

Example 3.

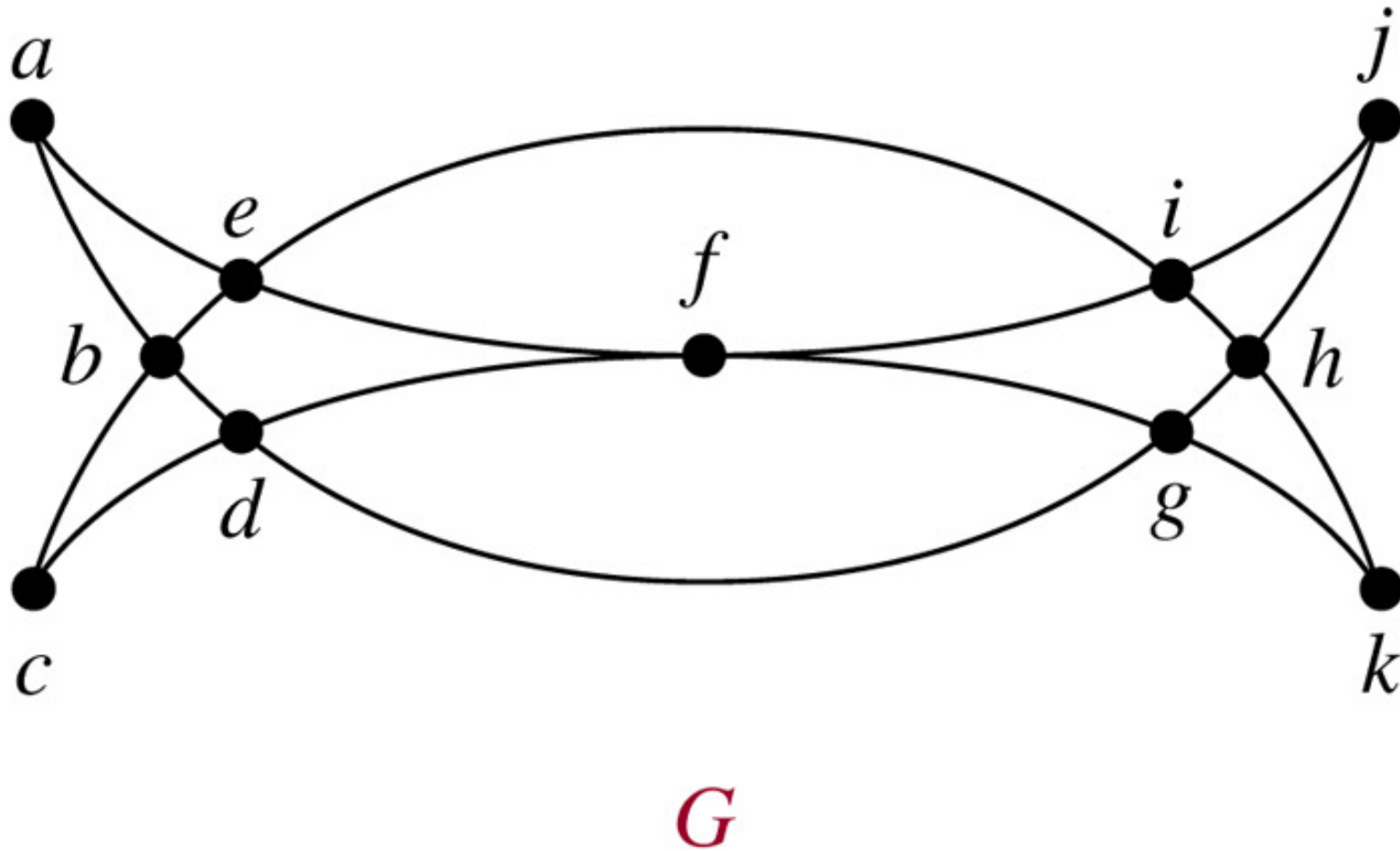


FIGURE 6 Mohammed's Scimitars.

Def 1:

An *Euler circuit* in a graph G is a simple circuit containing every edge of G .

An *Euler path* in G is a simple path containing every edge of G .

Thm. 1:

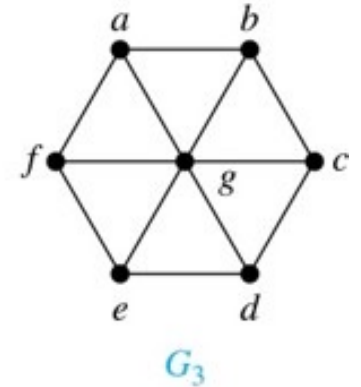
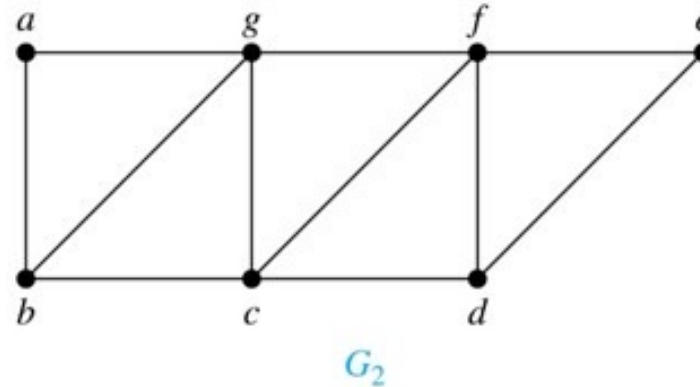
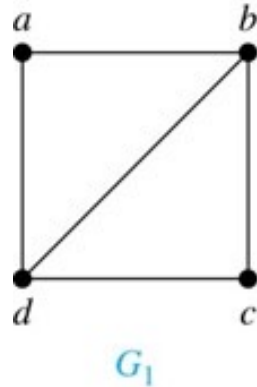
A connected multigraph with at least two vertices has an Euler circuit if and only if each of its vertices has even degree.

Thm. 2:

A **connected multigraph** has an **Euler path** (**but not an Euler circuit**) if and only if it has **exactly 2 vertices of odd degree**.

Euler Circuits and Paths

Example:



G_1 contains exactly two vertices of odd degree (b and d). Hence it has an Euler path, e.g., d, a, b, c, d, b .

G_2 has exactly two vertices of odd degree (b and d). Hence it has an Euler path, e.g., $b, a, g, f, e, d, c, g, b, c, f, d$.

G_3 has six vertices of odd degree. Hence, it does not have an Euler path.

Applications of Euler Paths and Circuits

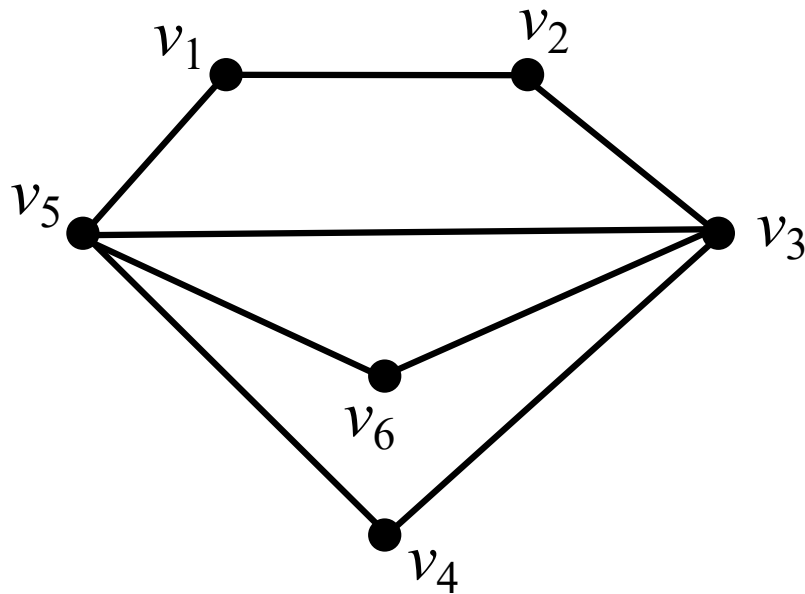
Euler paths and circuits can be used to solve many practical problems such as finding a path or circuit that traverses each

- street in a neighborhood,
- road in a transportation network,
- connection in a utility grid,
- link in a communications network.

Other applications are found in the

- layout of circuits,
- network multicasting,
- molecular biology, where Euler paths are used in the sequencing of DNA.

Example



Step 1: find the 1st circuit

$C: v_1, v_2, v_3, v_4, v_5, v_1$

Step 2: $H = G - C \neq \emptyset$,
find subcircuit

$SC: v_3, v_5, v_6, v_3$

Step 3:

$C = C \cup SC,$

$H = G - C = \emptyset$, stop

$C: v_1, v_2, v_3, v_5, v_6, v_3, v_4, v_5, v_1$

$\underbrace{\hspace{10em}}_{SC \text{ embedded}}$

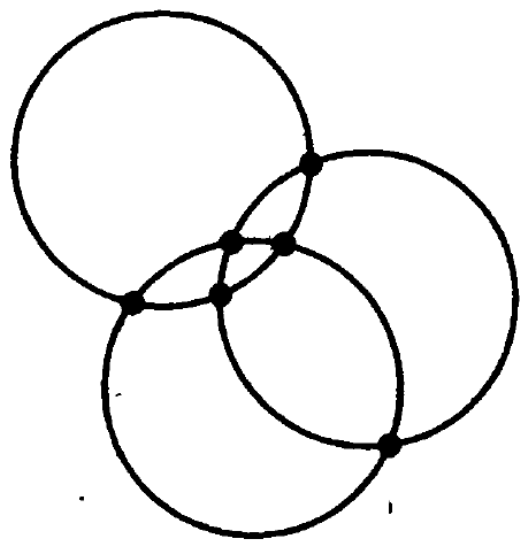


Figure 11.15

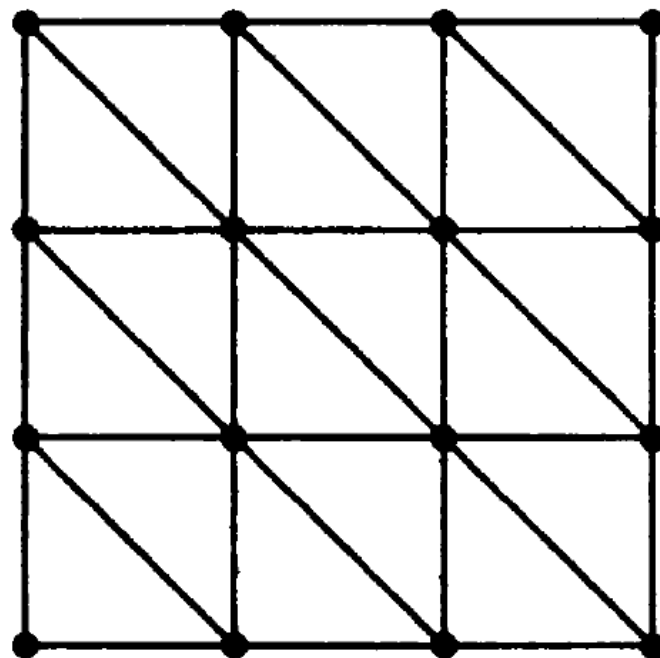


Figure 11.16

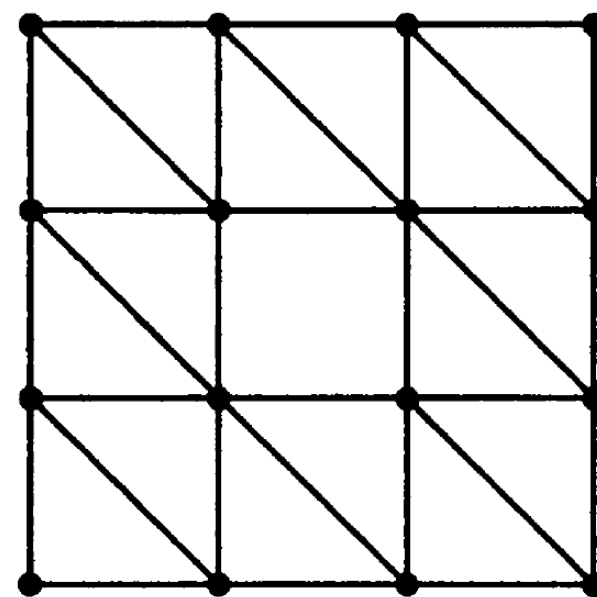
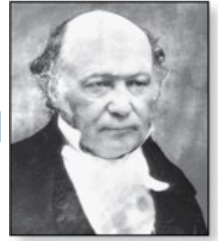


Figure 11.17

Hamilton Paths and Circuits



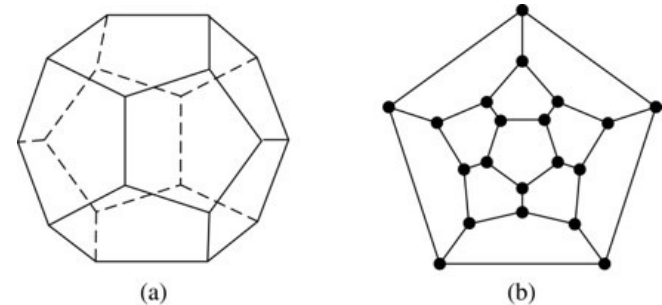
William Rowan
Hamilton
(1805- 1865)

Euler paths and circuits contained every edge only once. Now we look at paths and circuits that contain every vertex exactly once.

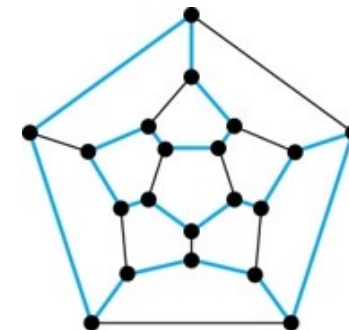
William Hamilton invented the *Icosian puzzle* in 1857.

It consisted of a wooden dodecahedron (with 12 regular pentagons as faces), illustrated in (a), with a peg at each vertex, labeled with the names of different cities. String was used to plot a circuit visiting 20 cities exactly once

The graph form of the puzzle is given in (b).



The solution (a Hamilton circuit) is given here.

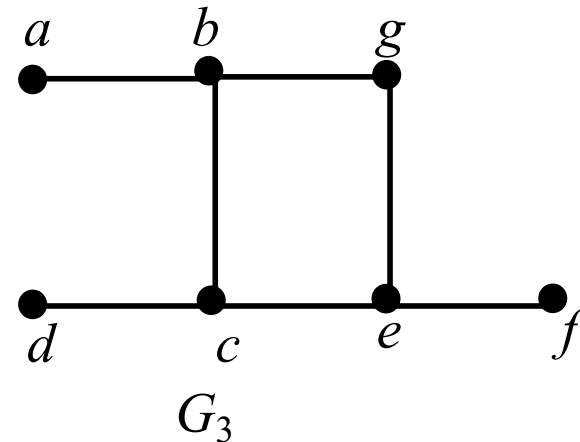
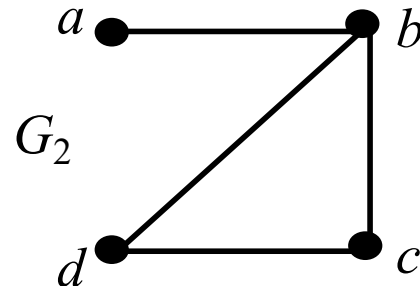
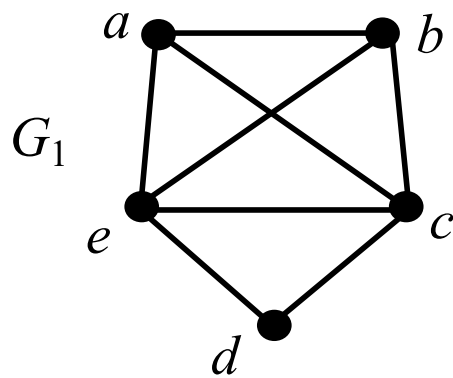


[Jump to long description](#)

Hamilton Paths and Circuits

Def. 2: A *Hamilton path* is a path that traverses each vertex in a graph G exactly once.
A *Hamilton circuit* is a circuit that traverses each vertex in G exactly once.

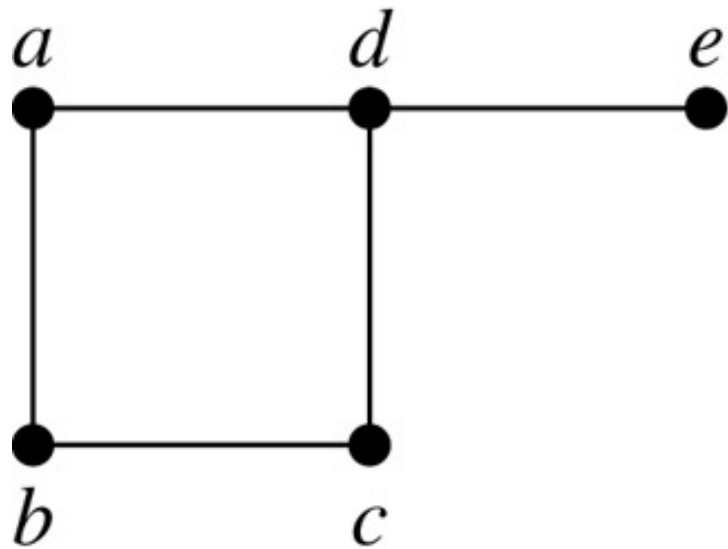
Example 5. Which of the following graphs have a Hamilton circuit or a Hamilton path?



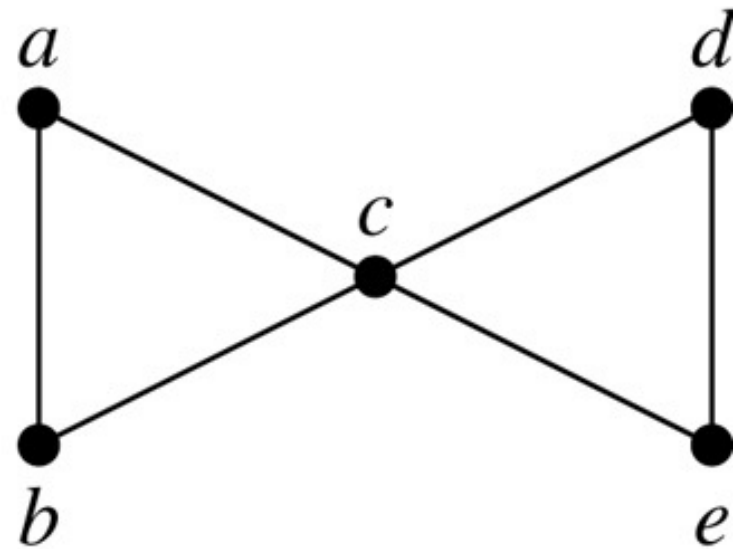
Hamilton circuit: G_1

Hamilton path: G_1, G_2

Example 6.



G



H

FIGURE 11 Two Graphs That Do Not Have a Hamilton Circuit.

Necessary Conditions for Hamilton Circuits



Gabriel Andrew
Dirac
(1925-1984)

Unlike for an Euler circuit, no simple necessary and sufficient conditions are known for the existence of a Hamilton circuit.

However, there are some useful necessary conditions. We describe two of these now.

Dirac's Theorem: If G is a simple graph with $n \geq 3$ vertices such that the degree of every vertex in G is $\geq n/2$, then G has a Hamilton circuit.

Ore's Theorem: If G is a simple graph with $n \geq 3$ vertices such that $\deg(u) + \deg(v) \geq n$ for every pair of nonadjacent vertices, then G has a Hamilton circuit.

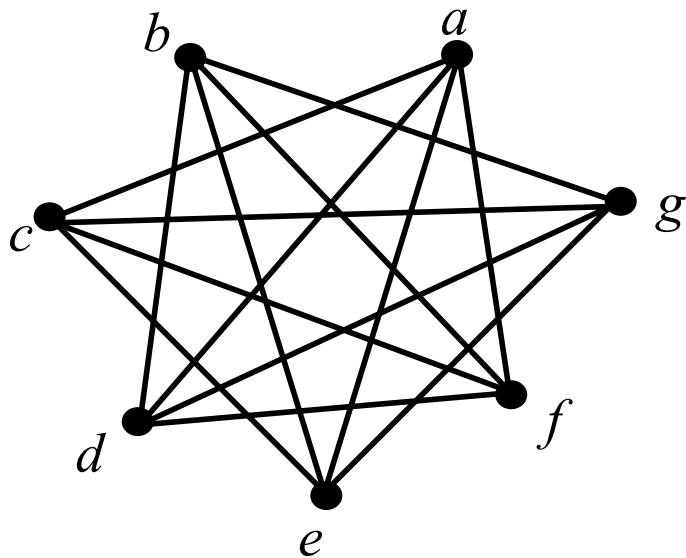
Oysten Ore
(1899-1968)



Thm. 3 (Dirac's Thm.):

If (but not only if) G is a simple graph with $n \geq 3$ vertices such that the degree of every vertex in G is at least $n/2$, then G has a Hamilton circuit.

Example



each vertex has $\deg \geq n/2 = 3.5$

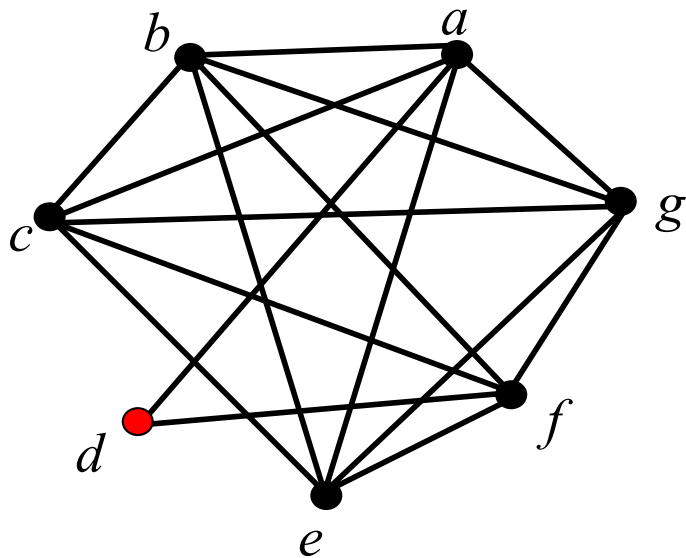
$\Rightarrow$ Hamilton circuit exists

如 : a, c, e, g, b, d, f, a

Thm. 4 (Ore's Thm.):

If G is a simple graph with $n \geq 3$ vertices such that $\deg(u) + \deg(v) \geq n$ for every pair of nonadjacent vertices u and v , then G has a Hamilton circuit.

Example



each nonadjacent vertex pair
has $\deg \text{ sum} \geq n = 7$

$\Rightarrow$ Hamilton circuit exists

如： a, d, f, e, c, b, g, a

Applications of Hamilton Paths and Circuits

Applications that ask for a path or a circuit that visits each intersection of a city, each place pipelines intersect in a utility grid, or each node in a communications network exactly once, can be solved by finding a Hamilton path in the appropriate graph.

The famous ***traveling salesperson problem (TSP)*** asks for the shortest route a traveling salesperson should take to visit a set of cities. This problem reduces to finding a Hamilton circuit such that the total sum of the weights of its edges is as small as possible.

A family of binary codes, known as *Gray codes*, which minimize the effect of transmission errors, correspond to Hamilton circuits in the n -cube Q_n . (See *the text for details*.)

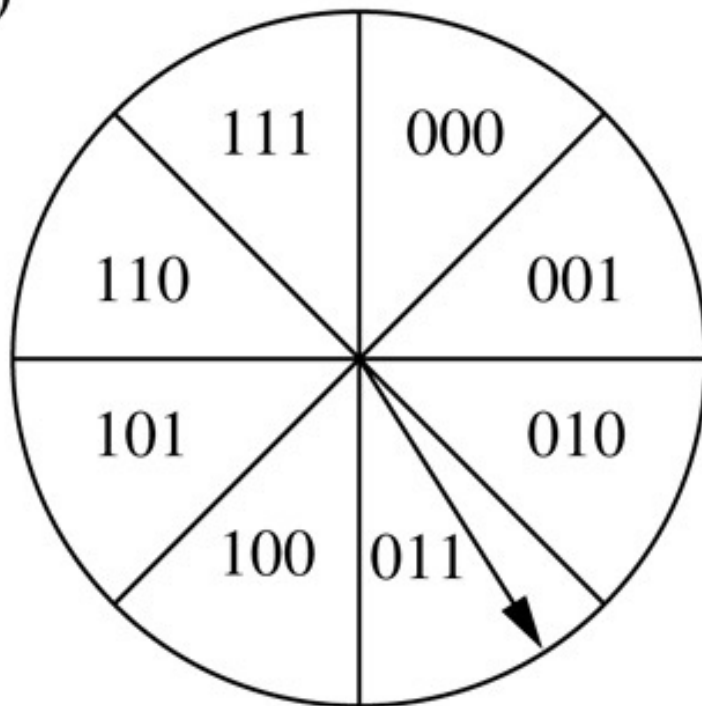
Example. *The knight's tour problem.* Consider an 8-by-8 chessboard and the chess piece known as a *knight*. A knight moves from its current location by moving two squares vertically and one square horizontally or one square vertically and two squares horizontally. Is it possible to place the knight on the board so that, with legal moves, the knight lands in each square exactly once? Such a tour is called a *knight's tour*, and we can ask for a knight's tour which has the property that the move from the last square to the first square is also a legal knight's move. A knight's tour with this property is called *reentrant*.

58	43	60	37	52	41	62	35
49	46	57	42	61	36	53	40
44	59	48	51	38	55	34	63
47	50	45	56	33	64	39	54
22	7	32	1	24	13	18	15
31	2	23	6	19	16	27	12
8	21	4	29	10	25	14	17
3	30	9	20	5	28	11	26

Example 8.

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(a)



(b)

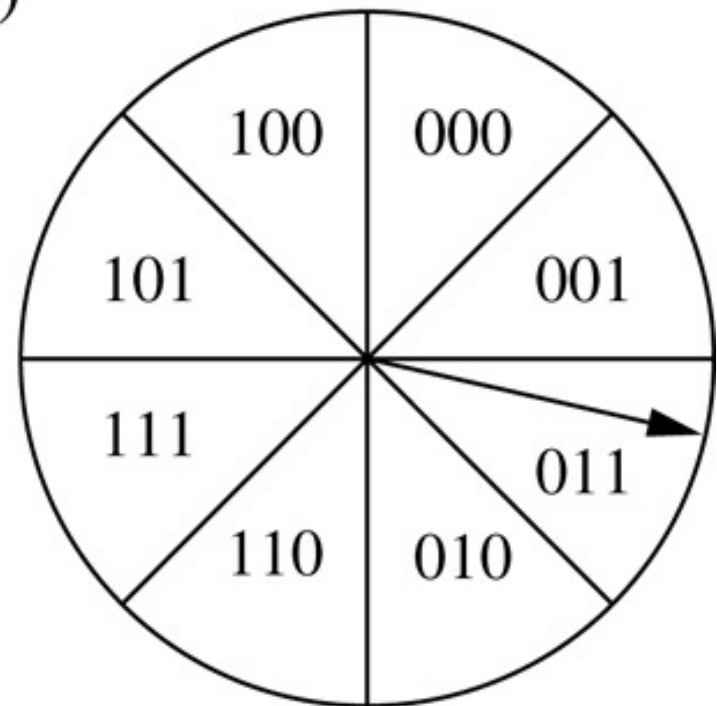


FIGURE 12 Converting the Position of a Pointer into Digital Form.

|0.5)

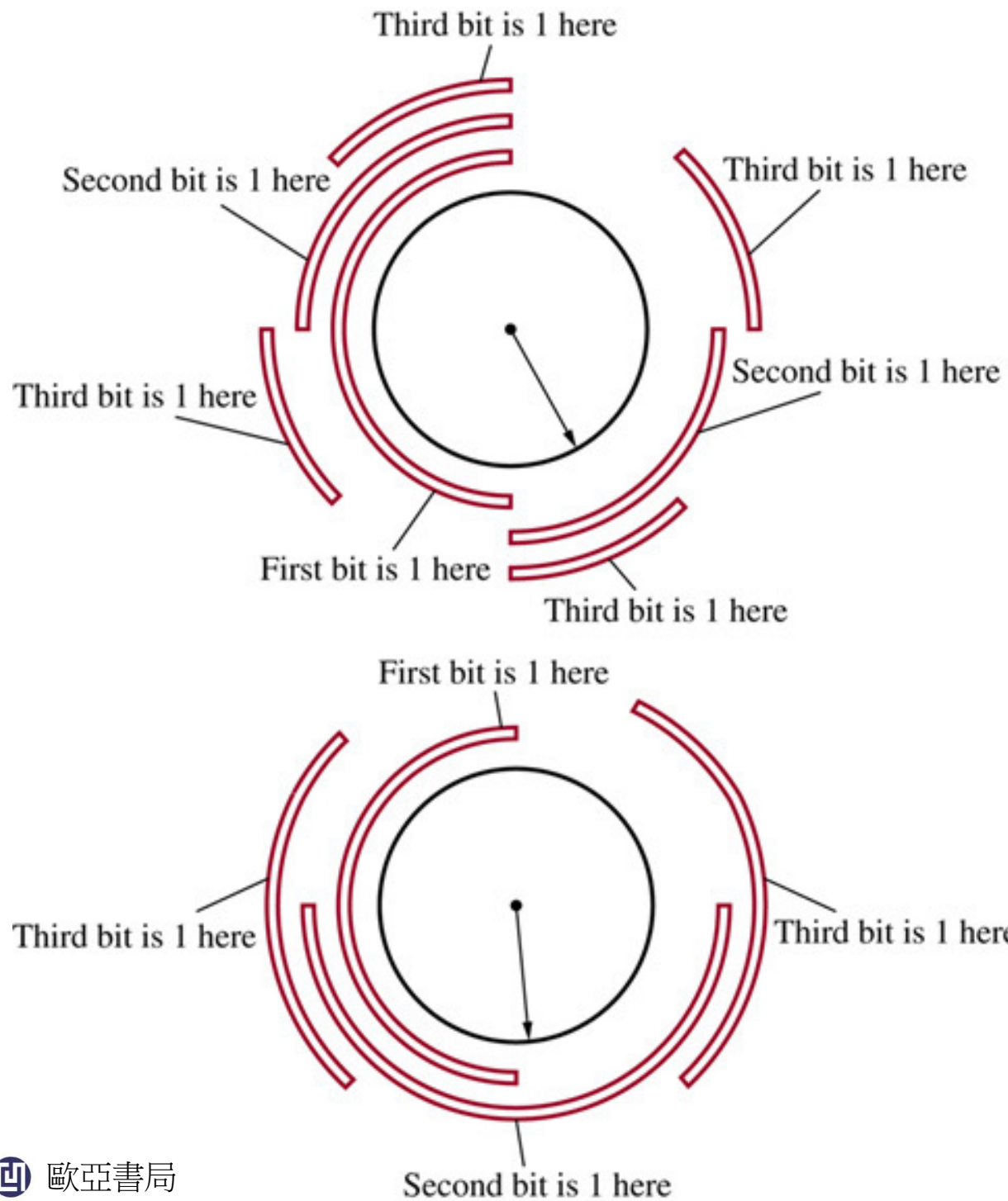


FIGURE 13 The Digital Representation of the Position of the Pointer.

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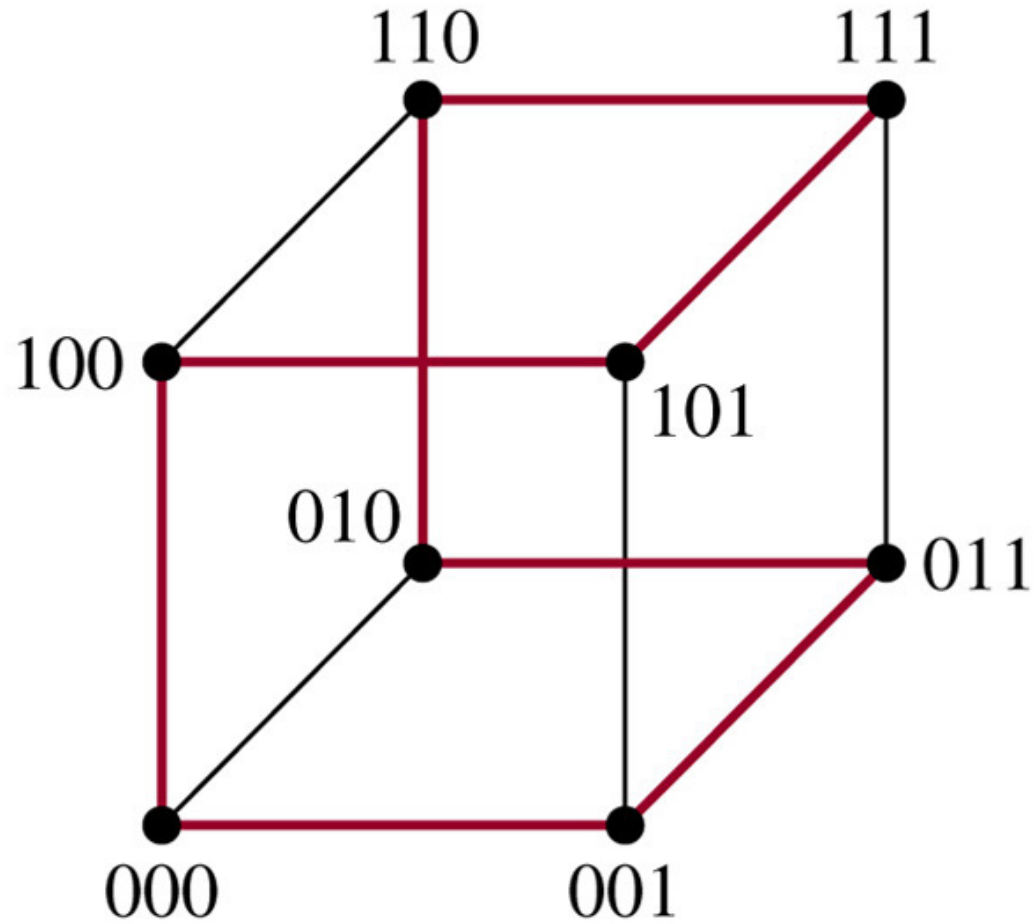
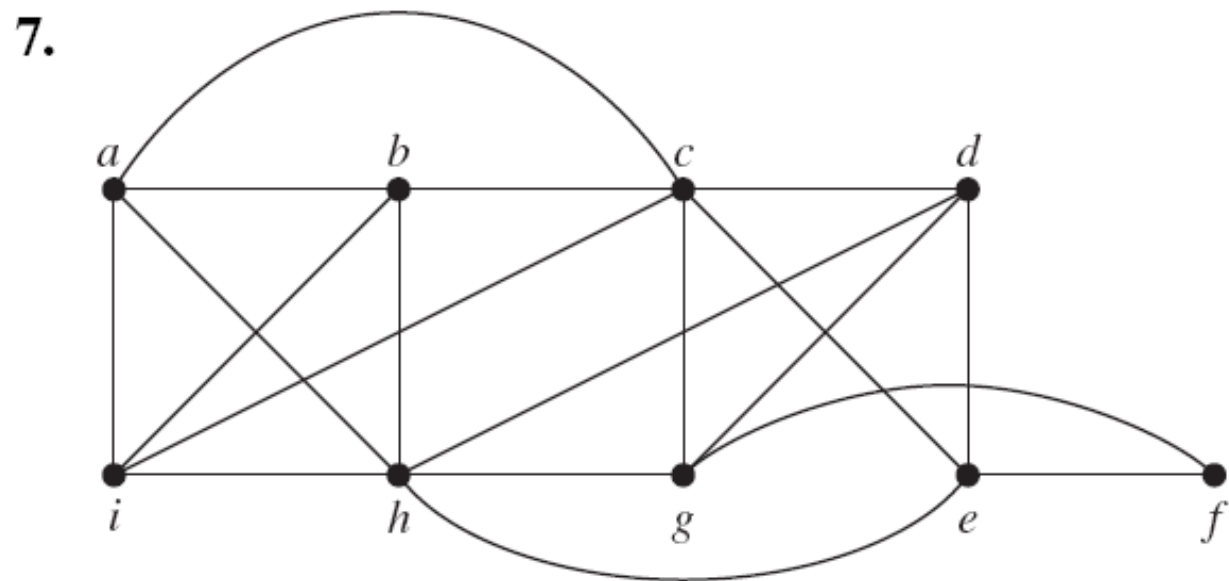
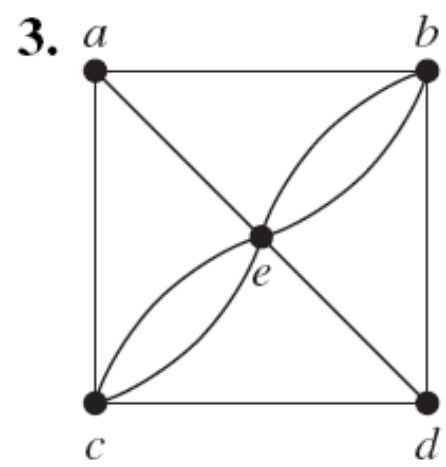
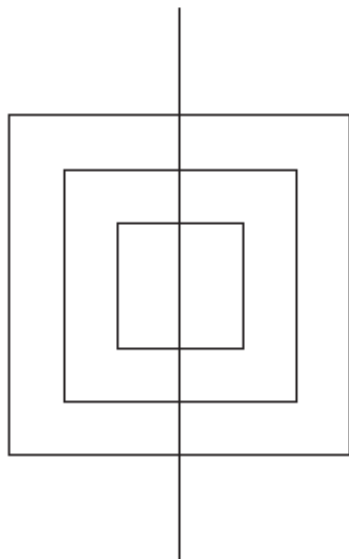


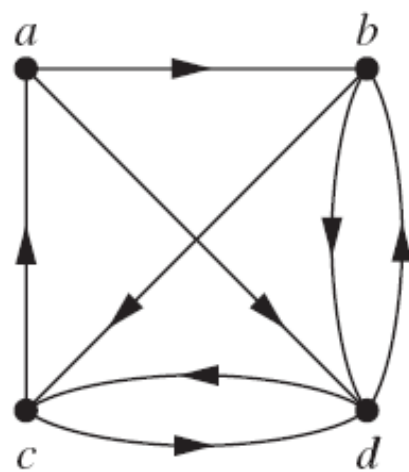
FIGURE 14 A Hamilton Circuit for Q_3 .



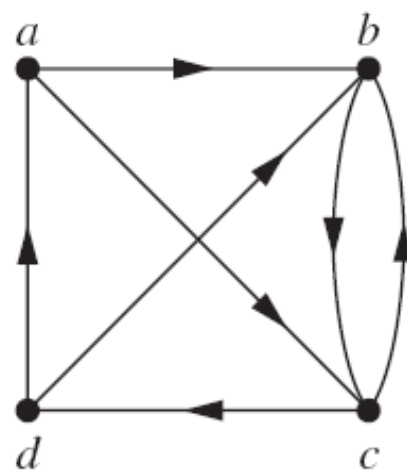
14.



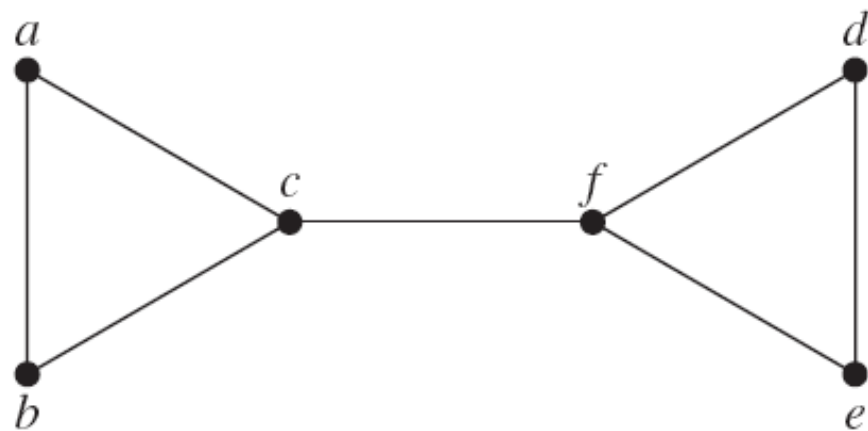
18.



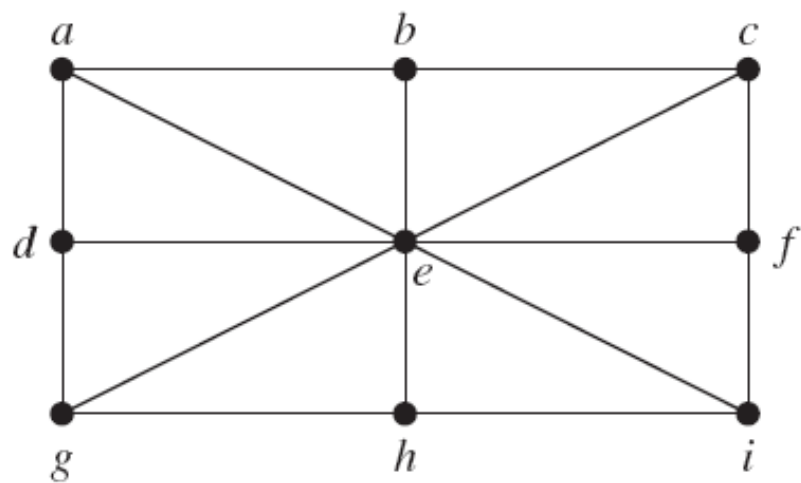
19.



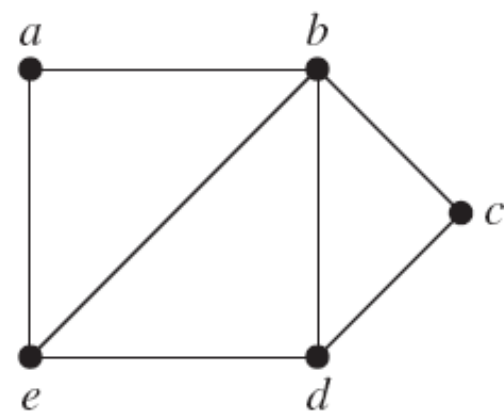
30.



36.



31.



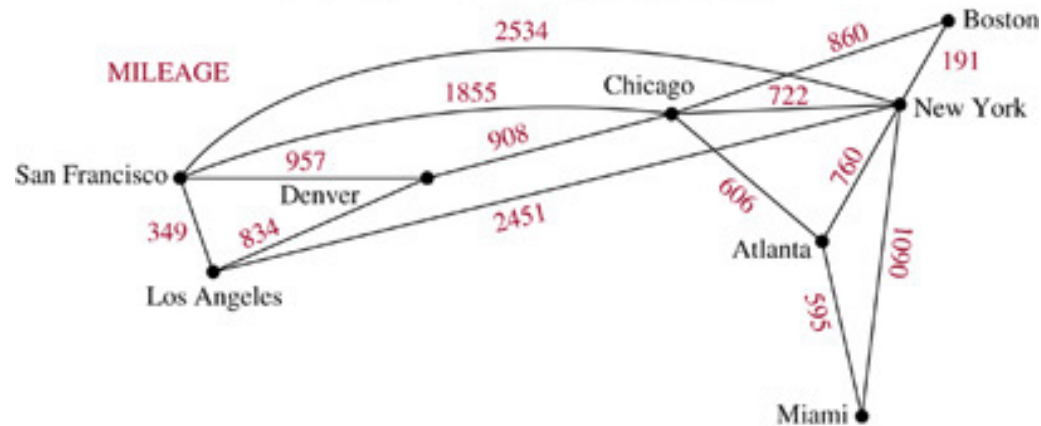
§10.6: Shortest-Path Problems

Def:

1. Graphs that have a number assigned to each edge are called *weighted graphs*.
2. The length of a path in a weighted graph is the sum of the weights of the edges of this path.

Shortest path Problem:

Determining the path of least sum of the weights between two vertices in a weighted graph.



(10.6)

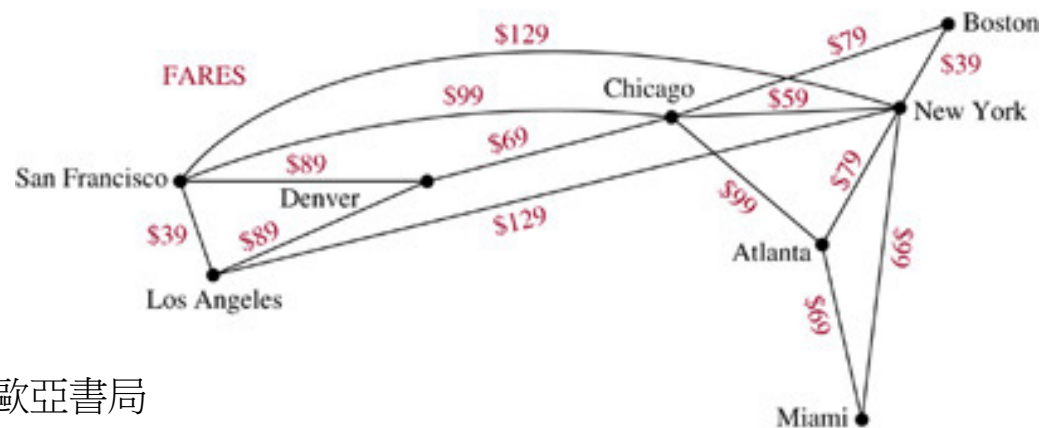
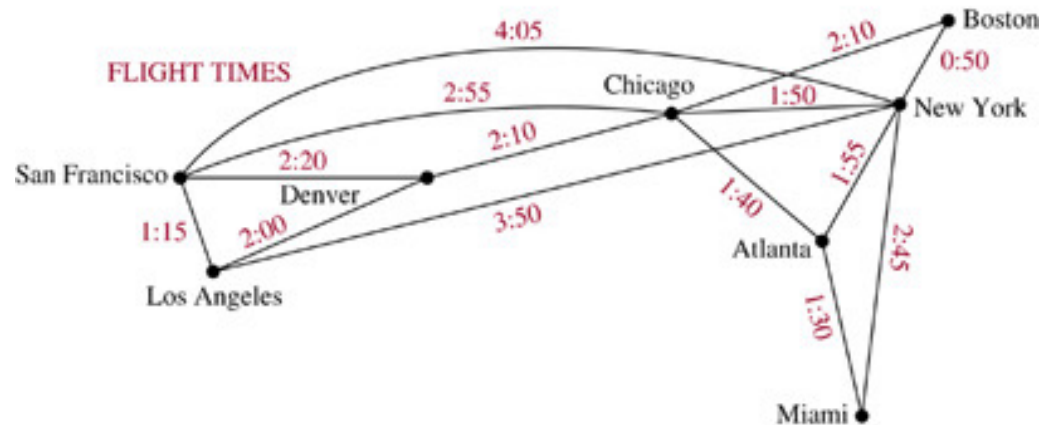
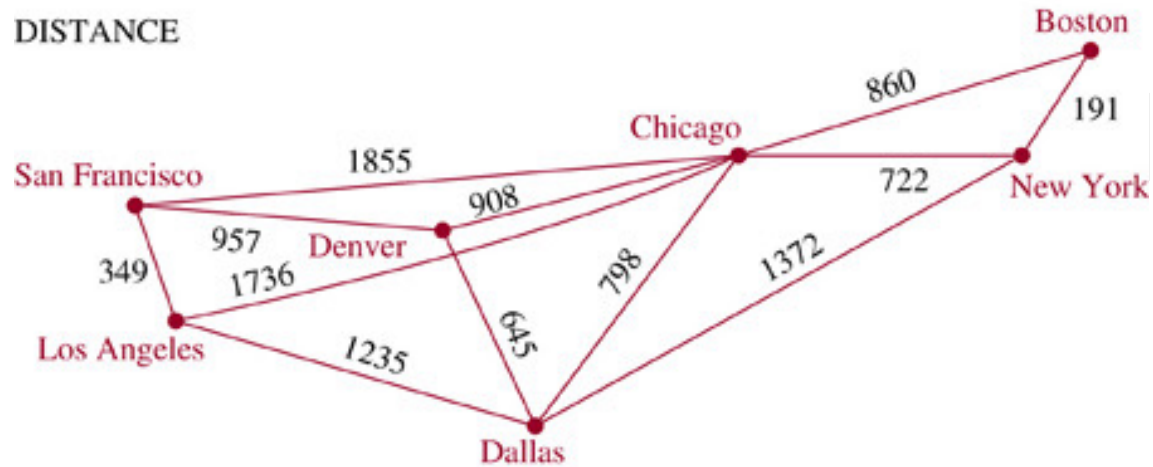


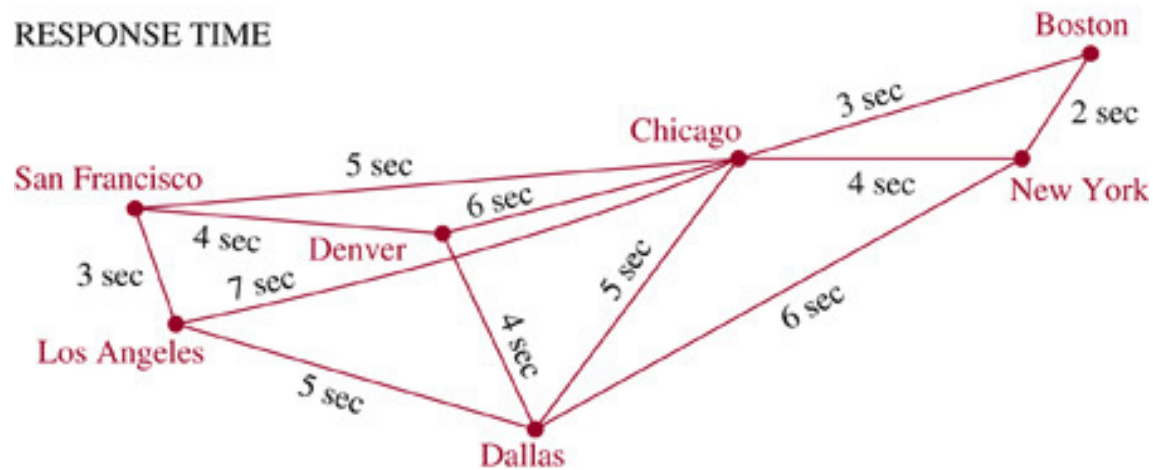
FIGURE 1 Weighted Graphs Modeling and Airline System.

DISTANCE



0.6)

RESPONSE TIME



LEASE RATES (PER MONTH)

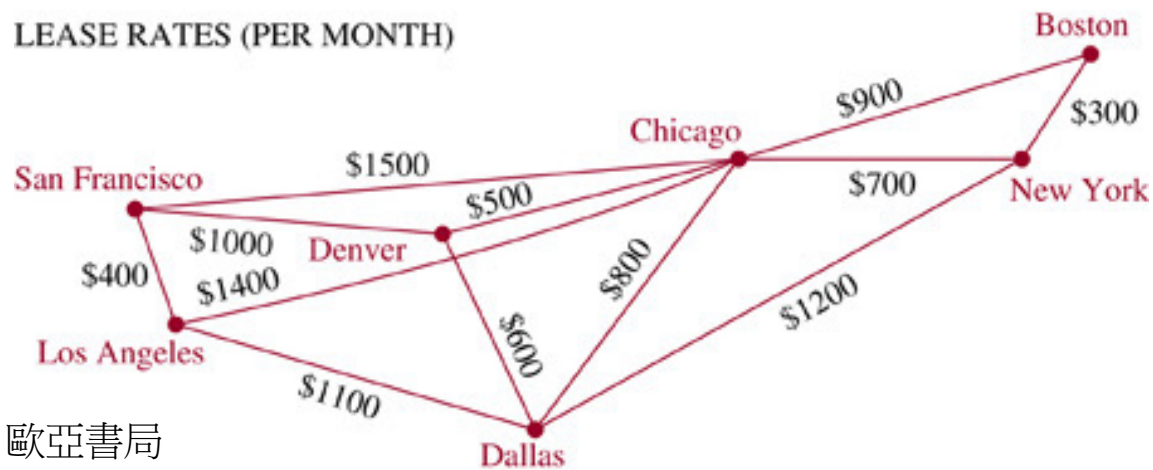


FIGURE 2 Weighted Graphs Modeling a Computer Network.

§10.6: Shortest-Path Problems

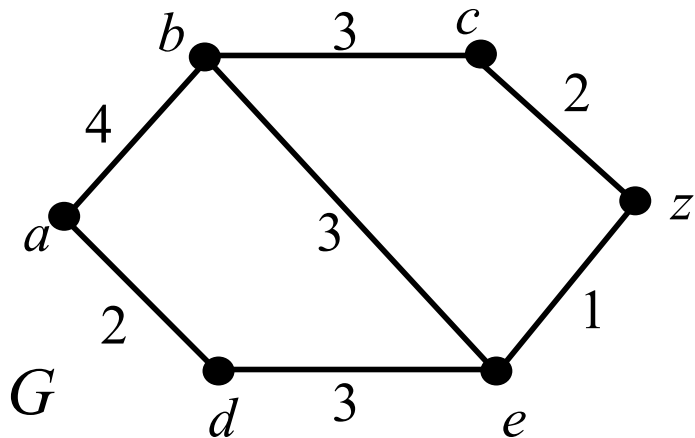
Def:

1. Graphs that have a number assigned to each edge are called *weighted graphs*.
2. The *length* of a path in a weighted graph is the sum of the weights of the edges of this path.

Shortest path Problem:

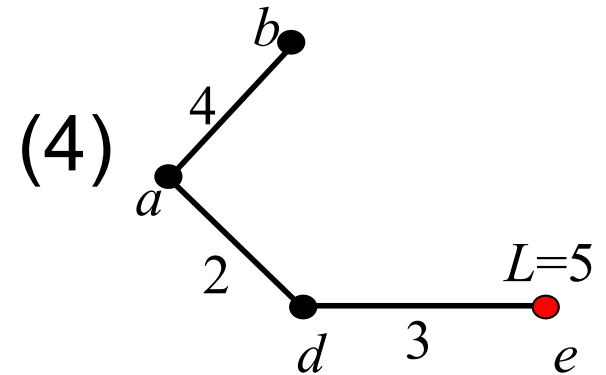
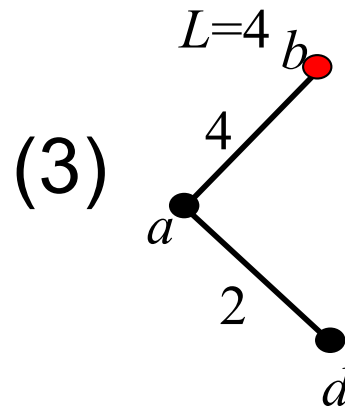
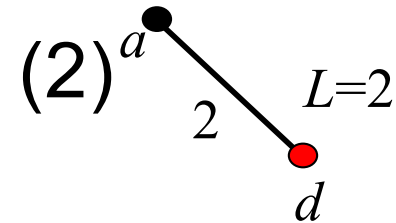
Determining the path of least sum of the weights between two vertices in a weighted graph.

Example 1. What is the length of a shortest path between a and z in the weighted graph G ?

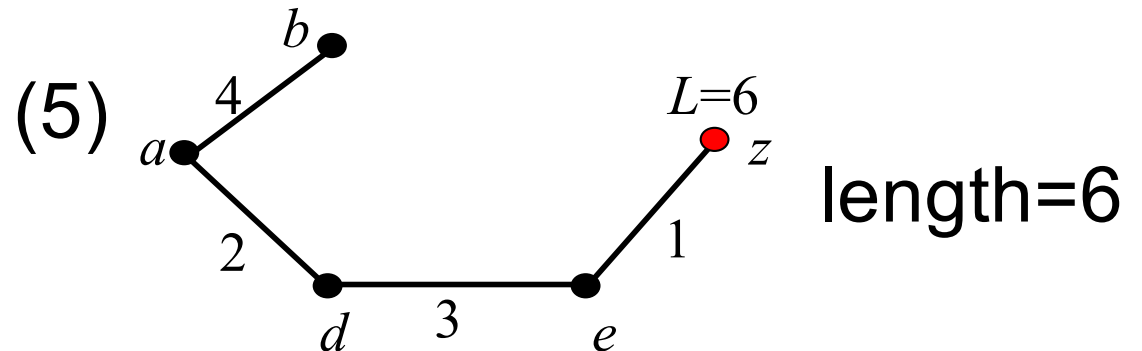


Sol. (1) $L=0$

a



每個step都從還未處理的點
找離 a 最近的，此時 a 至該點的
shortest path就找到了。



$$L_k(a, v) = \min \{L_{k-1}(a, v), L_{k-a}(a, u) + w(u, v)\}$$

Dijkstra's Algorithm (find the length of a shortest path from a to z)

Procedure *Dijkstra*(G : weighted connected simple graph,
with all weights positive)

{ G has vertices $a = v_0, v_1, \dots, v_n = z$ and weights $w(v_i, v_j)$
where $w(v_i, v_j) = \infty$ if $\{v_i, v_j\}$ is not an edge in G }

for $i := 1$ **to** n

$L(v_i) := \infty$

$L(a) := 0$

$S := \emptyset$

while $z \notin S$

begin

$u :=$ a vertex not in S with $L(u)$ minimal

$S := S \cup \{u\}$

for all vertices v not in S

if $L(u) + w(u, v) < L(v)$ **then** $L(v) := L(u) + w(u, v)$

end { $L(z)$ = length of a shortest path from a to z }

This algorithm can be extended
to construct a shortest path.

此處加一個變數
紀錄 v 的前一點是 u
 $previous(v) := u$
演算法最後再從 z 往前trace

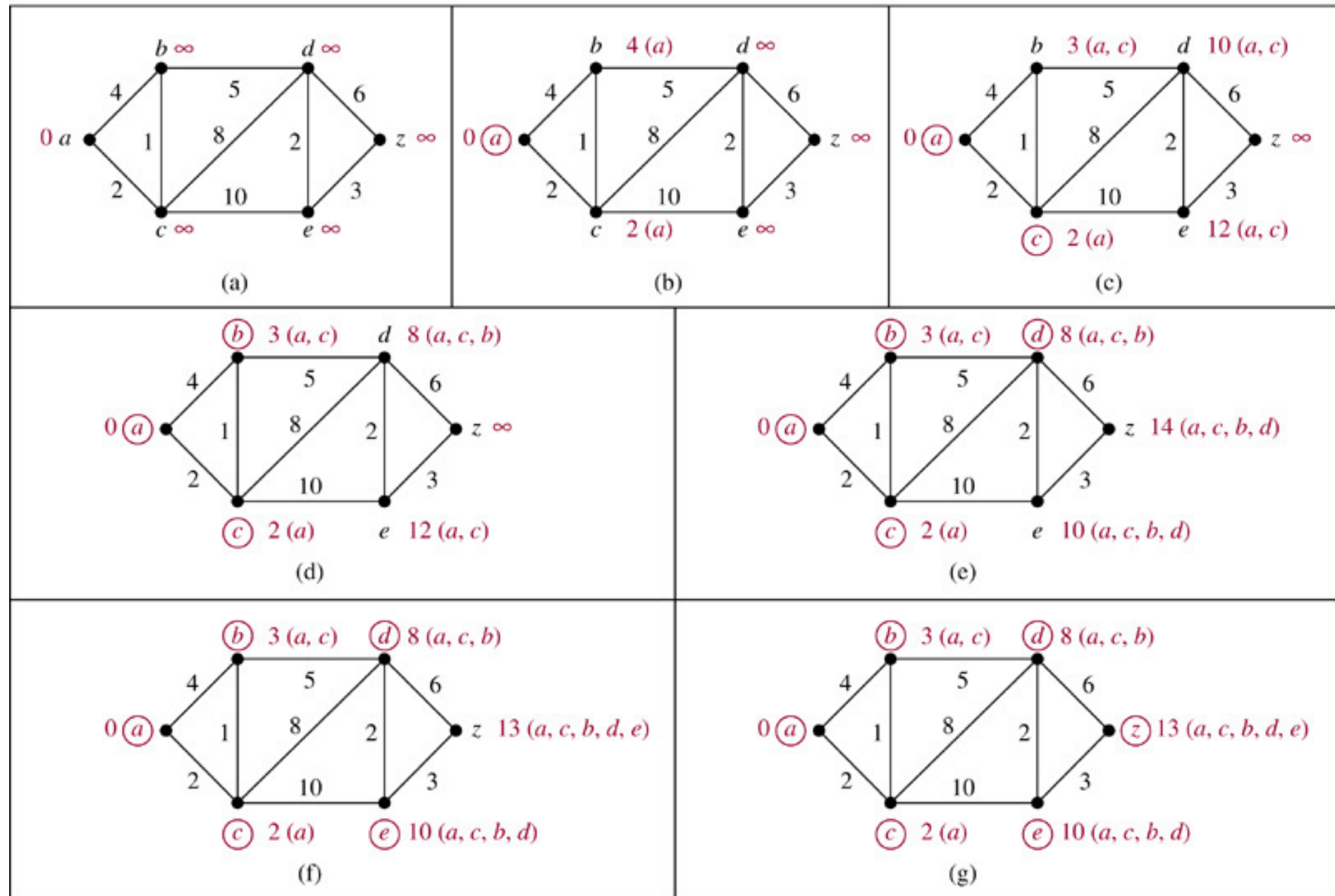
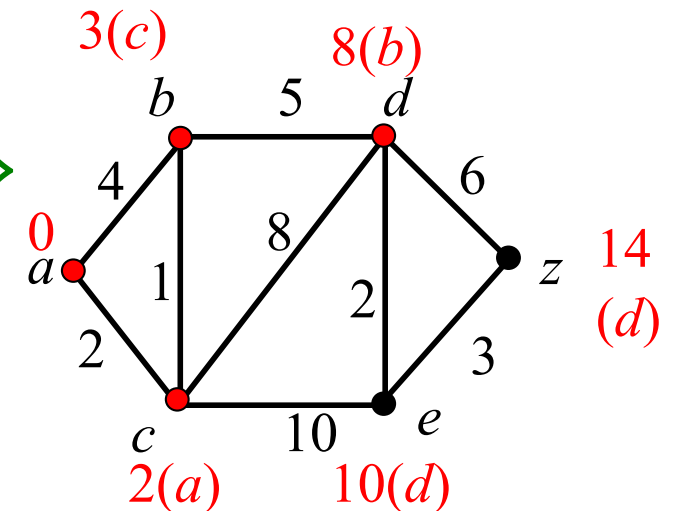
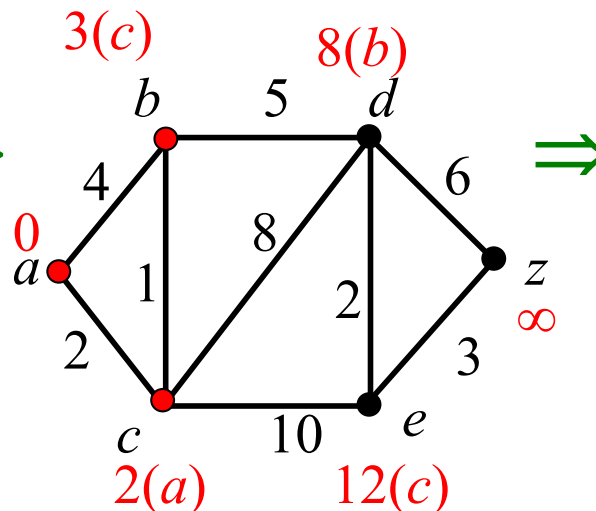
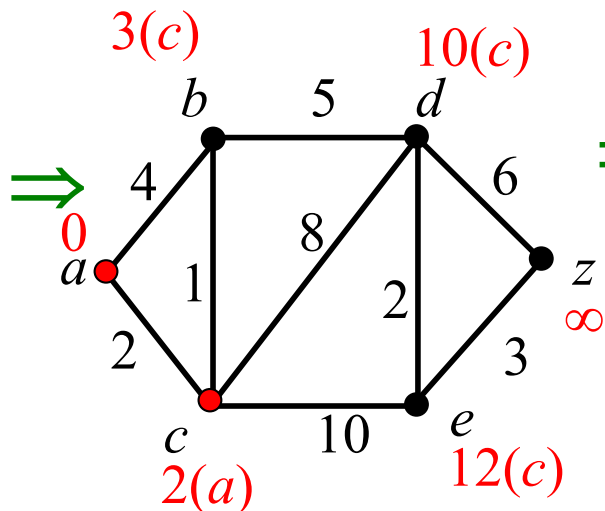
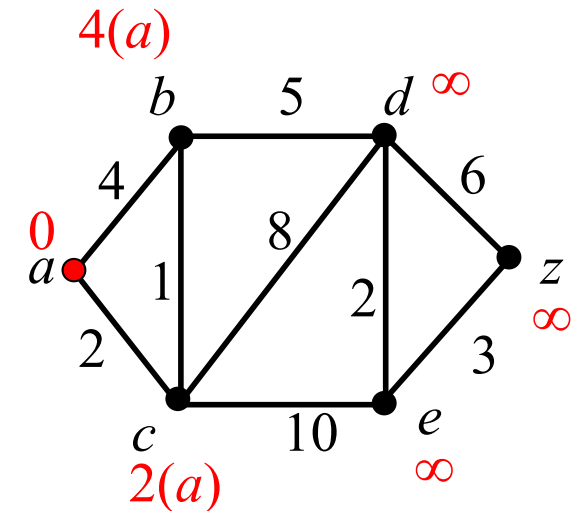
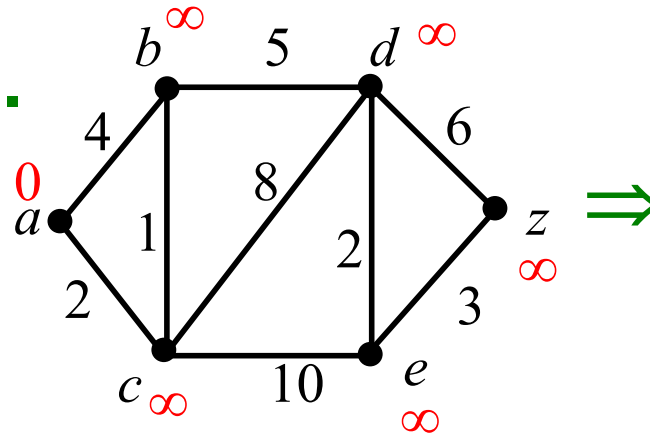
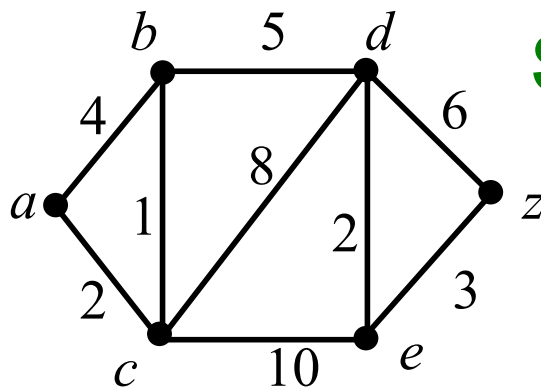


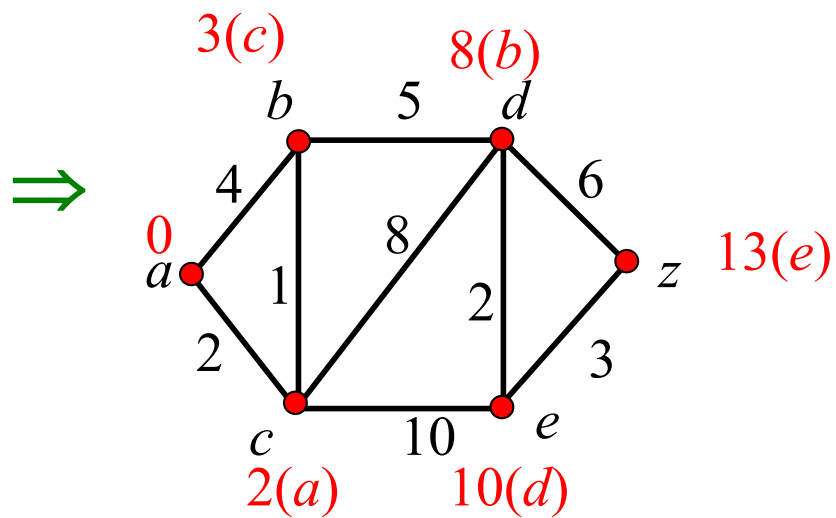
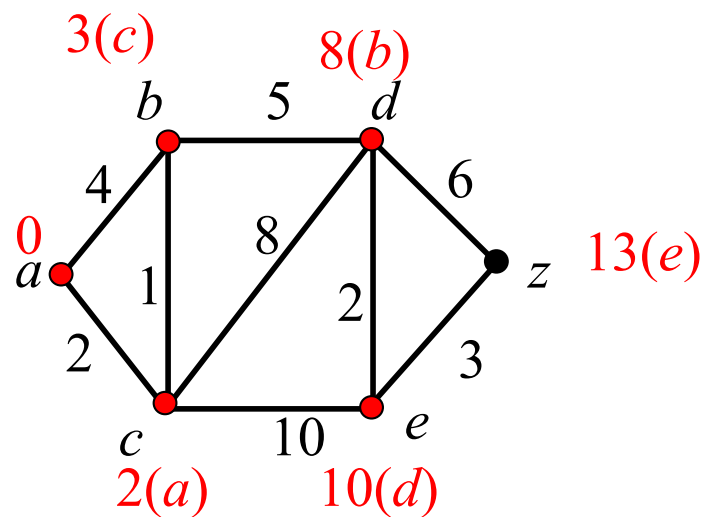
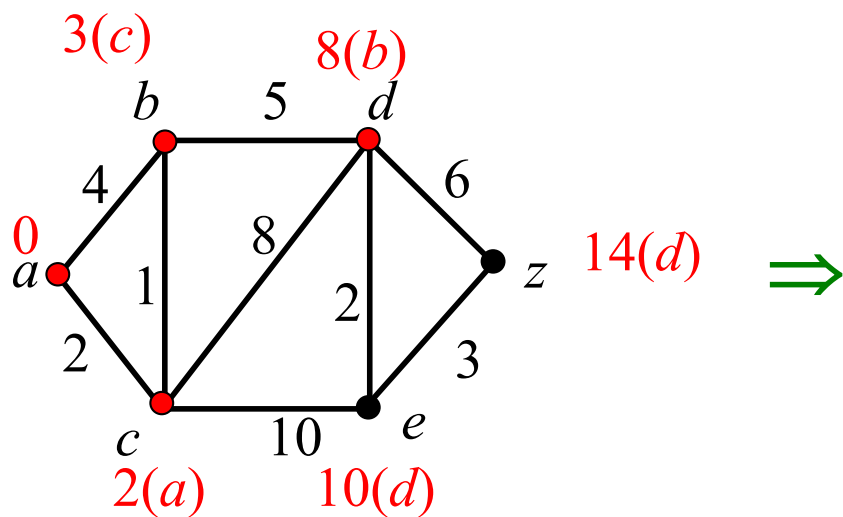
FIGURE 4 Using Dijkstra's Algorithm to Find a Shortest Path from a to z .

Example 2. Use Dijkstra's algorithm to find the length of a shortest path between a and z in the weighted graph.

Sol.



(前頁最後一個圖)



⇒ path: a, c, b, d, e, z
length: 13

Thm. 1

Dijkstra's algorithm finds the length of a shortest path between two vertices in a connected simple undirected weighted graph.

Thm. 2

Dijkstra's algorithm uses $O(n^2)$ operations (additions and comparisons) to find the length of a shortest path between two vertices in a connected simple undirected weighted graph with n vertices.

Floyd's Algorithm (find the distance $d(a, b) \forall a, b$)

Procedure *Floyd*(G : weighted simple graph)

$\{G$ has vertices $v_1, \dots, v_n$ and weights $w(v_i, v_j)$ with
 $w(v_i, v_j) = \infty$ if $\{v_i, v_j\}$ is not an edge $\}$

for $i := 1$ **to** n

for $j := 1$ **to** n

$d(v_i, v_j) := w(v_i, v_j)$

for $i := 1$ **to** n

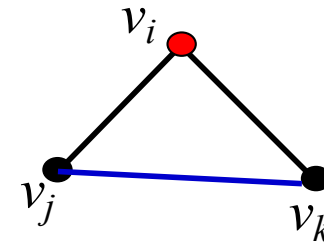
for $j := 1$ **to** n

for $k := 1$ **to** n

if $d(v_j, v_i) + d(v_i, v_k) < d(v_j, v_k)$

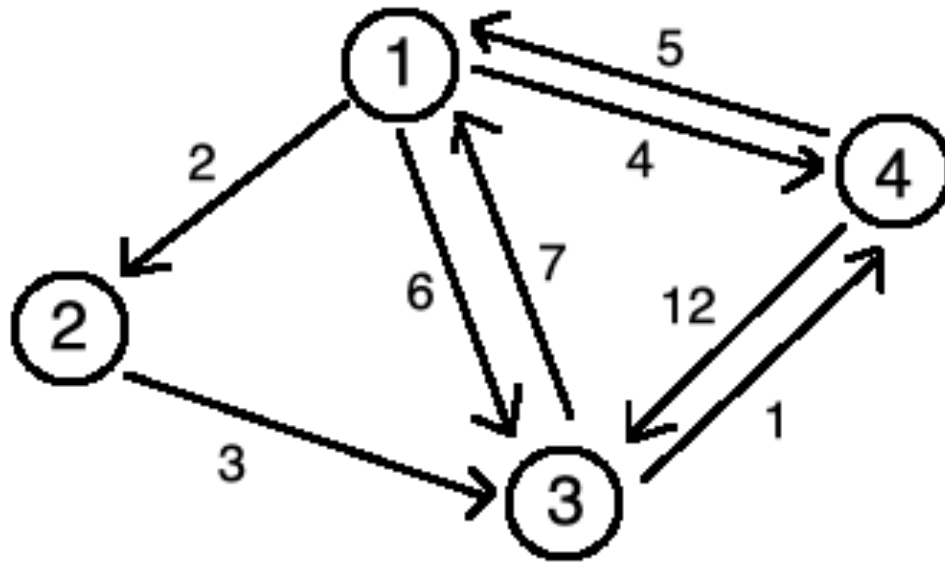
then $d(v_j, v_k) := d(v_j, v_i) + d(v_i, v_k)$

$\{d(v_i, v_j)$ is the length of a shortest path between v_i and $v_j\}$



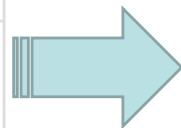
對每個 v_i ，檢查任何一對點 v_i, v_k 目前的距離會不會因為改走 v_i 而變小。

This algorithm cannot be used to construct shortest paths.

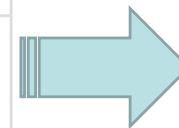


4 → 3 為 12
 4 → 1 → 3 為 11
 4 → 1 → 2 → 3 為 10

	1	2	3	4
1	0	2	6	4
2	∞	0	3	∞
3	7	∞	0	1
4	5	∞	12	0



	1	2	3	4
1	0	2	6	4
2	∞	0	3	∞
3	7	∞	0	1
4	5	∞	11	0

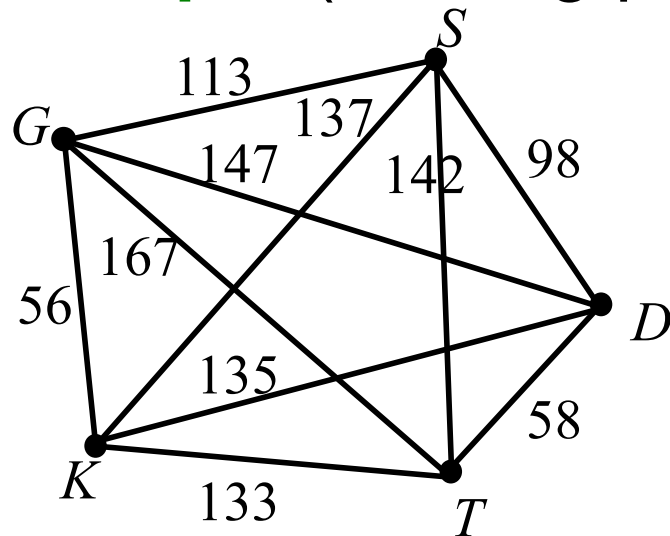


	1	2	3	4
1	0	2	6	4
2	∞	0	3	∞
3	7	∞	0	1
4	5	∞	10	0

The Traveling Salesman Problem: (簡介)

A traveling salesman wants to **visit each of n cities exactly once** and **return to his starting point**. In **which order** should he visit these cities to travel the **minimum total distance**?

Example (starting point D)



$$D \rightarrow T \rightarrow K \rightarrow G \rightarrow S \rightarrow D: 458$$

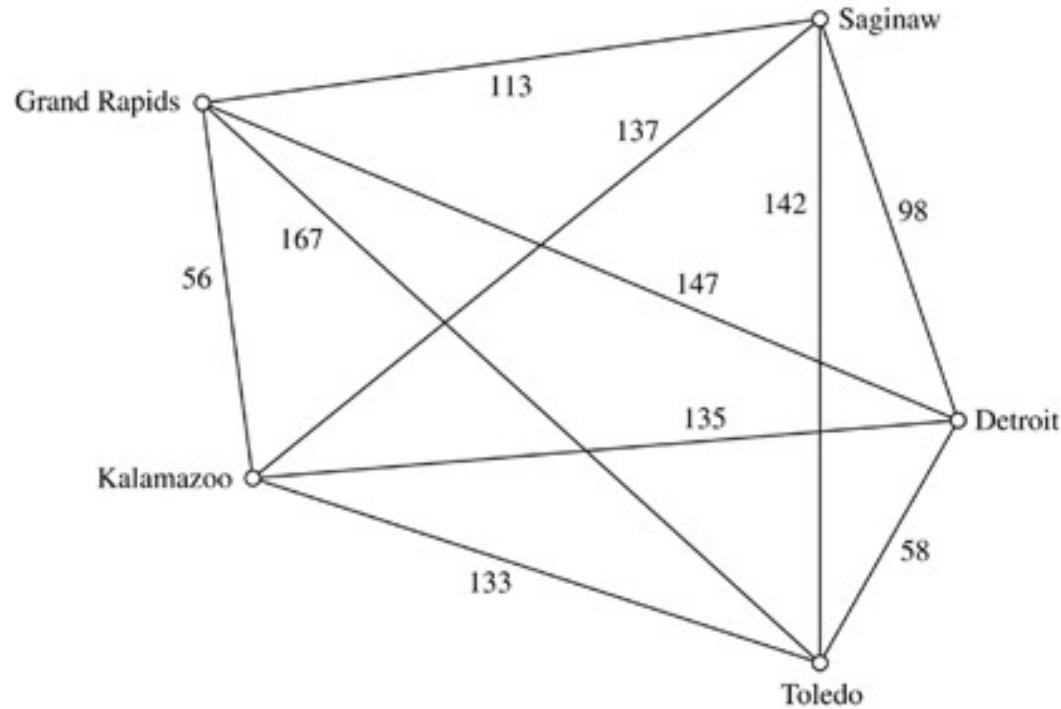
$$D \rightarrow T \rightarrow S \rightarrow G \rightarrow K \rightarrow D: 504$$

$$D \rightarrow T \rightarrow S \rightarrow K \rightarrow G \rightarrow D: 540$$

...

相當於在 complete weighted graph 上找 minimum weight Hamiltonian circuit.

不存在 polynomial time alg. 效率好的 algorithm 只能找近似解

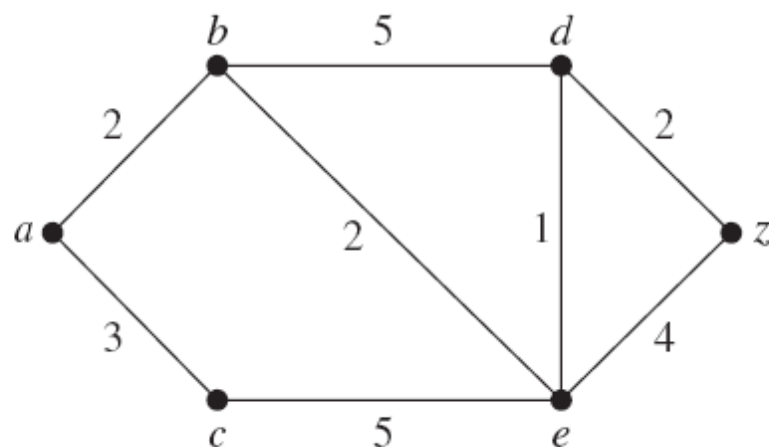
**FIGURE 5**

The Graph
Showing the
Distances
between Five
Cities.

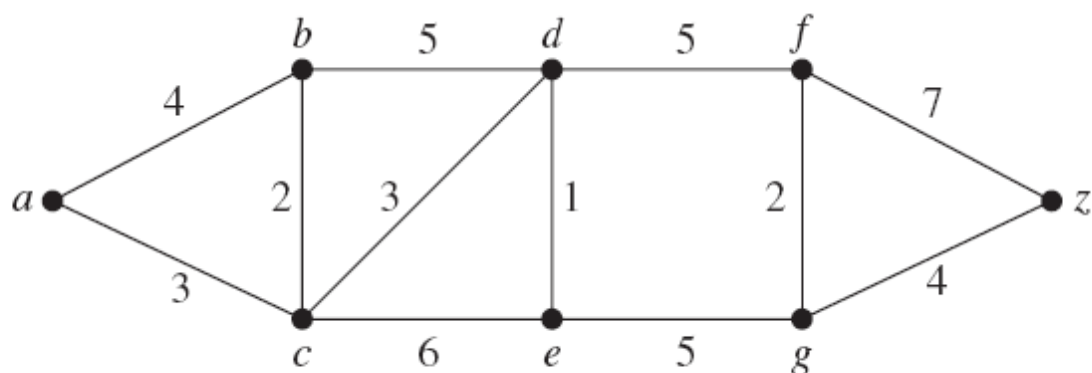
<i>Route</i>	<i>Total Distance (miles)</i>
Detroit–Toledo–Grand Rapids–Saginaw–Kalamazoo–Detroit	610
Detroit–Toledo–Grand Rapids–Kalamazoo–Saginaw–Detroit	516
Detroit–Toledo–Kalamazoo–Saginaw–Grand Rapids–Detroit	588
Detroit–Toledo–Kalamazoo–Grand Rapids–Saginaw–Detroit	458
Detroit–Toledo–Saginaw–Kalamazoo–Grand Rapids–Detroit	540
Detroit–Toledo–Saginaw–Grand Rapids–Kalamazoo–Detroit	504
Detroit–Saginaw–Toledo–Grand Rapids–Kalamazoo–Detroit	598
Detroit–Saginaw–Toledo–Kalamazoo–Grand Rapids–Detroit	576
Detroit–Saginaw–Kalamazoo–Toledo–Grand Rapids–Detroit	682
Detroit–Saginaw–Grand Rapids–Toledo–Kalamazoo–Detroit	646
Detroit–Grand Rapids–Saginaw–Toledo–Kalamazoo–Detroit	670
Detroit–Grand Rapids–Toledo–Saginaw–Kalamazoo–Detroit	728

In Exercises 2–4 find the length of a shortest path between a and z in the given weighted graph.

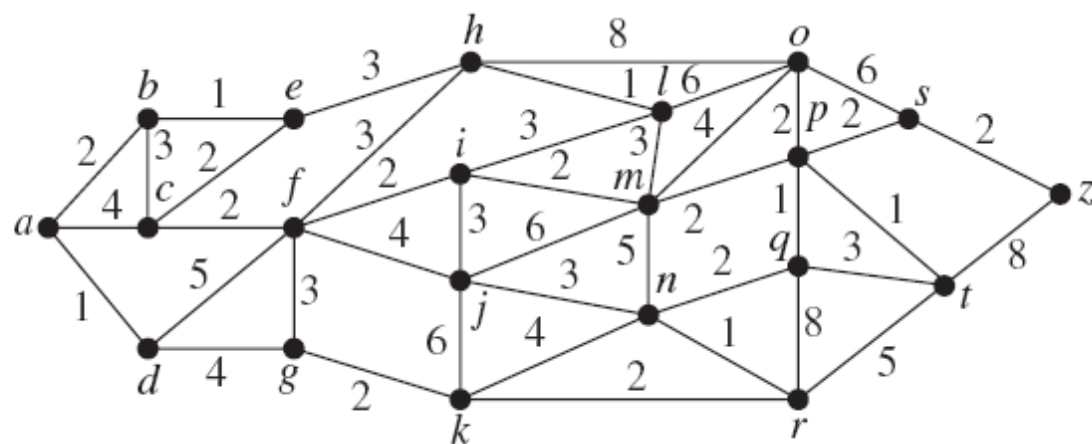
2.



3.



4.



8. Find a shortest path (in mileage) between each of the following pairs of cities in the airline system shown in Figure 1.

- New York and Los Angeles
- Boston and San Francisco
- Miami and Denver
- Miami and Los Angeles

9. Find a combination of flights with the least total air time between the pairs of cities in Exercise 8, using the flight times shown in Figure 1.

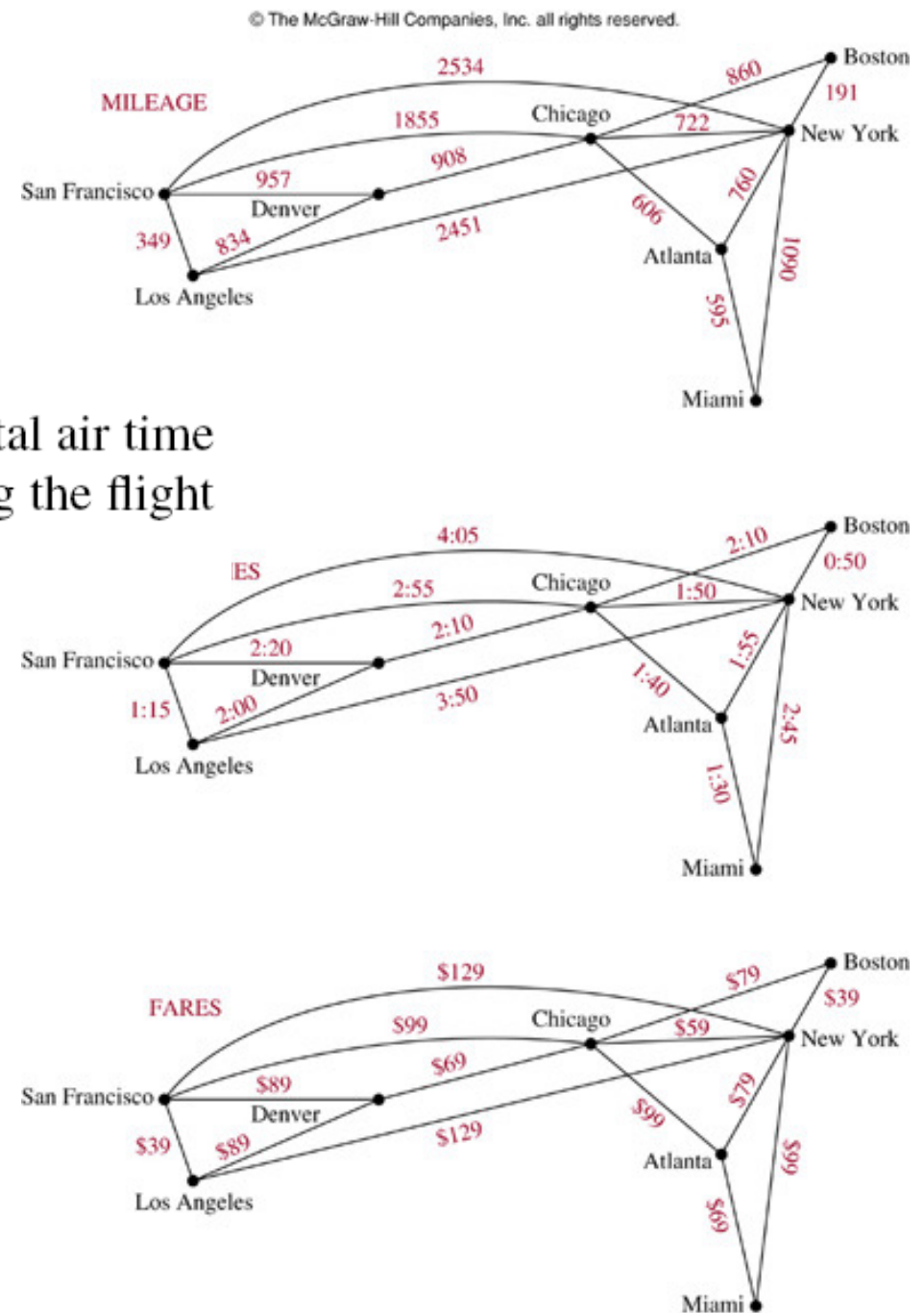


FIGURE 1 Weighted Graphs Modeling and Airline System.

§10.7: Planar Graphs

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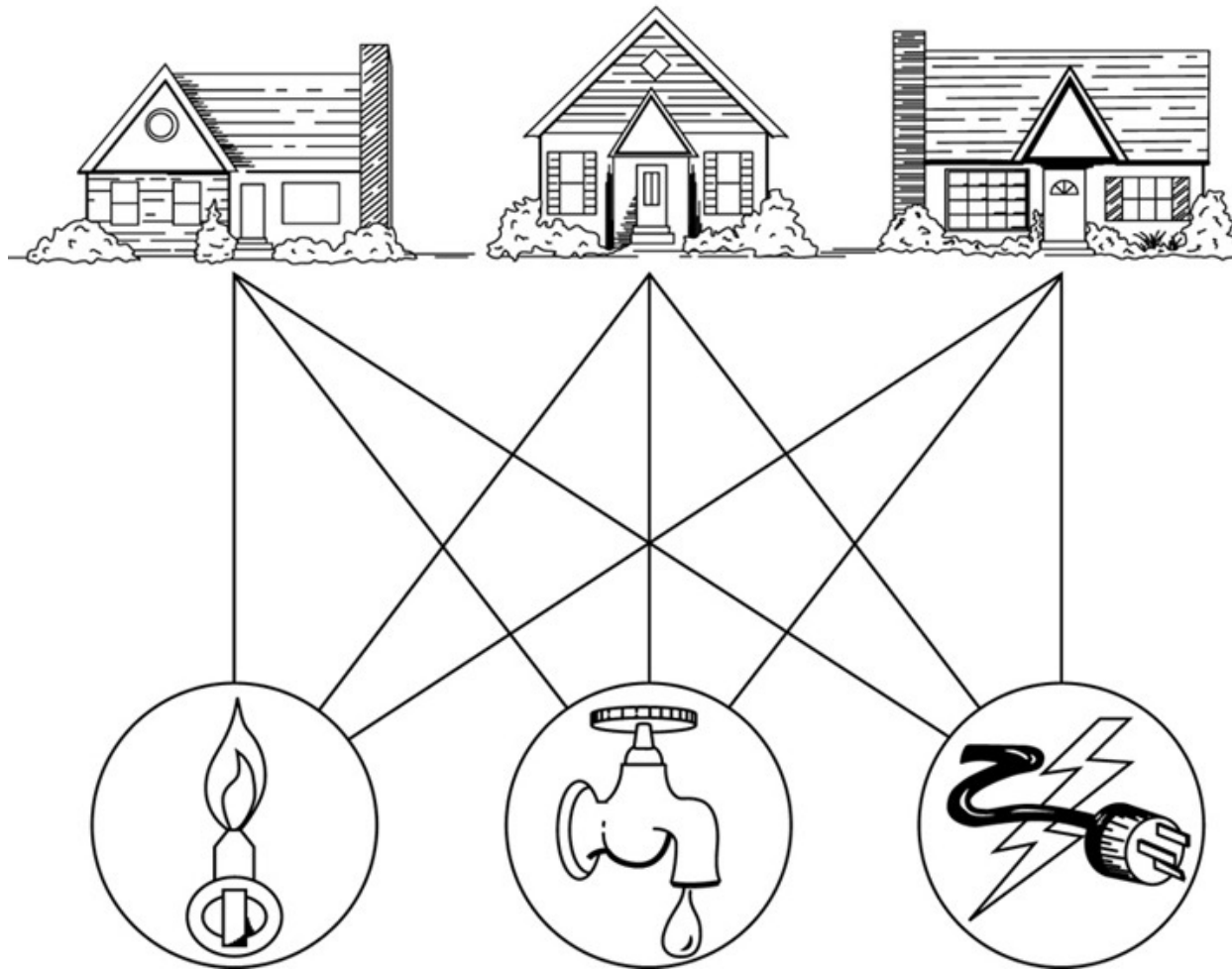


FIGURE 1 Three Houses and Three Utilities.

§10.7: Planar Graphs

Def 1.

A graph is called **planar** if it can be drawn in the plane **without any edge crossing**. Such a drawing is called a **planar representation** of the graph.

Example 1: Is K_4 planar?

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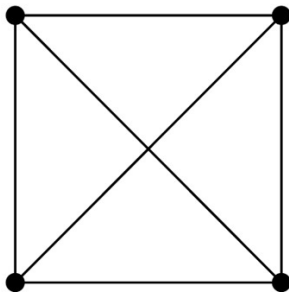
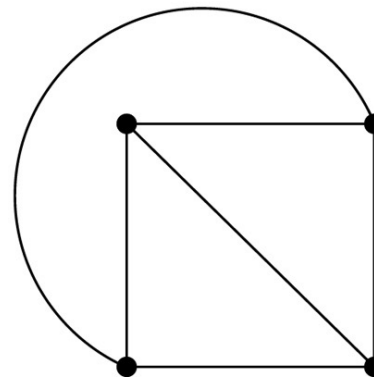
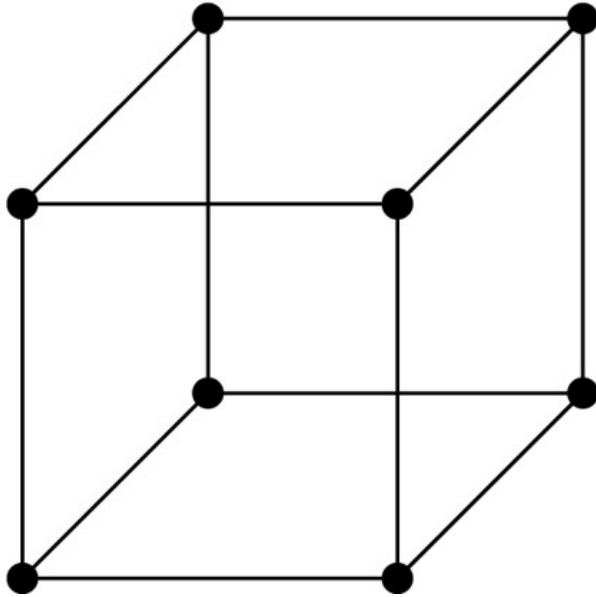


FIGURE 2 The Graph K_4 .



$\therefore K_4$ is planar

FIGURE 3 K_4 Drawn with No Crossings.



Example 2: Is Q_3 planar?

FIGURE 4 The Graph Q_3 .

$\therefore Q_3$ is planar

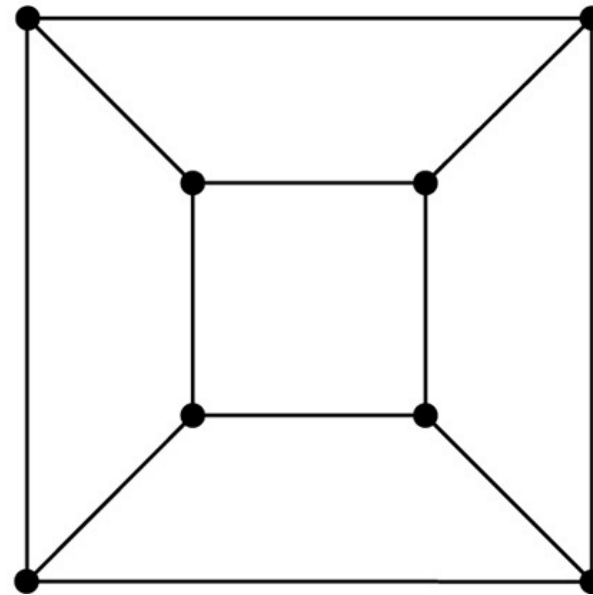
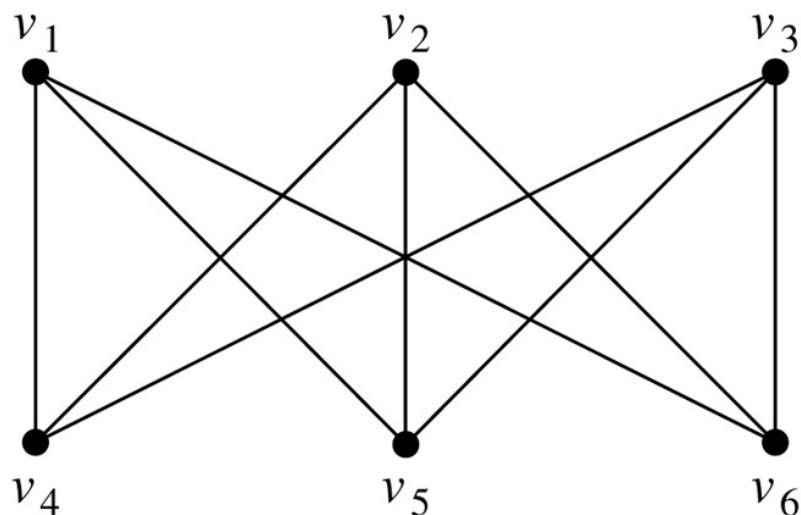


FIGURE 5 A Planar Representation of Q_3 .

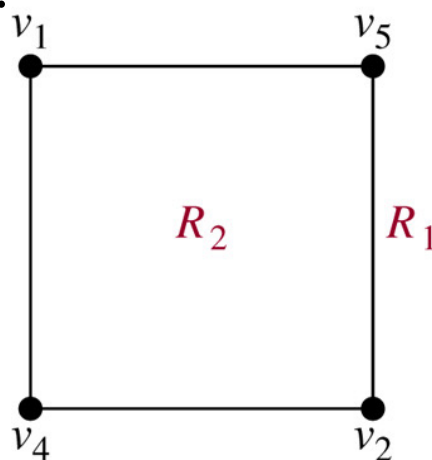
Example 3: Show that $K_{3,3}$ is nonplanar.



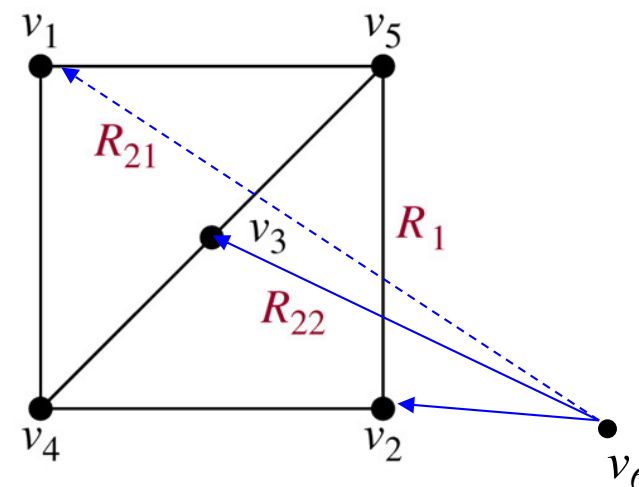
不管 v_3 在哪個region，都無法再將 v_6 放入而使邊不交錯

FIGURE 6 The Graph $K_{3,3}$.

任何畫法中， v_1, v_4, v_2, v_5 都是 cycle，並將平面切成兩個 region



(a)



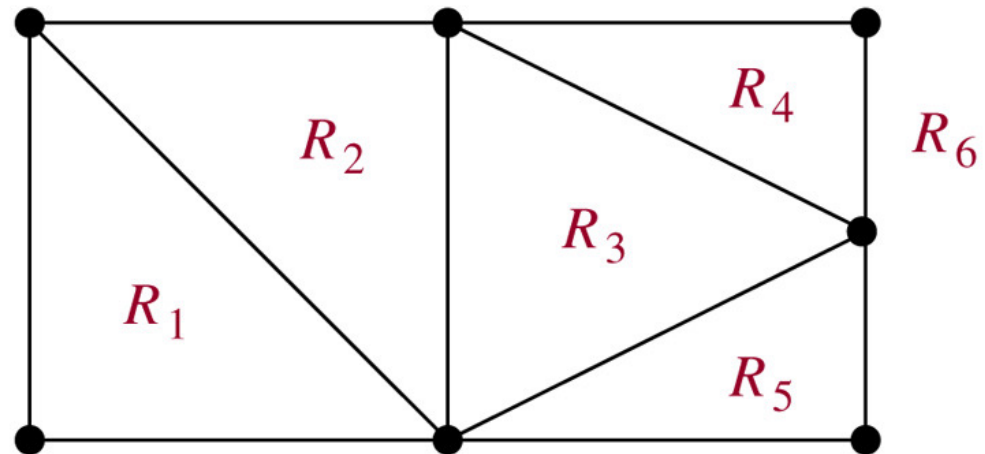
(b)

FIGURE 7 Showing that $K_{3,3}$ Is Nonplanar.

Euler's Formula

A planar representation of a graph **splits the plane into regions**, including an unbounded region.

Example : How many regions are there in the following graph?



Sol. $6 = 11 - 7 + 2$

Thm 1 (Euler's Formula)

Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of **regions** in a planar representation of G . Then $r = e - v + 2$.

Proof: First, we specify a planar representation of G . We will prove the theorem by constructing a sequence of subgraphs $G_1, G_2, \dots, G_e = G$, successively adding an edge at each stage. This is done using the following inductive definition. Arbitrarily pick one edge of G to obtain G_1 . Obtain G_n from G_{n-1} by arbitrarily adding an edge that is incident with a vertex already in G_{n-1} , adding the other vertex incident with this edge if it is not already in G_{n-1} . This construction is possible because G is connected. G is obtained after e edges are added. Let r_n, e_n , and v_n represent the number of regions, edges, and vertices of the planar representation of G_n induced by the planar representation of G , respectively.

The proof will now proceed by induction. The relationship $r_1 = e_1 - v_1 + 2$ is true for G_1 , because $e_1 = 1$, $v_1 = 2$, and $r_1 = 1$. This is shown in Figure 9.



FIGURE 9 The Basis Case of the Proof of Euler's Formula.

Now assume that $r_k = e_k - v_k + 2$. Let $\{a_{k+1}, b_{k+1}\}$ be the edge that is added to G_k to obtain G_{k+1} . There are two possibilities to consider. In the first case, both a_{k+1} and b_{k+1} are already in G_k . These two vertices must be on the boundary of a common region R , or else it would be impossible to add the edge $\{a_{k+1}, b_{k+1}\}$ to G_k without two edges crossing (and G_{k+1} is planar). The addition of this new edge splits R into two regions. Consequently, in this case, $r_{k+1} = r_k + 1$, $e_{k+1} = e_k + 1$, and $v_{k+1} = v_k$. Thus, each side of the formula relating the number of regions, edges, and vertices increases by exactly one, so this formula is still true. In other words, $r_{k+1} = e_{k+1} - v_{k+1} + 2$. This case is illustrated in Figure 10(a).

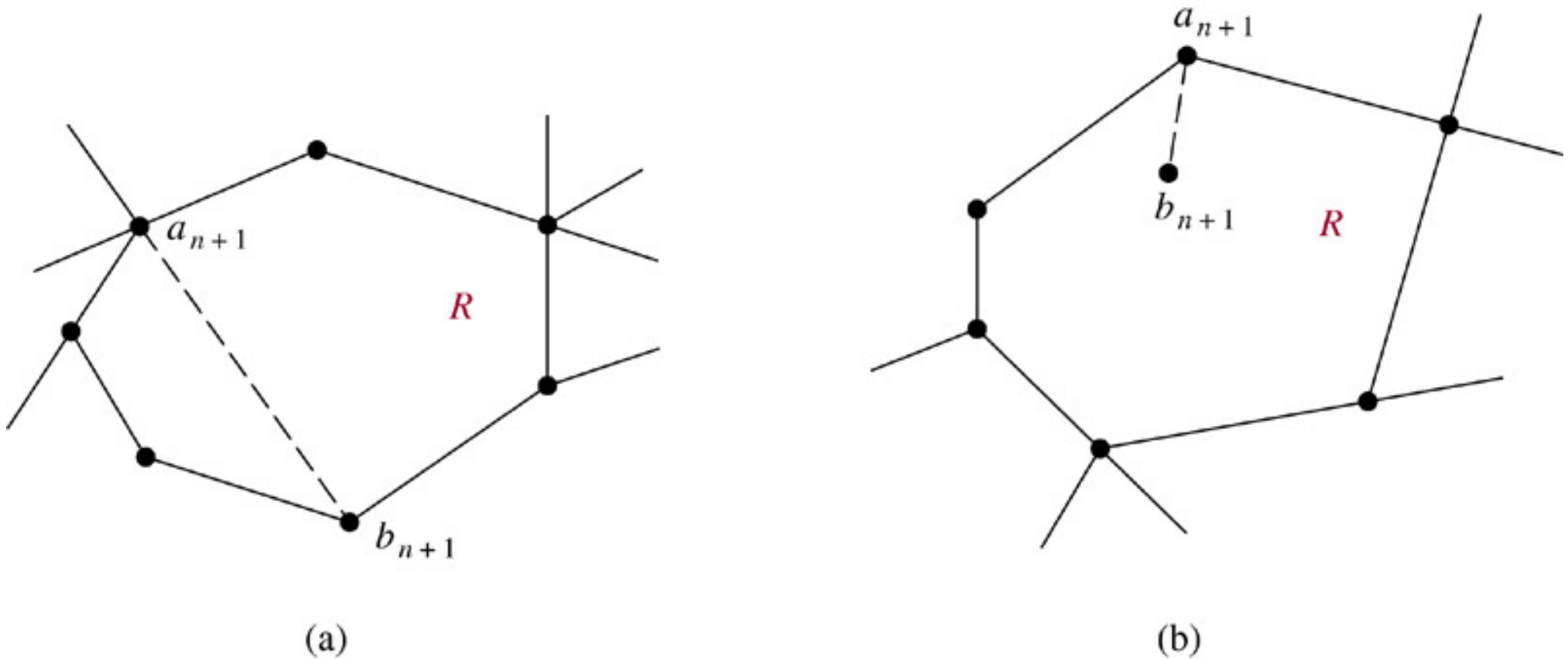
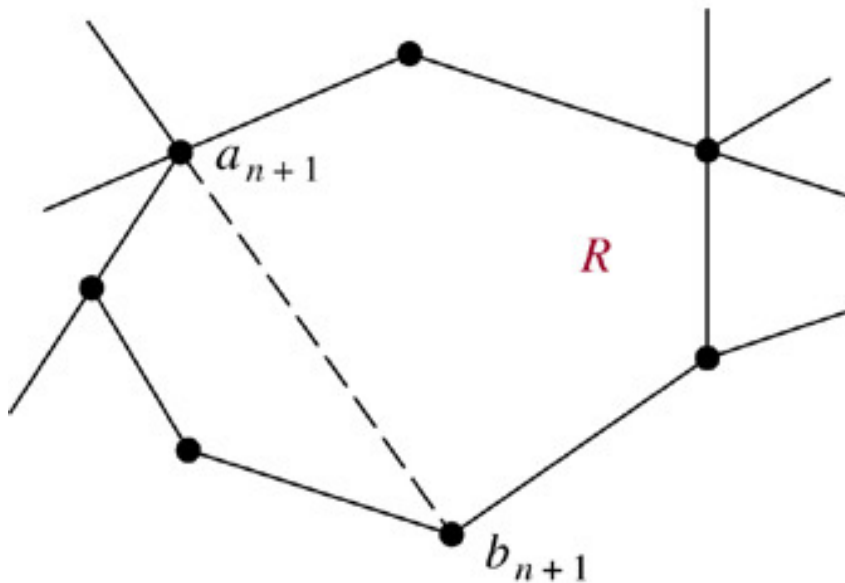


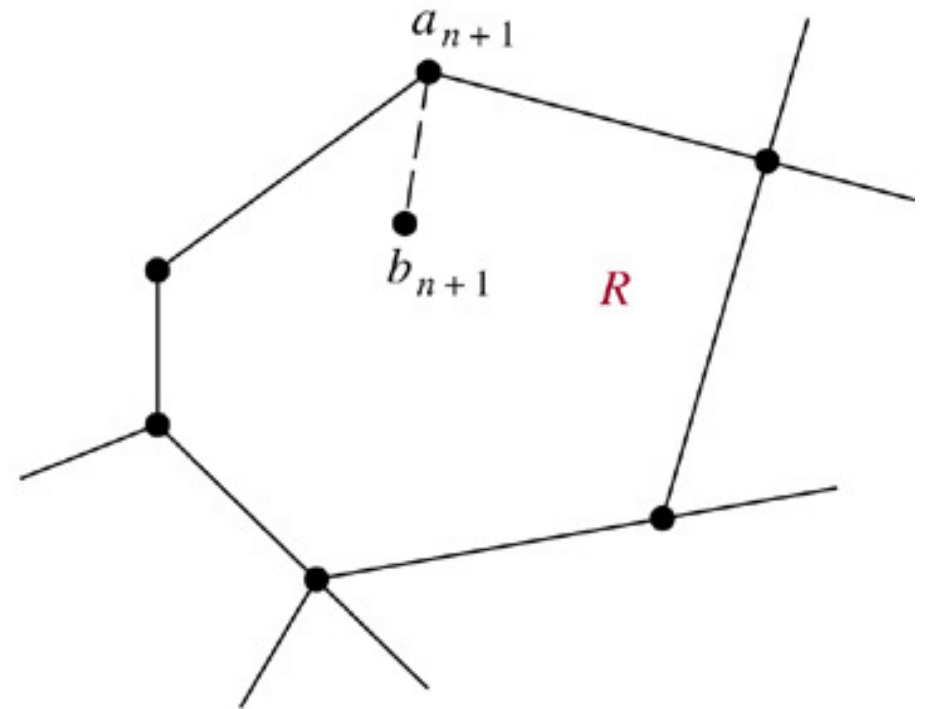
FIGURE 10 Adding an Edge to G_n to Produce G_{n+1} .

In the second case, one of the two vertices of the new edge is not already in G_k . Suppose that a_{k+1} is in G_k but that b_{k+1} is not. Adding this new edge does not produce any new regions, because b_{k+1} must be in a region that has a_{k+1} on its boundary. Consequently, $r_{k+1} = r_k$. Moreover, $e_{k+1} = e_k + 1$ and $v_{k+1} = v_k + 1$. Each side of the formula relating the number of regions, edges, and vertices remains the same, so the formula is still true. In other words, $r_{k+1} = e_{k+1} - v_{k+1} + 2$. This case is illustrated in Figure 10(b).

We have completed the induction argument. Hence, $r_n = e_n - v_n + 2$ for all n . Because the original graph is the graph G_e , obtained after e edges have been added, the theorem is true. ◀



(a)



(b)

FIGURE 10 Adding an Edge to G_n to Produce G_{n+1} .

Example 4: Suppose that a connected planar graph has 20 vertices, each of degree 3. Into how many regions does a representation of this planar graph split the plane?

Sol.

$$v = 20, 2e = 3 \times 20 = 60, e = 30$$

$$r = e - v + 2 = 30 - 20 + 2 = 12$$

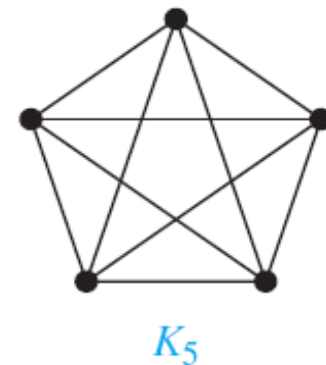
Corollary 1

If G is a connected planar simple graph with e edges and v vertices, where $v \geq 3$, then $e \leq 3v - 6$.

Example 5: Show that K_5 is nonplanar.

Sol.

$$v = 5, e = 10, \text{ but } 3v - 6 = 9.$$



Corollary 2

If G is a connected planar simple graph, then G has a vertex of degree ≤ 5 .

pf: Let G be a planar graph of v vertices and e edges.

If $\deg(v) \geq 6$ for every $v \in V(G)$

$$\Rightarrow \sum_{v \in V(G)} \deg(v) \geq 6v$$

$$\Rightarrow 2e \geq 6v \quad \rightarrow \leftarrow (e \leq 3v - 6)$$

FIGURE 11 (9.7)

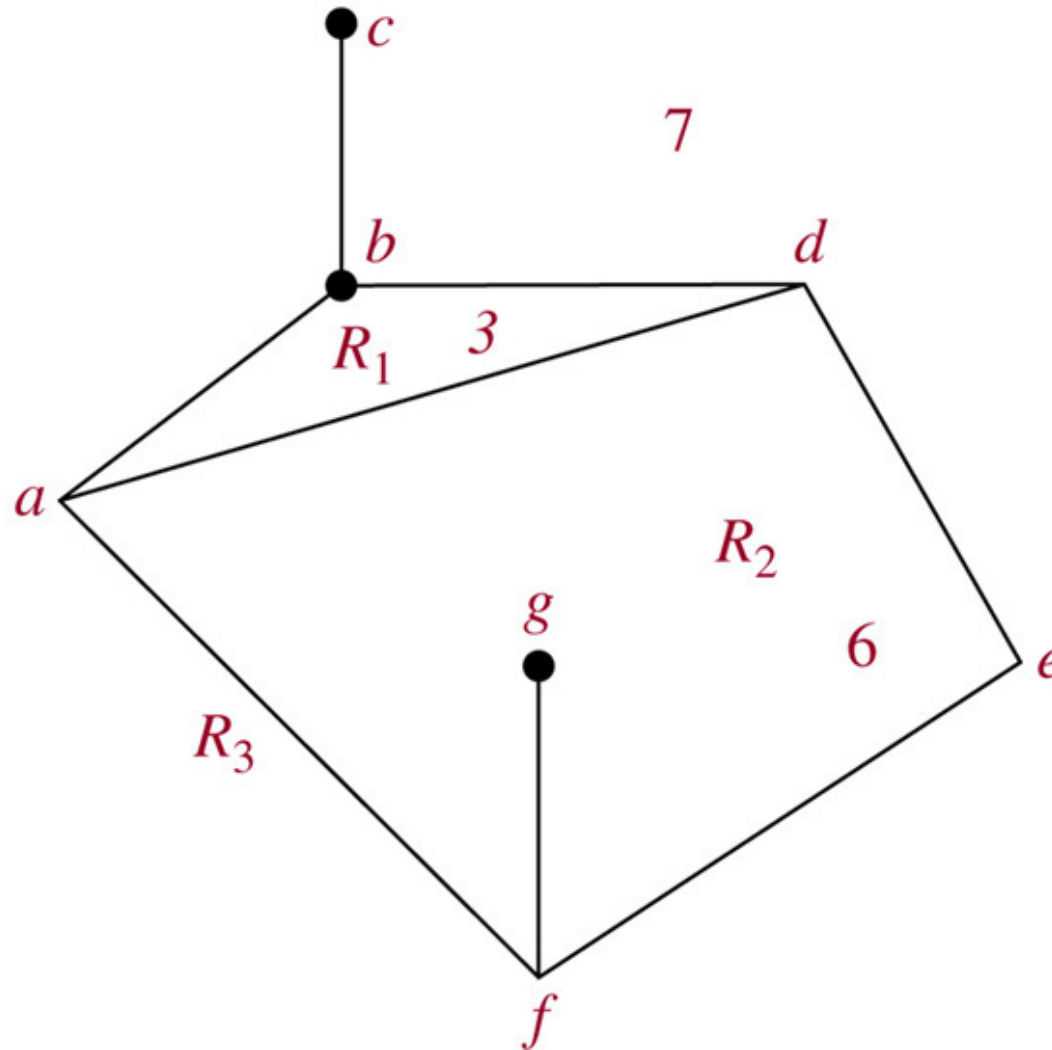


FIGURE 11 The Degrees of Regions.

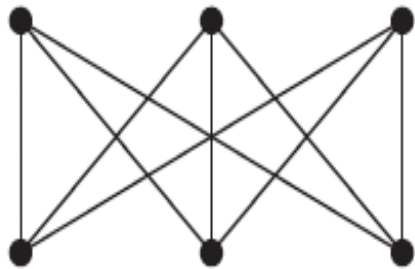
Corollary 3

If a connected planar simple graph has e edges and v vertices with $v \geq 3$ and no circuits of length three, then $e \leq 2v - 4$.

Example 6: Show that $K_{3,3}$ is nonplanar by Cor. 3.

Sol.

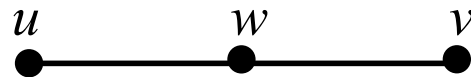
Because $K_{3,3}$ has no circuits of length three, and $v = 6$, $e = 9$, but $e = 9 > 2v - 4$.



$K_{3,3}$

Kuratowski's Theorem

If a graph is planar, so will be any graph obtained by removing an edge $\{u, v\}$ and adding a new vertex w together with edges $\{u, w\}$ and $\{v, w\}$.



Such an operation is called an **elementary subdivision**.

Two graphs $G_1 = (V_1, E_1)$, $G_2 = (V_2, E_2)$ are called **homeomorphic** if they can be obtained from the same graph by a sequence of elementary subdivisions.

Example 7: Show that the graphs G_1 , G_2 , and G_3 are all homeomorphic.

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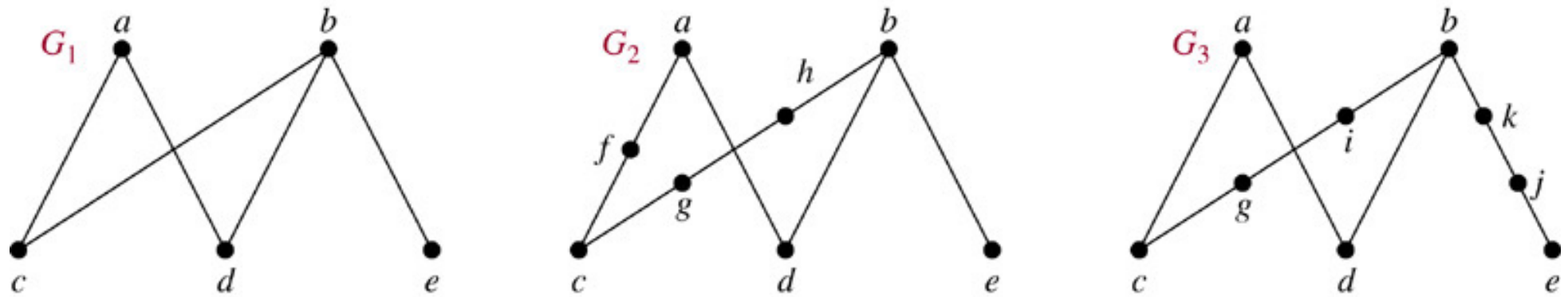


FIGURE 12 Homeomorphic Graphs.

Sol: all three can be obtained from G_1

Thm 2. (Kuratowski Theorem)

A graph is nonplanar if and only if it contains a subgraph homeomorphic to $K_{3,3}$ or K_5 .

FIGURE 13 (10.7)

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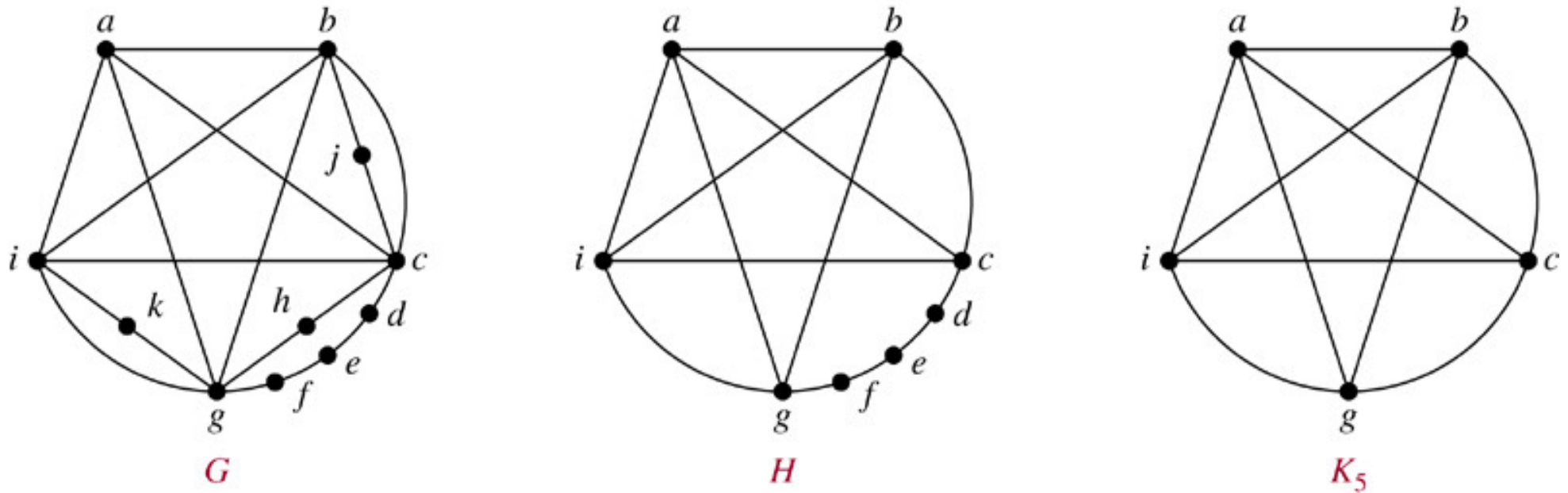
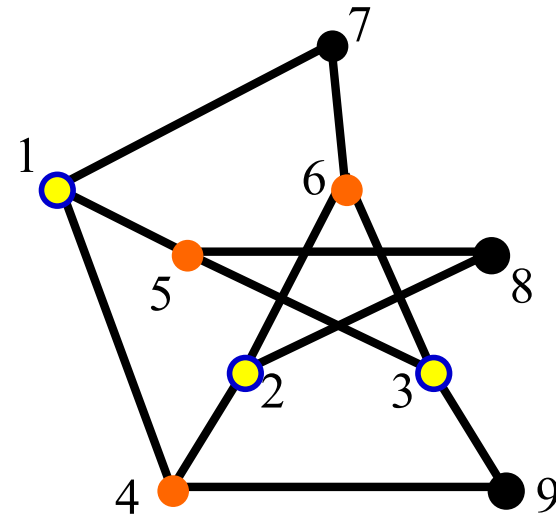
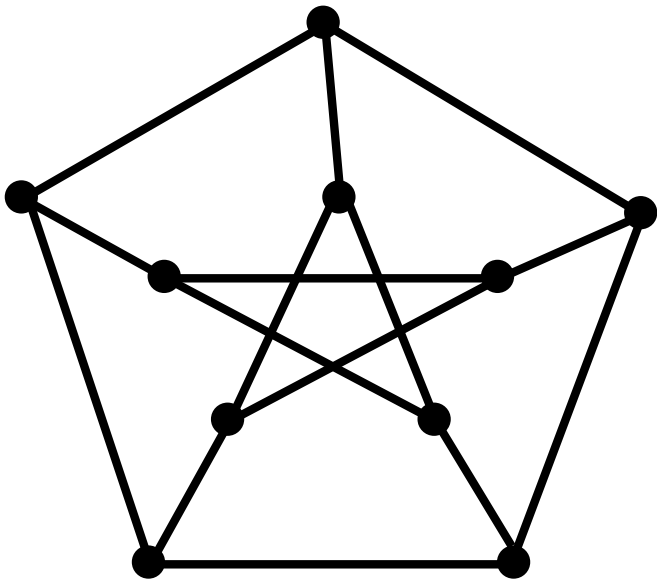


FIGURE 13 The Undirected Graph G , a Subgraph H Homeomorphic to K_5 , and K_5 .

Example 9: Show that the Petersen graph is not planar.

Sol:



It is homeomorphic to $K_{3,3}$.

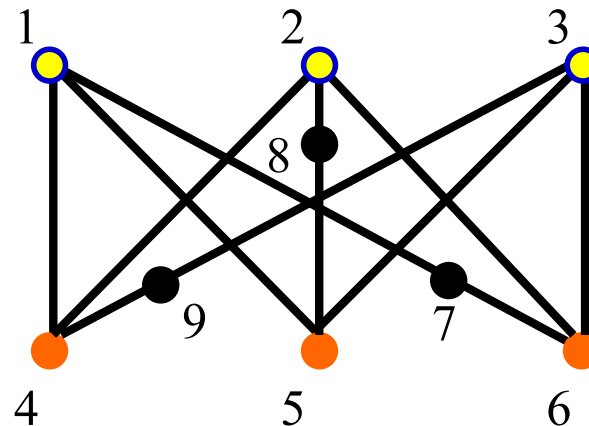


FIGURE 14 (10.7)

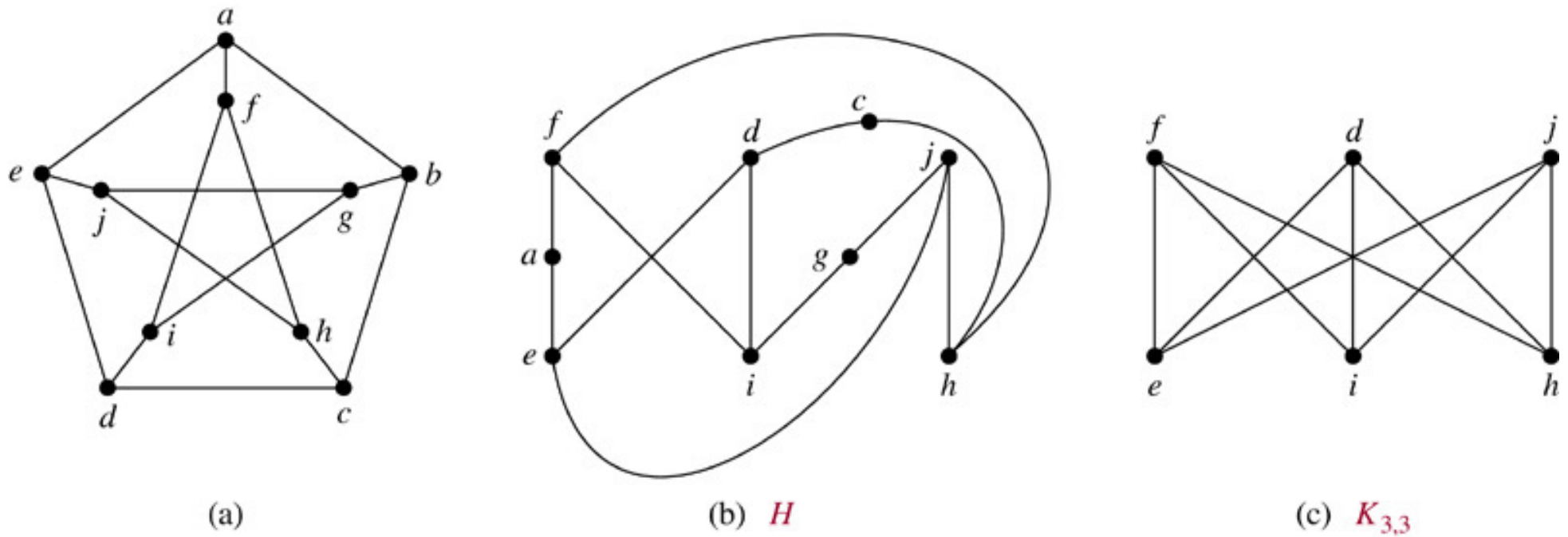


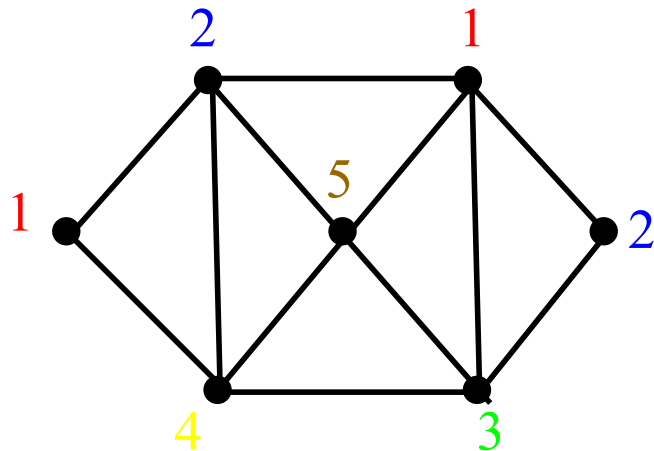
FIGURE 14 (a) The Petersen Graph, (b) a Subgraph H Homeomorphic to $K_{3,3}$, and (c) $K_{3,3}$.

§10.8: Graph Coloring

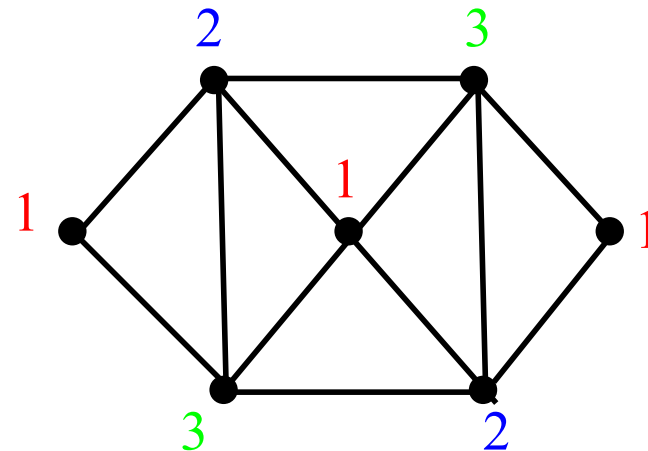
Def. 1:

A **coloring** of a simple graph is the assignment of a color to each vertex of the graph so that **no two adjacent vertices are assigned the same color**.

Example:



5-coloring



3-coloring

顏色數越少越好

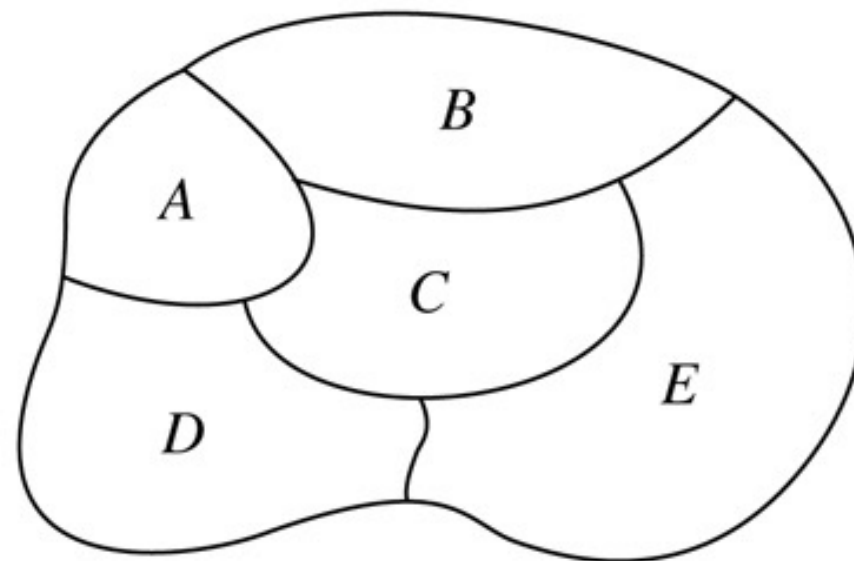
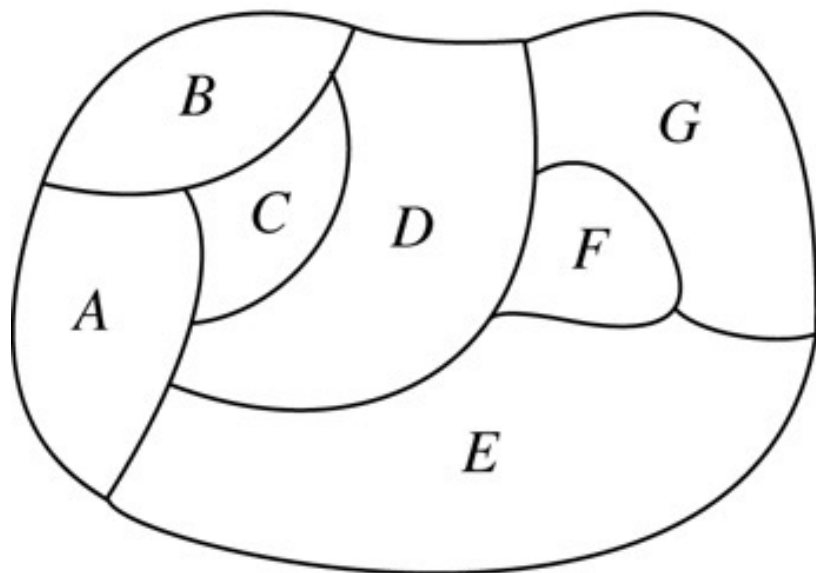


FIGURE 1 Two Maps.

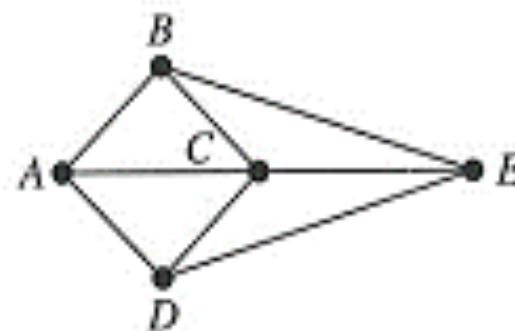
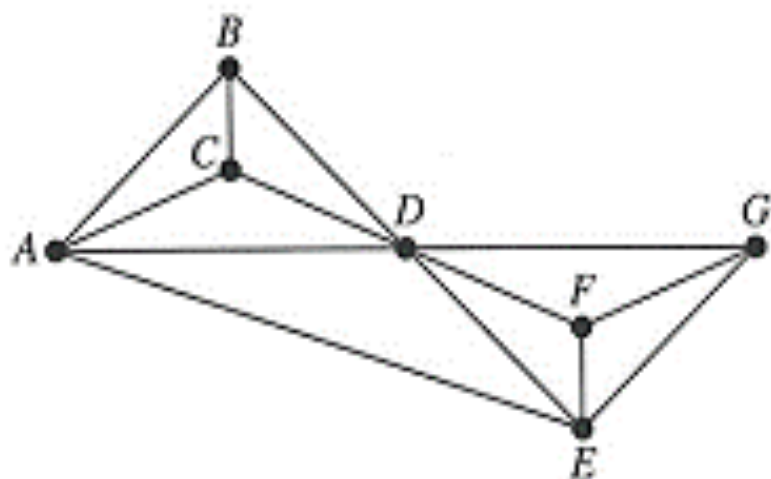
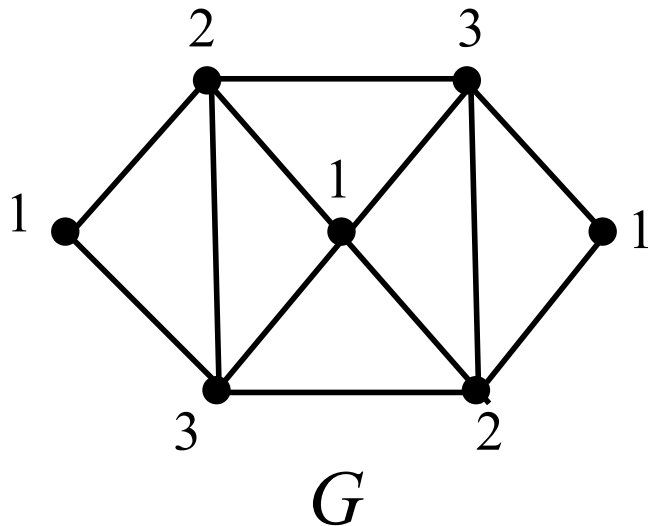
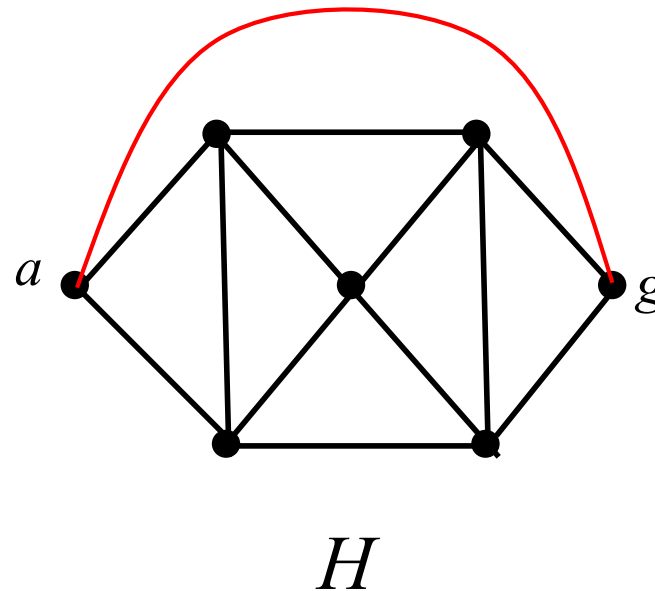


FIGURE 2 Dual Graph of the Maps in Figure 1.

Example 1: What are the chromatic numbers of the graphs G and H ?



Sol: G has a 3-cycle
 $\Rightarrow \chi(G) \geq 3$
 G has a 3-coloring
 $\Rightarrow \chi(G) \leq 3$
 $\Rightarrow \chi(G) = 3$



Sol: any 3-coloring for
 $H - \{(a, g)\}$ gives the
same color to a and g
 $\Rightarrow \chi(H) > 3$
4-coloring exists $\Rightarrow \chi(H) = 4$

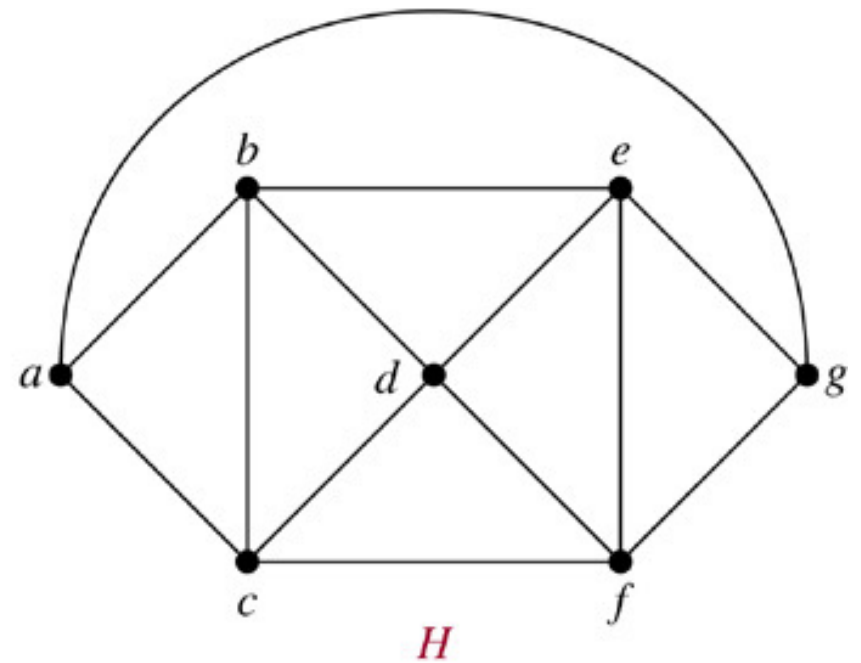
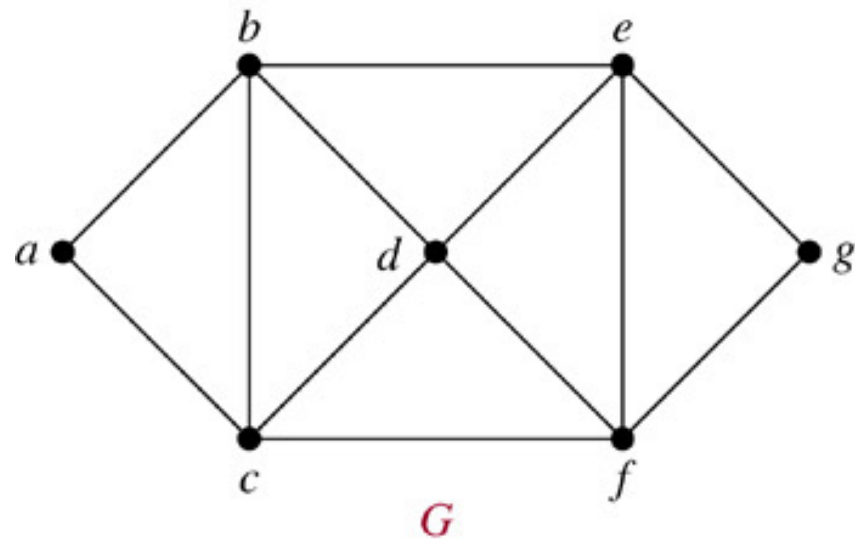


FIGURE 3 The Simple Graphs G and H .

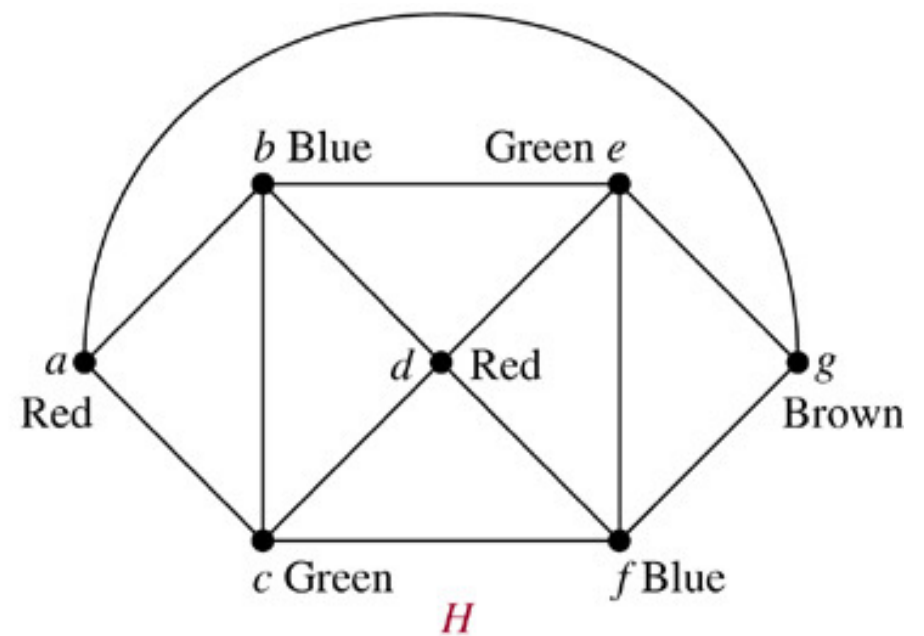
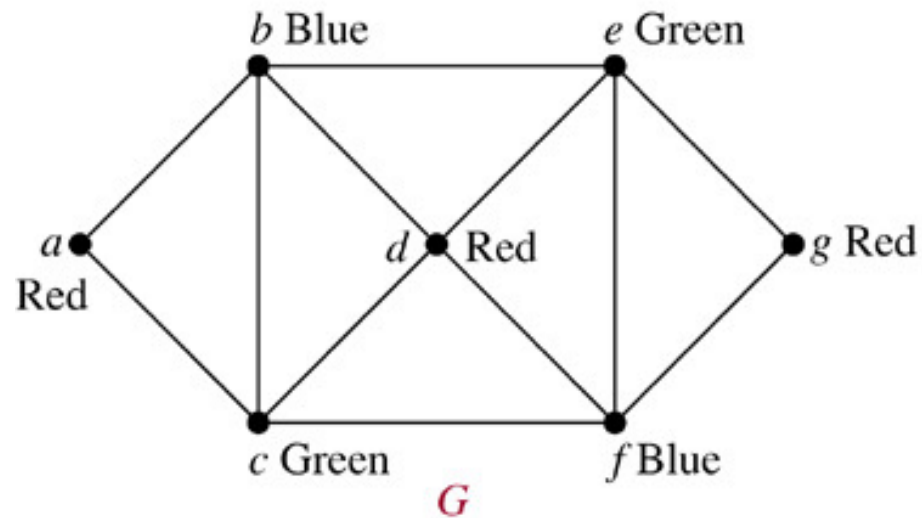


FIGURE 4 Colorings of the Graphs G and H .

Def. 2:

The **chromatic number** of a graph is the **least number of colors needed** for a coloring of this graph. (denoted by $\chi(G)$)

Example 2: $\chi(K_5)=5$

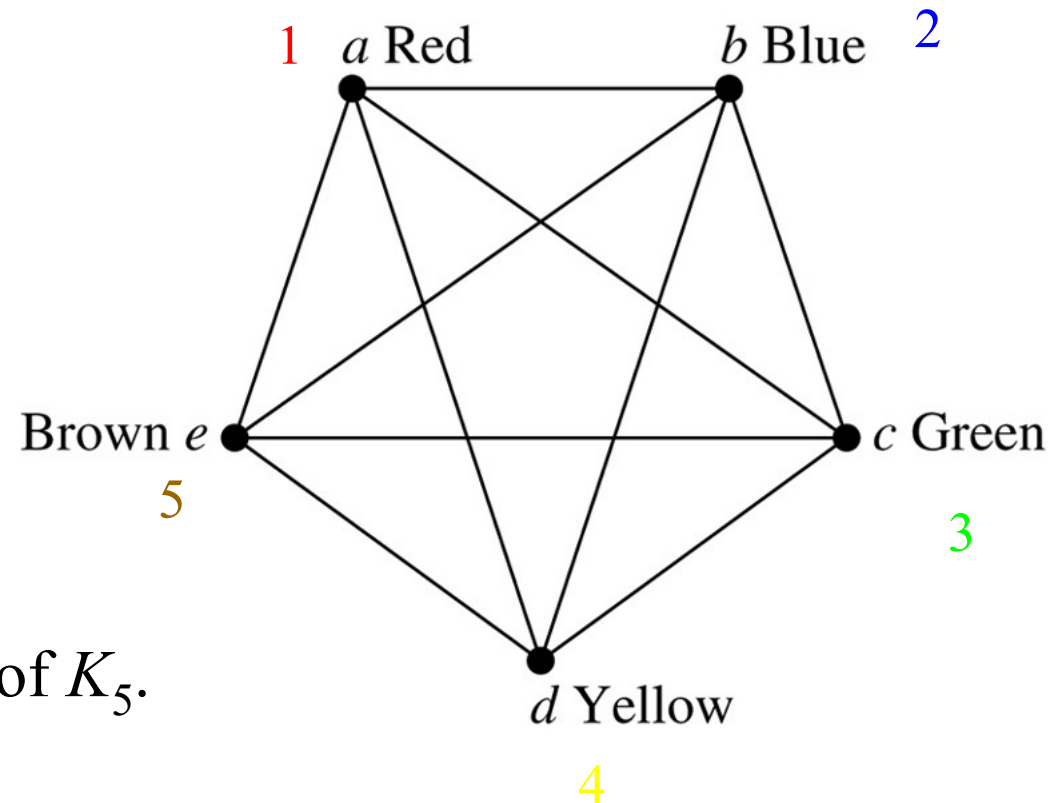


FIGURE 5 A Coloring of K_5 .

Note: $\chi(K_n)=n$

Example: $\chi(K_{3,4}) = 2$.

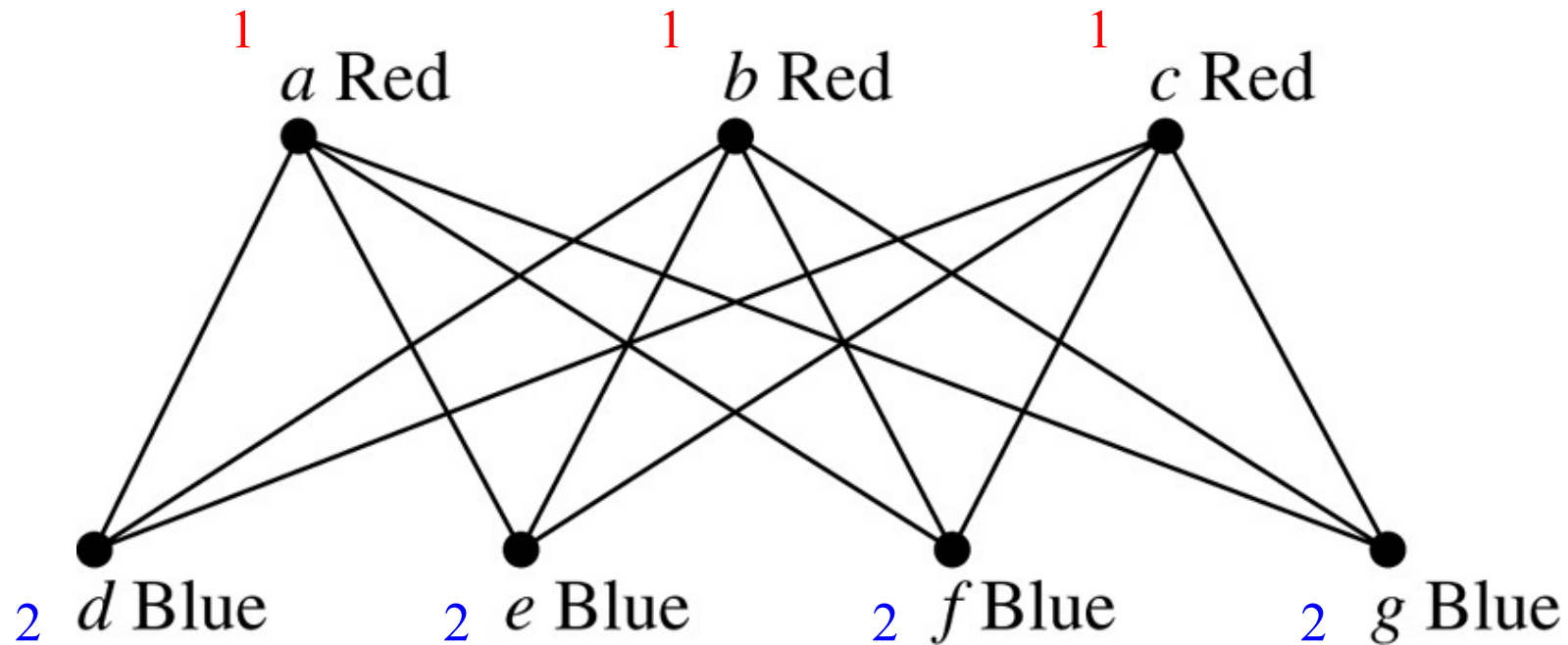


FIGURE 6 A Coloring of $K_{3,4}$.

Note: $\chi(K_{m,n}) = 2$

Note: If G is a bipartite graph, $\chi(G) = 2$.

Example 4: $\chi(C_n) = \begin{cases} 2 & \text{if } n \text{ is even,} \\ 3 & \text{if } n \text{ is odd.} \end{cases}$

C_n is bipartite
when n is even.

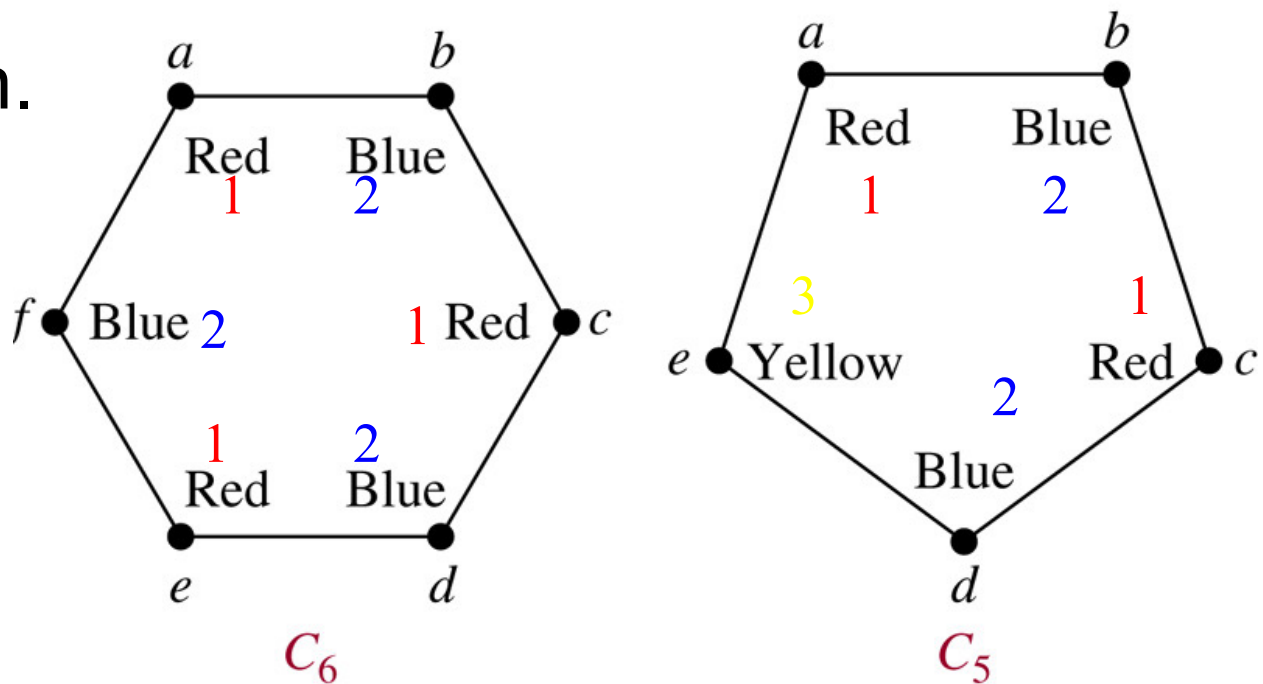


FIGURE 7 Colorings of C_5 and C_6 .

Example 4: $\chi(C_n) = \begin{cases} 2 & \text{if } n \text{ is even,} \\ 3 & \text{if } n \text{ is odd.} \end{cases}$

C_n is bipartite
when n is even.

Thm 1. (The Four Color Theorem)

The chromatic number of a planar graph is no greater than four.

Corollary

Any graph with chromatic number >4 is nonplanar.

Applications of Graph Colorings

Example 5: Scheduling Final Exams

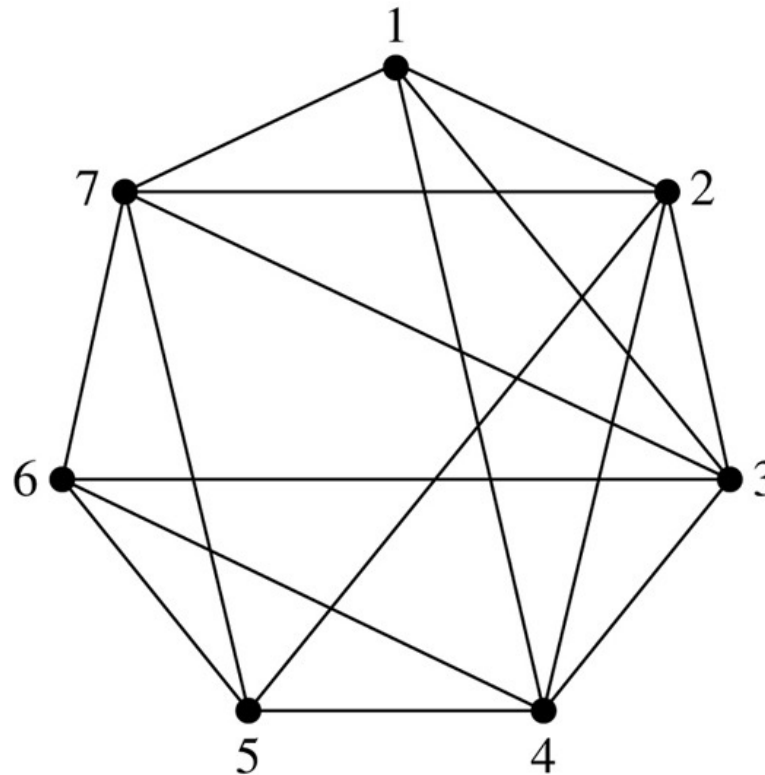


FIGURE 8 The Graph Representing the Scheduling of Final Exams.

Applications of Graph Colorings

Example 5: Scheduling Final Exams

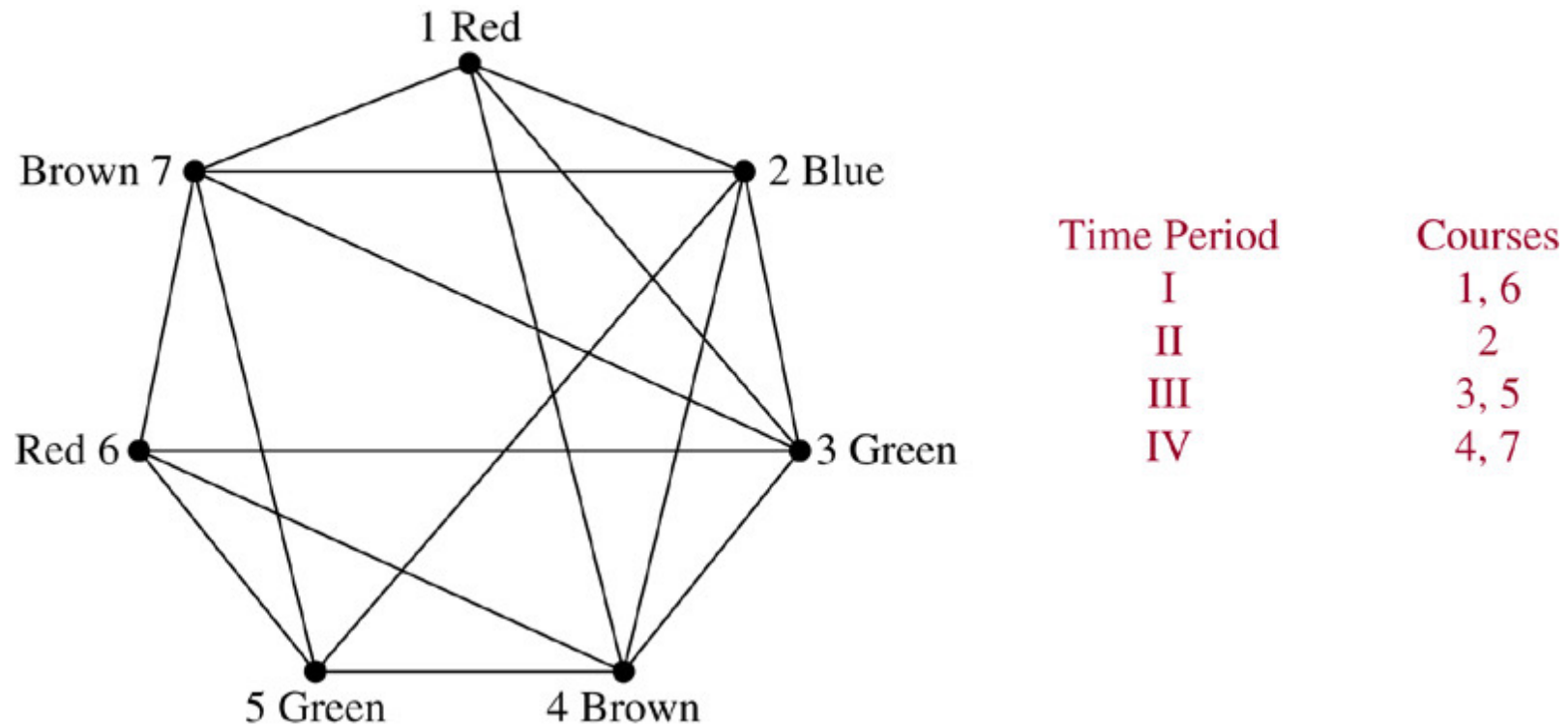
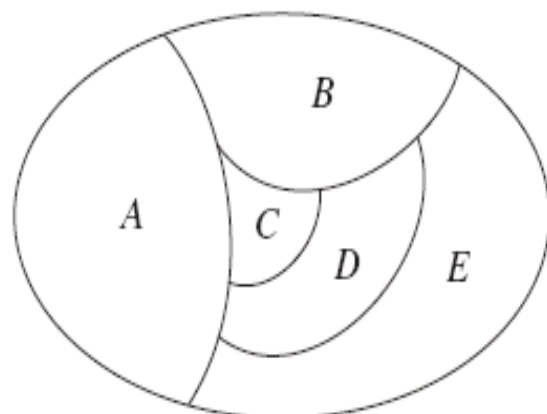


FIGURE 9 Using a Coloring to Schedule Final Exams.

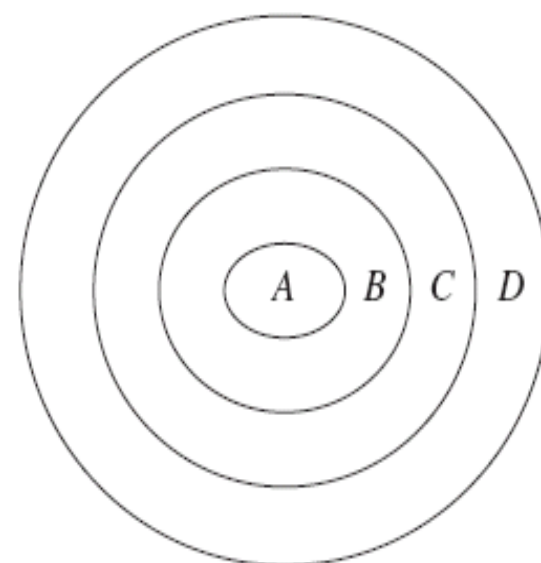
- Applications of Graph Colorings
 - Example 5. Scheduling Final Exams
 - Example 6. Frequency Assignments
 - Example 7. Index Registers

In Exercises 1–4 construct the dual graph for the map shown. Then find the number of colors needed to color the map so that no two adjacent regions have the same color.

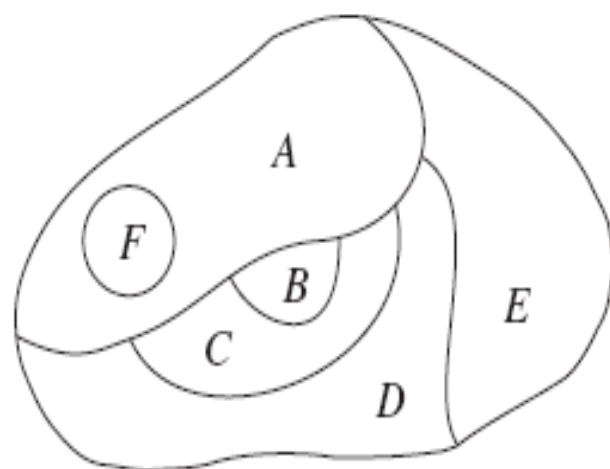
1.



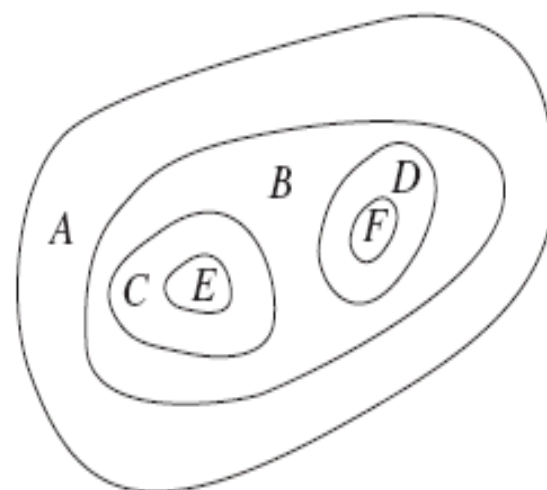
2.



3.



4.



In Exercises 5–11 find the chromatic number of the given graph.

